MouRa - Agri AI



FSN 111 – Principles of Food Science and Nutrition (1+1)

Theory Study Material

II - Semester

Course Teacher

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FSN 111 Principles of Food Science and Nutrition (1+1)

Theory Schedule

1. Food Science – definition, scope and classification, food pyramid.
2. Methods, merits and demerits of moist heat, dry heat and microwave cooking of foods.
3. Importance and scope of nutrition and the relation of nutrition to health.
4. Carbohydrate – classification, functions, digestion and absorption, deficiency symptoms, sources and requirements.
5. Protein – classification, functions, digestion and absorption, deficiency symptoms, sources and requirements. Protein quality – supplementary value of protein.
6. Fat - classification, functions, digestion and absorption, deficiency symptoms, sources and requirements. Rancidity – types. Determination of energy value of foods.
7. Fat soluble vitamins – A, D, E and K – functions, deficiency symptoms, sources and requirements.
8. Water soluble vitamins - thiamine, riboflavin, niacin, pyridoxine, folic acid, cyanacobalamin, biotin, pantothenic acid, ascorbic acid – functions, deficiency symptoms, sources and requirements.
9. **Mid Semester Examination**
10. Minerals – calcium, iron, phosphorus, iodine, magnesium, zinc, sodium, potassium, fluorine and chlorine – functions, sources, requirements and deficiency diseases.
11. Importance of water and maintenance of electrolyte balance. Health benefits of fibre.
12. Preservation of food by low and high temperature and food irradiation.
13. Processing of puffed, flaked and extruded products
14. Preservation by using sugar (jam, jelly, squash and marmalade), preservation by using salt (brining and pickling) and use of preservatives in food preservation.
15. Food packaging – importance, types of packaging materials and nutrition labeling.
16. Common food adulterants and their detection.
17. Food laws and regulations and quality control standards - FSSAI, ISO, EU standards, FDA, HACCP and Codex Alimentarius Commission.

LECTURE - 1

INTRODUCTION TO FOOD SCIENCE – NEED AND SCOPE OF FOOD SCIENCE

Food is the basic necessity of man. It is a mixture of different nutrients such as carbohydrate, protein, fat, vitamins and minerals. These nutrients are essential for growth, development and maintenance of good health throughout life. The selection of foods best suited for promoting health has been found out by trial and error by continued use. The knowledge of nutrition has been built upon the observation by several pioneers. **Lavoisier** is referred as **Father of Nutrition** due to his remarkable contribution in quantification of energy metabolism.

Food

Food is defined as any substance solid or liquid which when swallowed, digested and assimilated nourishes our body.

Food Science

Food Science can be defined as the application of the basic sciences and engineering to study the fundamental physical, chemical and biochemical nature of foods and the principles of food processing. It is a broad discipline which contains food chemistry, food analysis, food microbiology, food processing and food engineering.

Nutrition

Nutrition is defined as "the science of foods and nutrients, their action, interaction and balance in relationship to health and disease, the processes by which the organism ingests, digests, absorbs, transports and utilizes nutrients and disposes of their end product" (Robinson ,1982).

Health

Health is defined by the World Health Organization (WHO) as the "State of complete physical, mental and social well-being and not merely the absence of disease or infirmity".

Malnutrition as defined by World Health Organization (WHO) is a pathological state resulting from the intake of inadequate or excess quantity of food or nutrients.

Under-nutrition is the pathological state resulting from the intake of an inadequate quantity of food over an extended period of time.

Over-nutrition is the pathological state resulting from the intake of an excess quantity of food over an extended period of time.

Balanced diet

Balanced diet is a diet consisting of a variety of different types of food and providing adequate amounts of the nutrients necessary for good health. A balanced diet should provide around 60-70% of total calories from carbohydrate, 10-12% from protein and 20-25% of total calories from fat.

Objectives of studying Food Science

- To understand the functions of foods
- To know how to select foods to meet our need for nutrients
- To understand the composition of food and the changes that occurs during preparation
- To learn the methods of food preparation
- The economic management of food budget to meet family's needs efficiently.
- To increase variety and food availability through processing
- To avoid food borne diseases through sanitary handling and preparation.

Scope/Applications

- It is useful in developing palatable, nutritious, low cost foods.
- To develop new food products including those used in the space shuttle program.
- Food losses during storage and processing can be avoided by advanced methods of preservation
- The nutrient content of foods can be altered to reduce or increase the nutritive value.
- Fruits and vegetables can be stored for long time by controlled atmospheric packaging.
- Food Scientists are making the foods as safe as possible. The judicious application of food processing, storage and preservation methods helps to prevent outbreaks of food poisoning by pathogenic microorganisms and toxic chemicals such as pesticides.
- Food Scientists work with nutritionists to develop standards for the optimal nutritional content of the diet.
- Food Scientists are involved in establishing international food standards to promote and facilitate world trade and to assure the wholesomeness of foods.

Classification of Foods

Man must eat to live and what he eats will affect to a high degree his ability to keep well, to work and to live long. Food performs many vital functions in the body. There are different

food groups like cereals, pulses, fruits, vegetables, milk and meat products and nuts and oil seeds etc. Foods are classified based on their functions and nutritive value.

Functional Classification of Foods

In functional classification foods are classified based on their function in our body.

1. **Energy Yielding Foods**

Foods rich in carbohydrates, fats and protein are called energy yielding foods. They provide energy to carry out various professional, household and recreational activities. Each one gram carbohydrate and protein gives 4 Kcal. While one gram of fat gives 9 Kcal.

- Cereals, pulses, nuts and oil seeds roots and tubers
- Pure carbohydrates like sugars and fats and oils.

2. **Body Building Foods**

Foods rich in protein are called body building foods. Protein helps in building and repairing tissues. They are divided in to two groups.

- **Plant source:** Pulses, nuts and oil seeds
- **Animal source:** Egg, milk, meat, and fish etc.,

3. **Protective and Regulatory Foods**

Foods rich in protein, minerals and vitamins are known as protective and regulatory foods. They are essential for health and regulate activities such as maintenance of body temperature, muscle contraction, control of water balance, clotting of blood, removal of waste products. Protective foods are classified into two groups.

- Foods rich in vitamins, minerals and protein - milk, egg, fish, liver
- Foods rich in vitamins and minerals only - green leafy vegetables, vegetables and fruits

4. **Health Maintaining Foods**

Foods contain certain phytochemicals and antioxidants which help in preventing degenerative diseases like cancer, heart disease etc. Whole grains, soy bean, green leafy vegetables, fruits and vegetables and spices are rich in phytochemicals and antioxidants.

Food Groups / Nutritional Classification

Foods have been classified into different groups depending upon the nutritive value, for the convenience of planning diets. Food groups like ‘Basic four’, ‘Basic five’, ‘Basic seven’ or ‘Basic eleven’ can be used for planning diets as per the convenience.

1. **Basic Four**

Group	Nutrients
Cereals, millets and pulses	Energy, protein, B-vitamins, minerals
Vegetables and fruits	Vitamins, minerals and fibre
Milk, milk products, and animal foods	Protein, calcium, B-vitamins
Oils, fats, nuts and oilseeds	Energy, protein, fat and fat soluble vitamins

2. **Basic Five: ICMR Classification**

Food Group	Nutrients
1. Cereals, Grains and its Products: Rice, Wheat, Ragi, Bajra, Maize, Jowar, Barley, Rice Flakes, Wheat Flour.	Energy, protein, invisible fat, thiamine, folic acid, riboflavin, iron and fibre
2. Pulses and legumes: Bengal gram, Black gram, Green gram, Red gram, Lentil (whole as well as dhals) Cowpea, Peas, Rajmah, Soya beans, Beans.	Protein, invisible fat, thiamine, riboflavin, folic acid, calcium, iron and fibre
3. Milk and Meat products: Milk, Curd, Skimmed milk, Cheese Chicken, Liver, Fish, Egg, Meat	Protein, fat, riboflavin, calcium, iron, vitamin 'A'
4. Fruits and Vegetables: **Fruits:** Mango, Guava, Tomato Ripe, Papaya, Orange. Sweet Lime, Watermelon. **Vegetables (Green Leafy):** Amaranth, Spinach, Drumstick leaves, Coriander leaves, Mustard leaves, and fenugreek leaves. **Other Vegetables:** Carrots, Brinjal, Ladies fingers, Capsicum, Beans, Onion, Drumstick, Cauliflower.	Carotenoids, vitamin C, riboflavin, folic acid, iron, potassium, phosphorus and fibre
5. Fats and Sugars: **Fats:** Butter, Ghee, Hydrogenated oils, Cooking oils like Groundnut, Mustard, Coconut. **Sugars :** Sugar, Jaggery	Energy, essential fatty acids, iron

3. Basic Seven

Group	Nutrients
1.Green and yellow vegetables	Carotenoids, fibers
2. Oranges, grape fruit, tomatoes or raw cabbage	Vitamin c, carotenoids
3. Potatoes, other vegetables	Carbohydrate
4. Milk and milk products	Protein, fat, calcium, phosphorus
5. Meat, poultry, fish and eggs	Protein, fat, iron, vitamin A,
6. Bread, flour and cereals	Protein, thiamine, riboflavin, iron and fibre
7. Butter or fortified margarine	Fat and fat soluble vitamins.

IV. Basic Eleven

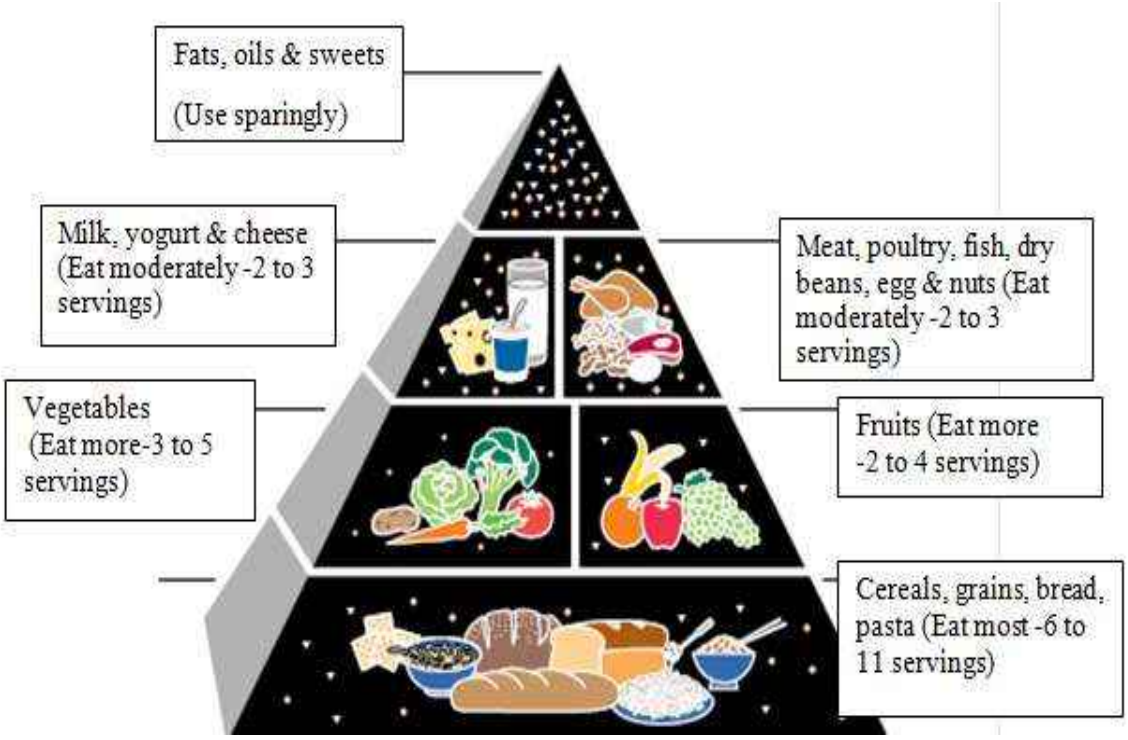
Group	Nutrients
1. Cereals	Carbohydrate (starch), thiamine, riboflavin, and fibre
2. Pulses	Protein, starch, fiber
3. Nuts and oil seeds	Fat, protein and minerals
4. Vegetables	Vitamins, minerals and fiber
5. Fruits	Vitamins, minerals and fiber
6. Milk and milk products	Protein, fat, calcium, phosphorus
7. Eggs	Protein, fat, vitamin D, vitamin A
8. Flesh foods, fish	Protein, iron, calcium, phosphorus, vitamin B_6, vitamin B_{12}, ω-3 fatty acids
9. Fats and oils	Fat and fat soluble vitamins
10. Sugar and jaggery	Energy and iron
11. Spices and condiments	Phytochemicals and antioxidants

In planning balanced diet food should be chosen from each group in sufficient quantity. Cereals and pulses should be taken adequately; fruits and vegetables liberally; animal foods moderately and oils and sugars sparingly

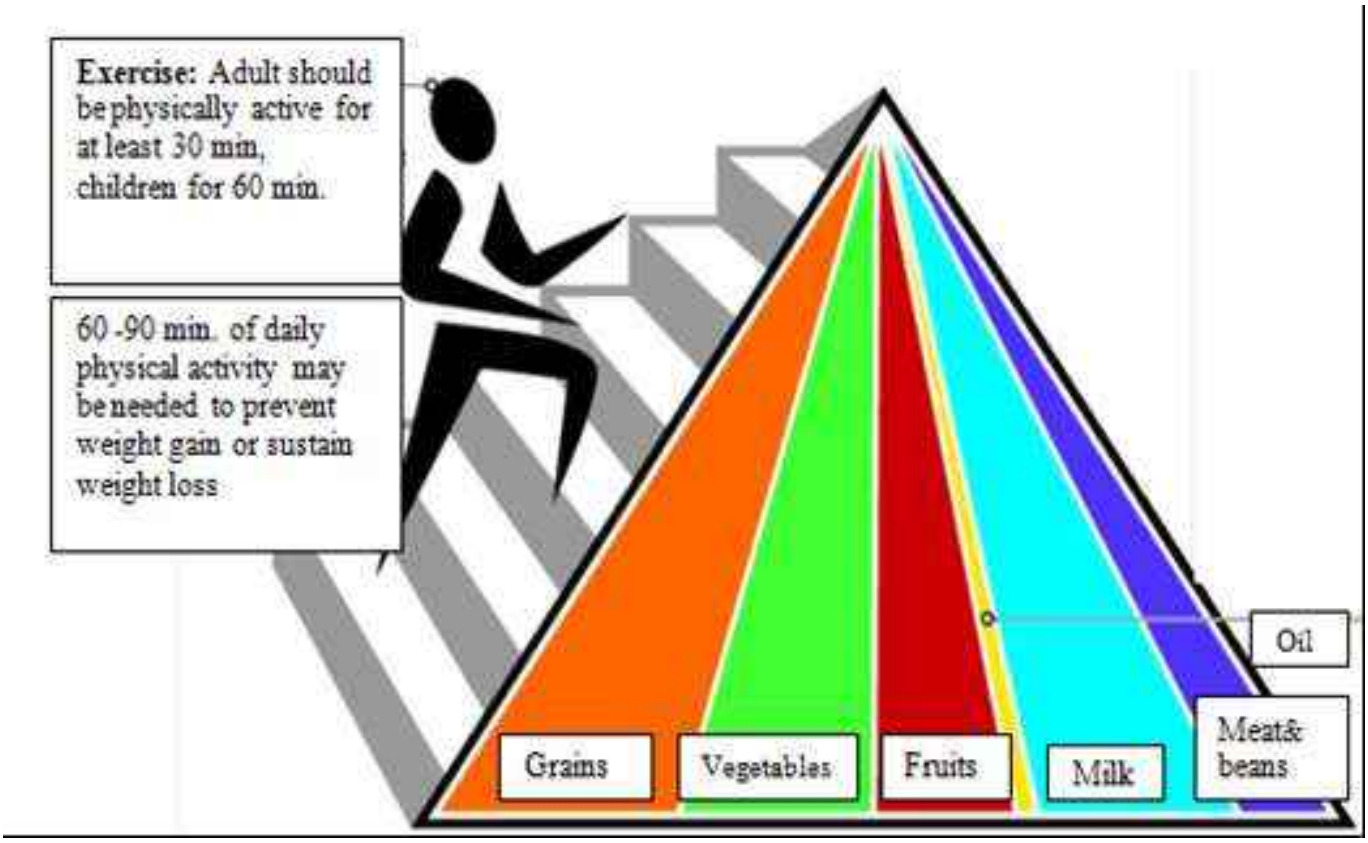
Food Pyramid

A food pyramid is a pyramid shaped diagram representing the amount and type of food to be included in our diet. It is an outline of what to eat each day. The first pyramid was published in Sweden in 1974. The USDA food guide pyramid was created in 1992 and divided into six horizontal sections containing depictions of foods from each section's food group. It was updated in 2005 with colorful vertical wedges replacing the horizontal sections and renamed My Pyramid. My Pyramid was often displayed with the food images absent, creating a more abstract design. In an effort to restructure food nutrition guidelines, the USDA rolled out its new My Plate program in June 2011. My Plate is divided into four slightly different sized quadrants, with fruits and vegetables taking up half the space, and grains and protein making up the other half. The vegetables and grains portions are the largest of the four.

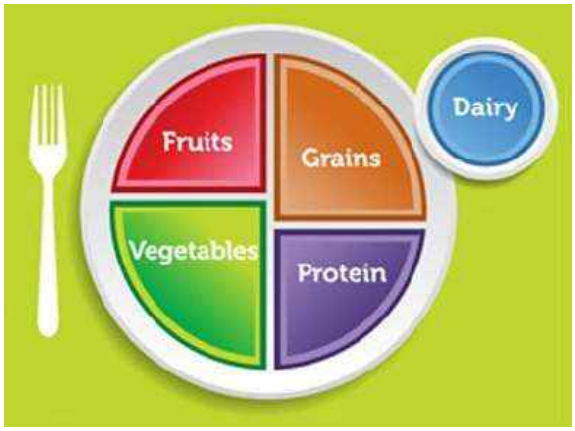
Food Guide Pyramid (USDA, 1992)



The USDA's updated Food Pyramid (2005)



My Plate (2011)



LECTURE –2
METHODS OF COOKING

Food preparation is an important step in meeting the nutritional needs of the family. Food has to be pleasing in appearance and taste in order to be consumed. Foods like fruits, vegetables and nuts can be eaten raw but most foods are cooked to bring about desirable changes. The process of subjecting food to the action of heat is termed as cooking.

Objectives / Merits of Cooking

- Improves the palatability, appearance, flavour, texture and taste of food quality
- Destruction of microorganisms
- Improves digestibility
- Increases variety
- Increase consumption of food
- Increase availability of nutrients
- Concentrates nutrients

Demerits of Cooking

- Heat sensitive nutrients (vitamins) may be lost during cooking.
- Water soluble vitamins are leached into the water during cooking.
- Vitamin A & C content may be reduced due to oxidation and heat.
- Quality of protein may be reduced due to destruction of certain amino acids during cooking e.g. bread

COOKING METHODS

Heat may be transferred to the food during cooking by conduction, convection, radiation or by the energy or microwaves – electronic heat transfer.

Water or steam and air or fat or combinations of these are used as cooking media. Moist heat involves water and steam. Air or fat are used in dry heat. Foods can also be cooked by microwaves.

Classification of cooking methods:

1. **Moist heat methods**

   Boiling, simmering, stewing, poaching, steaming and pressure cooking

2. **Dry heat methods**

   Grilling or broiling, toasting, pan broiling or roasting, baking, sautéing and frying

3. **Combination methods**

   Braising

I. MOIST HEAT METHODS

Boiling:

Boiling is a **method of cooking foods by just immersing them in water at 100ºC** and maintaining the water at that temperature till the food is tender. Foods that are cooked by boiling are rice, eggs, dhals, meat, roots and tubers.

Merits

- It is a simple method
- It does not require special skill and equipment.
- Uniform cooking can be done.

Demerits

- Continuous excessive boiling leads to damage in the structure and texture of food.
- Loss of heat labile nutrients and water soluble nutrients.
- Time consuming – Boiling takes more time to cook food and fuel may be wasted.
- Water soluble pigments may be lost.

Simmering

Foods are **cooked in a pan with a well-fitting lid at temperature just below the boiling point 82 – 92ºC of the liquid** in which they are immersed. It is a useful method when foods have to be cooked for a long time to make it tender as in the case of cheaper cuts of meat, fish, cooking custard, kheer, vegetables and carrot halwa. This method is also employed in making soups.

Merits

- Foods get cooked thoroughly
- Scorching or burning is prevented
- Losses due to leaching is minimum

Demerits

- There is loss of heat sensitive nutrients, due to long period of cooking.
- Takes more time and more fuel is required.

Stewing

Food is **cooked in a pan with a tight fitting lid, using small quantities of liquid to cover only half the food.** The liquid is brought to a boiling point and the heat applied is reduced to maintain the cooking at simmering temperature **(82 - 90ºC)**. The food above the liquid is cooked by the steam generated within the pan. Apple, meat along with some root vegetables and legumes are sewed.

Merits

- Loss of nutrients by leaching does not takes place.
- Falvour is retained.

Demerits

- This process is time consuming and there is wastage of fuel.

Poaching

This involves **cooking in the minimum amount of liquid at a temperature of 80-85°C** that is below the boiling point. Foods generally poached are eggs, fish and fruits.

Merits

- No special equipment is needed
- Very quick method of cooking and therefore saves fuel
- Easily digestible since no fat is used.

Demerits

- It is bland in taste.
- Food can be scorched if water evaporates due to careless monitoring.
- Water soluble nutrients may be leached into the water.

Steaming

Food is **cooked in steam generated from vigorously boiling water or liquid in a pan** so that the food is completely surrounded by steam and not in contact with the water or liquid. Foods prepared by steaming are idli, dhokla, rice (or) ragi puttu, idiappam, appam, kolukattai and custards. There are two types of steaming

Wet steaming

The steam is in direct contact with the food e.g. idli.

Dry steaming

Double boiler is used for cooking the food. Double boiling is cooking in a container over hot or boiling water. This process is used for such preparations as sauces and custards where temperatures below boiling point are desirable. The food is placed in a utensil which is kept in another utensil containing water. When the water is heated or boiled the food gets cooked.

Merits

- Cooking time is less and fuel wastage is less.
- Steamed foods are easily digestible contain less fat and are good for children, aged people and therapeutic diets.
- There is less chance for burning and scorching.
- Texture of the food is better and becomes light and fluffy.
- Steamed foods have good flavour.
- Nutrient loss is minimised

Demerits

- Steaming equipment is required.
- Many foods cannot be prepared by this method

Pressure cooking

When **steam under pressure is used the method is known as pressure cooking** and the equipment used is the pressure cooker. In pressures cooking, escaping steam is trapped and kept under pressure so that the temperature of the boiling water and steam can be raised **above 100ºC** and reduce cooking time. Rice, dhal, meat, roots and tubers are usually pressure cooked.

Merits

- It takes less time to cook.
- Different items may be cooked at the same time.
- Fuel is saved.
- Requires less attention.
- Nutrient and flavour loss may be less.
- Food is cooked thoroughly by this method.
- There is an indication for the completion of cooking.
- Less chances for scorching or burning.

Demerits

- Knowledge of the usage, care and maintenance of cooker is required to prevent accidents.
- Initial investment is not affordable to everybody
- Careful watch on the cooking time is required to prevent over cooking

II. DRY HEAT METHODS

In this method either air or fat is used as the medium of cooking.

Air as medium of cooking

Grilling (or) Broiling

Grilling or broiling refers to the **cooking of foods by exposing it to direct heat**. In this method **food is placed below (or) above (or) in between a red hot surface**. When under the heater, the food is heated by radiation only. Foods cooked by grilling are cob on the corn, papad, brinjal, phulkas, sweet potato and chicken.

Merits

- Enhance flavor, appearance and taste of the product
- It requires less time to cook
- Minimum fat is used

Demerits

- Constant attention is required to prevent charring.

Toasting

This is a method where **food is kept between the two heated elements to facilitate browning on both sides**. Bread slices are cooked by toasting.

Merits

- Quick method of cooking.
- Less or no fat is required.
- Flavour is improved.

Demerits

- Constant attention is required to prevent charring.
- Special equipment required

Pan broiling (or) Roasting:

In this method **food is cooked in a heated metal or frying pan without covering it**. Eg. Groundnut.

Merits

- Improves the appearance, colour, flavour and texture of the food.
- Reduces the moisture content of the food and improves the keeping quality e.g. rava.
- It is easy to powder e.g. cumin seeds and coriander seeds after roasting.
- It is one of the quick methods of cooking foods.

Demerits

- Constant attention is required otherwise the foods can be scorched
- Losses of nutrients like amino acids occur when the food becomes brown.

Baking

In this method the **food gets cooked in an oven by hot air**. The principle involved in baking is the air inside the oven is heated by a source of heat either electricity (or) gas and wood in the case of tandoori. The temperatures that normally maintained in the oven are **120°C – 260°C.** Foods prepared by baking are custards, pies, biscuits, pizzas, puffs, buns, bread, cakes, tandoori chicken, meat and fish.

Merits

- Baking lends a unique baked flavor to foods
- Foods become light and fluffy - cakes
- Flavour and texture are improved.
- Variety of dishes can be made.
- Uniform and bulk cooking can be achieved e.g. bun and bread.

Demerits

- Special equipment and skill are required.
- Careful monitoring needed to prevent scorching

Fat as a medium of cooking

Sauteing:

Sautéing is a method in which **food is tossed in just enough of oil to cover the base of the pan (greasing the pan)** e.g. dosa. Sometimes the pan is covered with lid, reducing the flame and allowing the food to be cooked till tender in its own steam. The product obtained in cooking

by this method is slightly moist, tender but without any liquid or gravy. Foods cooked by sauteing are generally vegetables used as side dishes in a menu. The heat is transferred to the food mainly by conduction.

Merits

- Take less time
- Simple technique
- Minimum oil is used

Demerits

- Constant attention is needed to prevent scorching

Frying:

In this method, the **food to be cooked is brought into contact with larger amount of hot fat.** Two types: Shallow fat frying and deep fat frying.

Shallow fat frying:

The food is cooked in only little fat and the food is turned in both sides in order to cook on both sides. Heat is transferred to the food partially by conduction by contact with the heated pan and partially by the convection currents of the foods. e.g. Paratha, Chaphti, Cashewnuts, potatoes, fish, cutlets.

Deep fat frying:

Food is totally immersed in hot oil and cooked by vigorous convection currents and cooking is uniform on all sides of the foods. Cooking can be rapidly completed in deep fat frying because the temperature used is **180°C - 220°C**. In most foods, this high temperature results in rapid drying out of the surface and the production of a hard crisp surface, brown in colour. Samosa, papad, chips, muruku, pakoda, bajji and bonda are made by deep fat frying.

Merits

- Taste is improved along with the texture.
- Increases the calorific value.
- Fastest method of cooking.
- In shallow fat frying the amount of oil consumption can be controlled.

Demerits

- Sometimes the food may become oily or soggy with too much absorption of oil.
- More attention is required as foods easily get charred and care should be taken to avoid

accidents.

- Fried foods are not easily digested.
- Repeated use of heated oils may produce harmful substances (acrolein, which causes irritation to the eyes and nose).

III. COMBINATION OF COOKING METHODS

Braising:

Braising is a **combined method of roasting and stewing in a pan with a tight fitting lid.** Flavourings and seasoning are added and allowed to cook gently.

Many food preparations are made by not only single method but by a combination of cooking methods.

Vermicelli payasam	-	Roasting and simmering
Vegetable curry	-	Sauteing and simmering
Uppma	-	Roasting and boiling
Meat cutlet	-	Boiling and deep frying
Vegetable pulav	-	Frying and simmering

Microwave cooking:

Microwaves are electromagnetic waves of radiant energy. These waves have **penetrating power** and they excite the molecules of the food. The **essential component of a microwave oven is a magnetron which converts electrical energy into microwave energy**. The electromagnetic energy is transformed into **heat energy**, this heat, cooks the food. These waves cannot pass through metal. They are reflected by metals and absorbed by food. The food is placed on a rotating tray, and when the oven starts, microwaves are emitted which cook the food the uniformly.

Merits

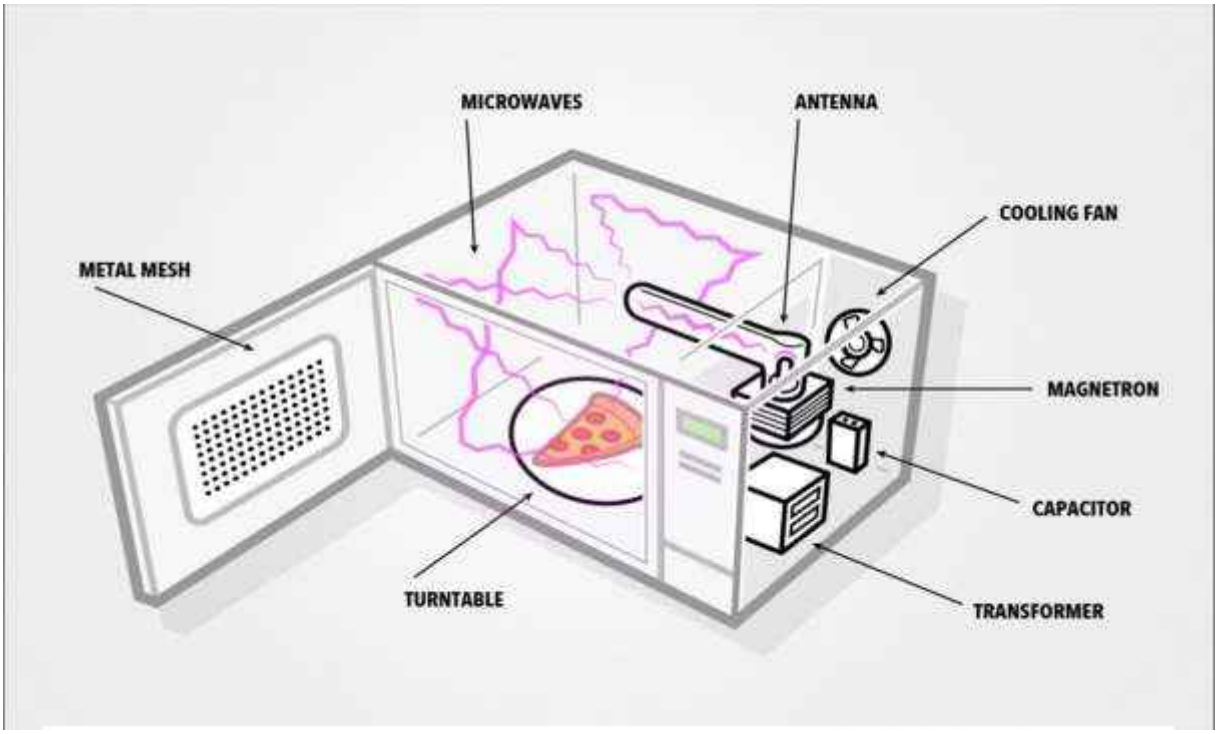
- Quick method – 10 times faster than conventional method.
- There is no wastage of energy.
- Loss of nutrients is minimized.
- Food gets cooked uniformly.
- Only the food gets heated and the oven does not get heated.
- Leftovers can be reheated without changing the flavour and texture of the product.
- It saves time in heating frozen foods. Thawing can be done in minutes or seconds.

- Preserves the natural colour of vegetables and fruits.

Demerits

- Due to short period of cooking, food does not become brown and crispness unless the microwave has a browning unit.
- It is not possible to make chapathi or tandoori rotis
- It cannot cook soft or hard boiled eggs.
- Deep frying cannot be done
- The short cooking time may not give a chance of blending of flavours as in conventional methods.
- The operator should be careful in operating the microwave oven since any exposure to microwave oven causes physiological abnormalities.

Microwave Oven



LECTURE 3

IMPORTANCE AND SCOPE OF NUTRITION AND THE RELATION OF NUTRITION TO HEALTH

Nature has provided a variety of foods for man to consume and be healthy. We consume food for maintenance of health, growth and to develop greater resistance against infections. Foods contain substances called nutrients in varying proportions, which are needed for proper growth and maintenance of life processes. Knowledge of the functions of these nutrients and major food sources is necessary for man to formulate a nutritious diet.

NUTRITION

Nutrition is defined as a science concerned with the role of food and nutrients in the maintenance of health.

Nutrition as defined by Robinson (1982) is " the science of foods and nutrients, their action, interaction and balance in relationship to health and disease, the processes by which the organism ingests, digests, absorbs, transports and utilizes nutrients and disposes of their end product".

NUTRIENTS

Nutrients are the constituents in food that must be supplied to the body in adequate amounts. These include Carbohydrates, Proteins, Fats, Minerals and Vitamins. Nutritional status is the condition of health of the individual as influenced by the utilization of the nutrients..

HEALTH

Health is defined by the World Health Organization (WHO) as the "State of complete physical, mental and social well-being and not merely the absence of disease or infirmity". To maintain good health and nutritional status one must eat a balanced food, which contains all the nutrients in the correct proportion.

RELATION BETWEEN GOOD NUTRITION AND HEALTH

- Nutrition is a process in which the food that is consumed is used for nourishing the body. Good nutrition is essential for good health.
- Nutrition is very important for a person to grow and develop normally and to remain healthy throughout life.

- When a person does not eat proper food, there are chances of his not developing normally- the likelihood that some organ of his body may start malfunctioning, or that he may get some disease?
- Poor nutrition may also influence his mental and social well-being.
- Thus, generally, the health of a person depends on the kind and amount of food he eats.
- The right kind of food eaten in the right quantity is necessary for good health.

The essential requisites of health would include the following:

- Achievement of optimal growth and development, reflecting the full expression of one' s genetic potential.
- Maintenance of the structural integrity and functional efficiency of body tissues necessary for an active and productive use.
- Mental well-being
- Ability to withstand the inevitable process of aging with minimal disability and functional impairment.
- Ability to combat diseases such as
  - resisting infections (immunocompetence)
  - preventing the onset of degenerative diseases
  - resisting the effect of environmental toxins/ pollutants

Till three decades ago the role of nutrition in growth and development and tissue integrity alone was clear, but now the persuasive role nutrition plays in the other dimensions of health is implicit. Hence an optimal nutritional status is an indication of good health. This recent advance has brought about a large-scale change in dietary habits and practices of the population.

CONCEPTS OF MALNUTRITION – UNDER NUTRITION AND OVER NUTRITION

Malnutrition as defined by World Health Organisation (WHO) is a pathological state resulting from a relative or absolute deficiency or excess of one or more essential nutrients, this state being clinically manifested or detected only by biochemical, anthropometric or physiological tests.

Four forms can be distinguished:

1. Undernutrition – the pathological state resulting from the consumption of an inadequate quantity of food over an extended period of time.

2. Marasmus is synonymous with severe undernutrition. Starvation implies total elimination of food and hence the rapid development of under nutrition and marasmus.
3. Specific deficiency – the pathological state resulting from a relative or absolute lack of an individual nutrient.
4. Over nutrition – the pathological state resulting from a disproportion of essential nutrients with or without the absolute deficiency of any nutrient as determined by the requirement of a balanced diet.

IMPORTANCE OF NUTRITION

- Eating a balanced diet is vital for good health and wellbeing. Food provides our bodies with the energy, protein, essential fats, vitamins and minerals to live, grow and function properly. We need a wide variety of different foods to provide the right amounts of nutrients for good health.
- Nutrition, nourishment, or aliment, is the supply of materials - food - required by organisms and cells to stay alive. In science and human medicine, nutrition is the science or practice of consuming and utilizing foods.
- In hospitals, nutrition may refer to the food requirements of patients, including nutritional solutions delivered via an IV (intravenous) or IG (intragastric) tube.
- Nutritional science studies how the body breaks food down (catabolism) and repairs and creates cells and tissue (anabolism) - catabolism and anabolism = metabolism. Nutritional science also examines how the body responds to food.
- Nutrition also focuses on how diseases, conditions and problems can be prevented or lessened with a healthy diet.
- In addition, nutrition involves identifying how certain diseases, conditions or problems may be caused by dietary factors, such as poor diet (malnutrition), food allergies, metabolic diseases, etc.,

CLINICAL SIGNS OF NUTRITIONAL DEFICIENCY DISORDERS

Clinical examination is an important practical method for assessing the nutritional status of a community. Essentially, the method is based on examination for changes, believed to be related to inadequate nutrition that can be seen or felt in the superficial epithelial tissues especially the skin, eyes, hair and buccal mucosa or in organs near the surface of the body such as the parotid and thyroid glands.

Clinical assessment must always be carried out by individuals with adequate training. The following simple guide is employed to interpret the following deficiencies.

Guide for the interpretation of deficiencies and identifying the clinical signs.

Condition	Clinical Signs
(i)Protein Energy Malnutrition **:**	Odema, depigmentation, sparseness and easy pluckability of hair, moon face, enlarged liver, muscle wasting
(ii) Vitamin A deficiency **:**	Night blindness, Bitot' s spots in the eye, Xerosis of skin.
(iii) Riboflavin deficiency **:**	Angular stomatitis, cheilosis.
(iv) Thiamine deficiency **:**	Oedema, sensory loss, calf muscle tenderness.
(v) Niacin deficiency **:**	Raw tongue, pigmentation of the skin
(vi) Vitamin C deficiency **:**	Spongy and bleeding gum.
(vii) Vitamin D deficiency **:**	Rickets, beading of ribs, Knock – knees, bowed legs.
(viii) Iron deficiency **:**	Pale conjunctiva, spoon – shaped nails.
(ix) Iodine deficiency **:**	Enlargement of thyroid gland.

Dietitian's physicians can observe physical signs that suggest good or poor nutrition as listed in the table

	GOOD NUTRITION	BAD NUTRITION
Body size	Normal weight for height body frame and age, normal rate of growth	Recessive obesity or thinness, sudden loss or gain in weight failure to grow in stature to gain weight.
Behaviour	Alert ,good attention span, regular attendance, cooperative, cheerful, interested ,has endurance	Apathetic, short attention span, frequent absence. Irritable, easily fatigue, inability to concentrate, poor work capacity
Skin	Firm, clear, slightly moist, healthy, pink mucous membrane, firm pink nails	Dry, pale scaly around ears and nose, bed sores, brittle ridged nails
Hair	Soft, glossy, healthy scalp	Dry, brittle then easily pulled out, change in pigmentation
Eyes	Clear bright not unduly sensitive to light	Red, swollen or dry, itching or burning. Poor vision in dim light, excessive, sensitivity to bright light.
Mouth	Moist smooth, pink tongue with	Chopped tissues at corners of lips.

	surface papillae, pink firm gums. Even teeth, well-formed jaw.	Swollen tongue – scarlet or magenta smooth in appearance. Bleeding gums, receding from tooth line. Decayed or missing teeth. Inability to chew
Skeleton	Erect posture, arms and legs straight abdomen in, chest chin in	Poor posture, deformities of long bones, spines, pelvis, hump back, bow legs or knock ness
Neuromuscular system	Firm, strong muscles with moderate padding of fat good muscular co-ordination.	Flabby under developed muscles, lack of fat padding or excessive fat; poor muscle coordination reduced knee and ankle reflexes, burning and tingling of hands and feet.
Gastro intestinal system	Good appetite & digestion, regular elimination	Poor appetite, in digestion, diarrhea or constipation
Glands	No enlargement of thyroid	Enlarged thyroid
Immune system	Resistance to infections	Frequent colds & other infections, longer covalence from illness slower wound healing

LECTURE 4

CARBOHYDRATES

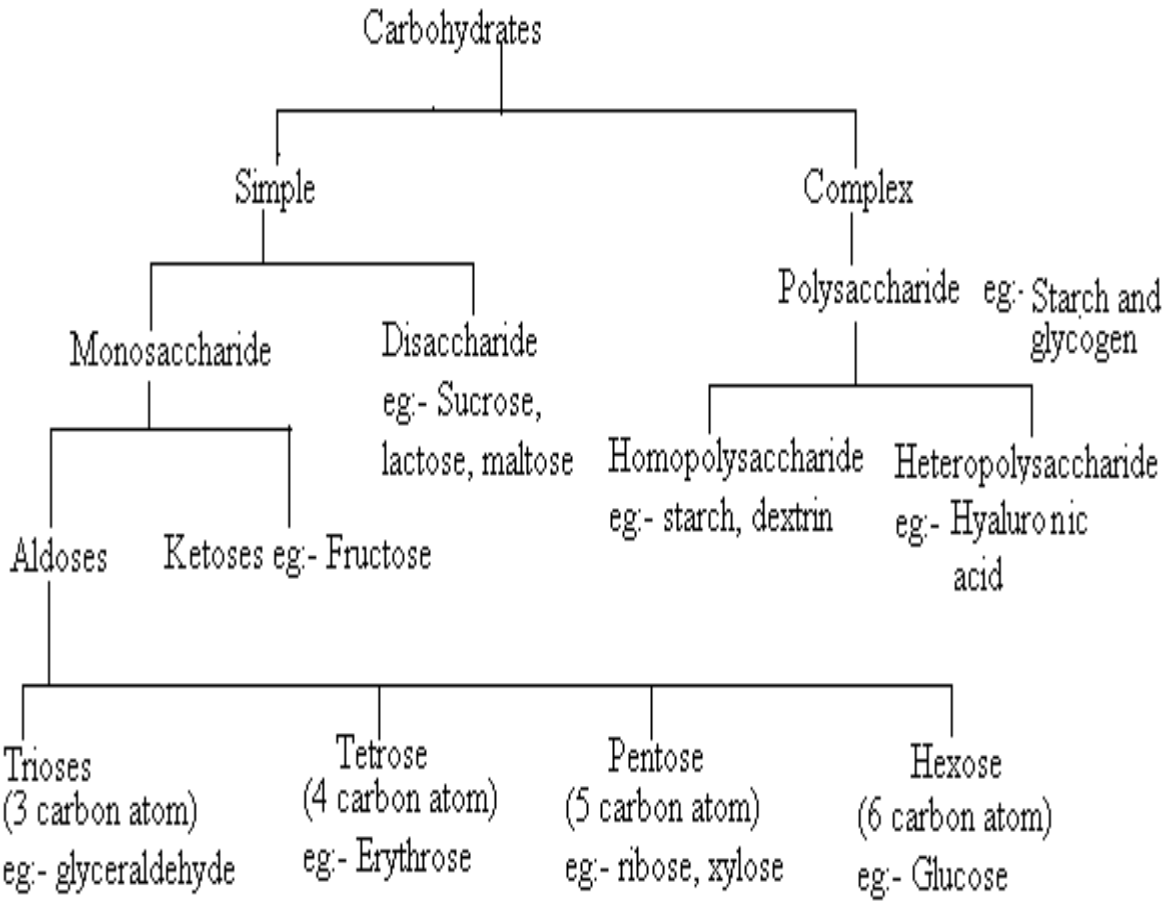
Energy that is needed to move, perform work and live is chiefly consumed in the form of carbohydrates. Carbohydrates, primarily starches, are least expensive, easily obtained and readily digested form of fuel.

COMPOSITION

Carbohydrates are organic compounds composed of carbon, hydrogen and oxygen, with the later elements in the ratio of 2:1. The general formula is $(CH_2O)n$. They are viewed as hydrated carbon atoms.

CLASSIFICATION – SIMPLE AND COMPLEX:

Carbohydrates are classified, depending on the number of sugar units they contain, as simple carbohydrate and complex carbohydrates.



SIMPLE CARBOHYDRATES

Monosaccharides and disaccharides are called simple sugars containing one and two sugar units respectively.

Monosaccharides

There are two main classes of monosaccharides based on the carbonyl group present in them. They are aldoses and ketoses, aldoses (eg; glucose) containing the aldehyde group (CHO) and ketoses, (eg;- fructose) containing the ketone group (C=O). Aldoses are further divided into trioses, tetroses, pentoses and hexoses based on the number of carbon atoms.

Glucose (Dextrose, Blood Sugar)

Glucose serves as the main source of energy in the body. Normal human blood contains about 80-120 mg of glucose per 100ml of blood. This level is maintained constantly in healthy subjects.

Fructose (Levulose, Fruit sugar)

Fructose occurs in the free state along with glucose in many fruits (2-5 per cent) and in honey 30-40 per cent). It is utilised by the body as a source of energy.

Galactose

It does not occur in the free state but occurs as a constitute of lactose present in milk.

Disaccharides

Disaccharides are formed by the combination of two monosaccharides by the elimination of one molecule of water.

Sucrose (Table sugar/Cane sugar/Beet sugar)

Sucrose occurs in sugar cane (10-12 per cent) and beet root (12-18 per cent). It is manufactured on a large scale from sugar cane or beet root. Sucrose is hydrolysed to glucose and fructose by dilute mineral acids or by the enzyme sucrase present in intestinal juice.

Maltose (Malt sugar)

Maltose is present in malt. It is formed in cereals grains (barley, jowar, ragi etc) during germination by the hydrolysis of starch by the enzyme diastase (Amylase)

Starch $\xrightarrow{\textbf{Amylase}}$ Maltose $\xrightarrow{\textbf{Maltase}}$ **Glucose (2 molecules)**

Lactose (Milk sugar)

Lactose occurs in the milk of all mammals but has not so far been found in plant foods. Lactose is hydrolysed to glucose and galactose by the enzyme lactase present in the intestinal juice.

$$\text{Lactose} + H_2O \xrightarrow{\textbf{Lactase}} \text{Glucose} + \text{Galactose}$$

COMPLEX CARBOHYDRATES

Oligosaccharides and Polysaccharides are called Complex Carbohydrates.

Oligosaccharides

Oligosaccharides are short carbohydrate chains of 3 – 10 monosaccharides, which cannot be broken down by human enzymes, though can be digested by colonic bacteria. Found in legumes and human milk Examples: Raffinose and Stachyose.

Polysaccharides

Polysaccharides are structurally larger and more complex than simple sugars. Polysaccharides have high molecular weight and are insoluble in water. They are in the form of long chains either branched or unbranched. Examples include storage polysaccharides such as starch and glycogen, and structural polysaccharides such as cellulose and chitin.

Storage Polysaccharides

Starch (Plant storage form of carbohydrate)

Starch is a glucose polymer in which glucopyranose units are bonded by alpha-linkages. It is made up of a mixture of amylose (15–20%) and amylopectin (80–85%). Amylose is a polysaccharide made of α-D-glucose units, bonded to each other through α (1→4) glycosidic bond. It consists of a linear chain of several hundred glucose molecules. In amylopectin, glucose units are linked in a linear way with α (1→4) glycosidic bond. Branching takes place with α (1→6) bonds occurring every 24 to 30 glucose units. Starches are insoluble in water. They can be digested by breaking the alpha-linkages (glycosidic bonds). Both humans and animals have amylases, so they can digest starches. Potato, rice, wheat, and maize are major sources of starch in the human diet.

Glycogen (Animal storage form of carbohydrate)

Glycogen is analogous to starch, referred to as animal starch having a similar structure to amylopectin but more extensively branched and compact than starch. Glycogen is a polymer of α (1→4) glycosidic bonds linked, with α (1→6)-linked branches. Glycogen forms an energy reserve in liver and the muscles that can be quickly mobilized to meet a sudden need for glucose.

Structural polysaccharides

Cellulose

Cellulose is a polysaccharide consisting of a linear chain of several hundred to many thousands of β(1→4) linked D-glucose units. It is a polymer made with repeated glucose units bonded together by beta-linkages. Cellulose is an important structural component of plants is formed primarily from cellulose. Humans and many animals lack an enzyme to break the beta-linkages, so they do not digest cellulose.

Chitin

Chitin is one of many naturally occurring polymers. It forms a structural component of many animals, such as exoskeletons. Its breakdown may be catalyzed by enzymes called chitinases, secreted by microorganisms such as bacteria and fungi, and produced by some plants.

The polysaccharides are further classified into groups depending upon the products they yield on hydrolysis. **Homopolysaccharides** yield only one type of monosaccharide units on hydrolysis eg:- starch, dextrin, cellulose, glycogen. **Heteropolysaccharides** yield more than one type of monosaccharide units on hydrolysis eg:- Heparin, Hyaluronic acid. Heparin is an anticoagulant found in the liver, spleen, lungs and blood. Hyaluronic acid is found in the umbilical cord, synovial fluid and vitreous humour. It has a lubricating action. In tissues it forms an important part of the cementing ground substance.

FUNCTIONS OF CARBOHYDRATES

Carbohydrates have a variety of functions in the animal and human body.

i. They supply energy for body functions and for doing work.

ii. They are essential for the oxidation of fats

iii. They have a sparing action on proteins.

iv. They provide the carbon skeleton for the synthesis of some non-essential amino acids, glycoproteins and glycolipids.

v. Some carbohydrates are present in some tissue constitutents.

vi. They add flavour to the diet.

vii. composite carbohydrates can absorb water and give bulk to the intestinal contents which aids in the elimination of waste products

Protein Sparing Action:

Carbohydrates exert a protein sparing action. If sufficient amounts of carbohydrates are not available in the diet, the body will convert protein to glucose in order to supply energy. Hence to spare proteins for tissue building, carbohydrates must be supplied in optimum amounts in the diet. This is called the protein sparing action of carbohydrates.

DIGESTION AND ABSORPTION OF CARBOHYDRATES

The first stage in the digestion of carbohydrates takes place in the mouth when the food is chewed. The saliva contains an **alpha amylase called ptyalin**. This enzyme acts on starch spliting it into dextrin and maltose amylase acts best at neutral pH. As soon as the food reaches the stomach it mixes with the acidic gastric juice and amylase activity is inhibited. The digestion of carbohydrates is mainly accomplished in the small intestines where they are subjected to the action of pancreatic amylases and intestinal amylase, sucrase, lactase, maltase and isomaltase present in the intestinal juice.

Starch $\xrightarrow[\text{and pancreatic \& intestinal juice}]{\text{Amylase from saliva}}$ Maltose + Isomaltose

Maltose $\xrightarrow{\text{Maltase}}$ glucose

Isomaltose $\xrightarrow{\text{Isomaltase}}$ Glucose

Sucrose $\xrightarrow{\text{Sucrase}}$ Glucose + Fructose

Lactose $\xrightarrow{\text{Lactase}}$ Glucose + Galactose

The ultimate products of digestion of carbohydrate are glucose, fructose and galactose. These are absorbed in the small intestine (brush border of the epithelium covering the villi - small hair-like structure). Only monosaccharaides are absorbed. The rate of absorption is

Galactose > Glucose > Fructose. The non-digestible carbohydrates present in the food. Viz., cellulose, hemicellulose, pentosans, galactans. Fructosane etc., are not acted upon by the digestive juices. They add bulk to the contents of large intestines is excreted in the faeces. Some of these are fermented by bacteria present in the large intestine.

Metabolism of carbohydrates

Glucose, galactose and fructose absorbed in the intestines pass through the portal circulation to the liver. In the liver a part of the glucose and the entire galactose and fructose are converted into glucose and enter into the general circulation and to the various tissues for being oxidized and used as energy. A small part of glucose is stored in liver and muscle as glycogen and some portion of the glucose is converted into fat stored in adipose tissue. The oxidation of glucose in the issues occurs into 2 stages as indicated below:-

i. Glycogen $\rightleftharpoons$ glucose $\longrightarrow$ pyruvic acid $\rightleftharpoons$ lactic acid

ii. Pyruvic acid $\xrightarrow{\text{Oxidation}}$ $CO_2 + H_2O$

The first stage is called glycolysis. The oxidation of pyruvic acid takes place through a series of reactions known as tricarboxylic acid cycle (krebs cycle)

Regulation of Blood Glucose

The body needs glucose to make ATP (via cell respiration), however the amount required will fluctuate according to demand. **Normal Blood Glucose level 80 - 120 mg/100 ml**. High levels of glucose in the blood can damage cells (creates hypertonicity) and hence glucose levels must be regulated. Two antagonistic hormones are responsible for regulating blood glucose concentrations – **insulin and glucagon**. These hormones are released from pancreatic pits (called the islets of Langerhans) and act principally on the liver.

When blood glucose levels are high (e.g. after feeding):

- Insulin is released from **beta (β) cells** of the pancreas and cause a decrease in blood glucose concentration
- This may involve stimulating **glycogen synthesis** in the liver **(glycogenesis),** promoting glucose uptake by the liver and adipose tissue, or increasing the rate of glucose breakdown (by increasing cell respiration rates)

When blood glucose levels are low (e.g. after exercise):

- Glucagon is released from **alpha (α) cells** of the pancreas and cause an increase in blood glucose concentration
- This may involve stimulating **glycogen breakdown** in the liver **(glycogenolysis),** promoting glucose release by the liver and adipose tissue, or decreasing the rate of glucose breakdown (by reducing cell respiration rates).

DIABETES MELLITUS

Diabetes mellitus is a metabolic disorder that results from a high blood glucose concentration over prolonged period results in **hyperglycemia**. It is a chronic disease in which blood glucose level is raised above 180 per 100 ml of blood and glucose is excreted in urine.

Causes

1. Insufficient production of insulin (Type I or Juvenile diabetes or insulin-dependent diabetes)
2. Failing to respond to insulin production (Type II or non - insulin-dependent diabetes)

Symptoms

- Acute - increased thirst, urine production, hunger
- Chronic - risk of heart disease, kidney disease, blindness, neural damage

Type I Insulin-Dependent Diabetes Mellitus (IDDM)	Type II Non Insulin Dependent Diabetes Mellitus (NIDDM)
Usually occurs during childhood (early onset)	Usually occurs during adulthood (late onset)
Body does not *produce* sufficient insulin	Body does not *respond* to insulin production
Caused by the destruction of β-cells (autoimmune)	Caused by the down-regulation of insulin receptors
Requires insulin injections to regulate blood glucose	Controlled by managing diet and lifestyle

DEFICIENCY SYMPTOMS OF CARBOHYDRATES

Acidosis:

In carbohydrate starvation, there is shift from glycolysis (breakdown of glucose) to lipolysis (breakdown of lipids) and ketogenesis for energy needs. The resultant production of ketoacids increases acidity in the blood and other body tissues. These changes in the pH of arterial blood outside 7.35 pH - 7.45 pH result in irreversible cell damage.

Ketosis:

During prolonged carbohydrate fasting or starvation, acetyl-CoA in the liver is used to produce ketone bodies formed by the breakdown of fatty acids and by the deamination of amino acids, leading to a state of ketosis.

Hypoglycemia:

The non availability of glucose due to severe lack of carbohydrate causes drop in the blood sugar levels. Hypoglycemia occurs when blood glucose, levels drop under 70 mg/dL with typical symptoms like giddiness, fatigue, distress and delirium.

Fatigue and decreased energy levels:

The immediate non availability of glucose in the blood for energy production, result in dip in the energy levels and fatigue.

Muscle wasting:

As the fat reserves and amino acids are getting used up for energy production, there will be general loss of muscle mass and impairment of growth.

Unhealthy weight loss:

The loss of fat and muscle mass leads to emaciation and weight loss.

Dehydration and reduced body secretions:

As there is loss of fluids from the body due to ketosis, a state of chronic dehydration is reached. This results in reduction in mucus secretion, dry eyes and compromised mucus production in tear glands, salivary glands, sinuses, airways, and gastrointestinal tract.

Loss of sodium:

Excess of ketone in the blood leads to fluid loss and excretion of sodium ions(Na+) from the body. This may lead to muscle cramps, exhaustion and lassitude.

Weakened immune system:

With increased fluid loss and degradation, vitamin C loss from the body is increased. Adding to this, the chronically dehydrated condition leads to weakened immune system and susceptibility to infections.

Constipation:

Dietary fiber is an essential component of carbohydrate food, which is known to prevent recto-colon cancer and help digestion. The absence of dietary fiber can cause constipation.

Mood swing:

Dietary carbohydrate exclusion causes the brain to stop regulating serotonin hormone. Low serotonin supply causes mood swing and depression.

FOOD SOURCES OF CARBOHYDRATES

Cereal grains, roots and tubers are the major sources of starch. Fruits and vegetables contain varying amounts of monosaccharides and dissaccharides. Sugar is obtained from sugarcane. Types and sources of Carbohydrates are given in the table

Types and Sources of Carbohydrates

Carbohydrate	Food Source
Monosaccharides	
Glucose	Fruits, honey, corn-syrup.
Fructose	Fruits and honey.
Galactose, Maltose	These do not occur in free form in foods.
Dissaccharides	
Sucrose	Cane and beet sugar.
Lactose	Milk and milk products.
Maltose	Malt and Cereal products.
Polysaccharides	
Digestible:	
Starch & Dextin	Grains, vegetables especially roots& tubers and legumes
Glycogen	Meat products and sea foods
Indigestible:	
Cellulose	Stalks and leaves of vegetables, outer coat of seeds
Pectins, Gums	Fruits, Plant secretions and seeds.

REQUIREMENTS

As carbohydrate is utilized as main source of energy, at least 40 percent of the total energy in the food should come from Carbohydrates. In our country 60 – 80 percent of a day's energy needs are met from carbohydrates in the form of starch furnished by cereals and pulses. In developed countries only 30 – 40 percent of days energy needs are met from carbohydrates. The percentage of total energy intake for adults and Adolescents (50 - 70%), Pregnant and lactating women (40 -60 %), Infants and Preschool children (40 -50 %).

LECTURE 5

PROTEIN

The name protein was suggested by **Mulder in 1838**. One fifth of an adults total body weight is protein. Protein is found in every cell of our body. All the tissues in our body such as muscle, blood, bone, skin and hair are made up of proteins. Many hormones and enzymes are either protein or protein derivatives. The nucleic acids in the cell nucleus occur in combination with proteins as nucleoproteins.

COMPOSITION

Proteins contain carbon, hydrogen, oxygen and nitrogen. They are distinguished from carbohydrates and fats by the presence of nitrogen. **Protein is synthesized from basic units called amino acids**. Protein molecules, which contain up to hundred amino acids, are much larger than carbohydrates or lipid molecule.

Amino acid

Amino acids are organic compounds containing amine (-NH2) and carboxyl (-COOH) functional groups, along with a side chain (R group) specific to each amino acid

CLASSIFICATION OF AMINO ACIDS

Amino acids are classified based on a variety of features, including whether people can acquire them through diet. Three amino acid types:

1. Essential
2. Nonessential

Essential Amino acids

An essential amino acid is one that **cannot be synthesized by the body** to meet the physiological needs. So it **should be supplied by the diet**. Eight of these amino acids are essential (or **indispensable**) and cannot be produced by the body. Eight amino acids (Leucine, Isoleucine, Valine, Threonine, Methionine, Phenylalanine, Tryptophan and Lysine) are considered essential for adults and nine (those mentioned above plus Histidine) for children.

Nonessential amino acids

Nonessential amino acids are **produced by the human body** either from essential amino acids or from normal protein breakdowns. Nonessential amino acids include:

1. Asparagine
2. Alanine
7. Glutamine
8. Proline

3. Arginine
4. Aspartic acid
5. Cysteine
6. Glutamic acid
9. Glycine
10. Tyrosine
11. Serine

Amino acids' classification depends upon the side chain structure

- **Sulfur containing amino acids** - Cysteine and Methionine
- **Neutral amino acids** - Asparagine, Serine, Threonine, and Glutamine
- **Acidic amino acids** - Glutamic acid and Aspartic acid
- **Basic amino acids** - Arginine and Lysine
- **Aliphatic amino acids** - Leucine, Isoleucine, Glycine, Valine, and Alanine
- **Aromatic amino acids** - Phenylalanine, Tryptophan, and Tyrosine

CLASSIFICATION OF PROTEINS

I. BASED ON THE SOURCE OF PROTEIN MOLECULE

Proteins have been traditionally divided into two well defined groups animal proteins and plant proteins

Animal proteins are the proteins derived from animal sources such as eggs, milk, meat and fish. They are usually called **higher quality proteins** because they contain adequate amounts of all the essential amino acids.

Plant proteins are the proteins derived from plant sources. They are called **lower quality proteins** since they have a low content (limiting amount) of one or more of the essential amino acids. **The four most common limiting amino acids are methionine, lysine, threonine and tryptophan.** Although plant proteins have limiting amounts of some (but not all) amino acids, it should not be construed that they are poor protein sources. Most plant foods tend to have too little of one or more particular essential (indispensable) amino acids.

Limiting amino acid in some plant proteins

Food	Amino acids
Cereal grains and millets	Lysine , Threonine
Rice and soy bean	Methionine
Legumes	Methionine, Tryptophan
Groundnuts	Methionine, Lysine, Threonine
Sunflower seeds	Lysine
Green leafy vegetables	Methionine

II. BASED ON THE COMPOSITION

Simple protein

Proteins containing **only amino acids** are called simple proteins. On hydrolysis they yield only amino acids.

Example: albumins, globulins, glutelins, prolamins, histones, protamins and fibrous protein

Conjugated or complex proteins

The protein which **combined with non-protein substances** like carbohydrate, lipid, phosphorus, nucleic acid are called as conjugated protein.

Example: glycoprotein, lipoprotein, nucleoprotein, haemoglobins

- Glycoprotein - Carbohydrate & Protein
- Lipoprotein - Lipid & Protein
- Phosphoprotein - Phosphorus & Protein
- Haemoglobin - Haem & Protein

Derived proteins

These proteins are **derived by partial to complete hydrolysis of simple or conjugated proteins** by the action of acids, alkalies or enzymes.

Types of derived proteins

- **Primary derived proteins** are derivatives of proteins in which the size of protein molecule is not altered materially - Proteans, Metaprotein, Coagulated Protein (boiled egg)
- **Secondary derived proteins** are derivatives of proteins in which the hydrolysis has certainly occurred - Proteoses, Peptones, Peptides

BASED ON THE NUTRITIONAL VALUE

Complete Protein

A complete protein (or whole protein) is a source of protein that contains an **adequate proportion of all nine of the essential amino acids** necessary for the dietary needs of humans. Sources of complete proteins are mostly foods of animal origin such as milk, yogurt, cheese, eggs, meat, fish, and poultry. The exceptions are gelatin, which is of animal origin but does not have the indispensable amino acid tryptophan, and soy protein, which is of plant origin but is a complete protein.

Partially Complete Protein

It is **partially lacking in one or more essential amino acids**, e.g., Wheat protein (gluten).

Incomplete Protein

It is any protein that **lacks one or more essential amino acids in correct proportions.** Incomplete proteins or low-quality proteins are derived from plant foods such as legumes, vegetables, cereals, and grain products. Examples, Zein, gelatin. Incomplete protein containing foods and their limiting amino acid(s) are listed in Table.

Complementary proteins

To ensure that the body receives all the essential amino acids, certain proteins can be ingested together or combined so that their amino acid patterns become complementary. This practice or strategy is called mutual supplementation. For example, legumes, with their high content of lysine but low content of sulfur-containing amino acids, complement the grains, which are more than adequate in methionine and cysteine but limited in lysine.

FUNCTIONS/IMPORTANCE OF PROTEIN

1. It helps in growth and repairs the body tissues.
2. Enzymes pepsin, trypsin, amylase etc., are made up of proteins.
3. It act the as catalyze in digestion of foods and other biochemical reactions.
4. Hormones such as insulin, growth hormones etc are also made up of protein. It regulates metabolic functions of cells.
5. It aids in transmitting nerve signals from one nerve cell to another.
6. It aids in transporting nutrients from intestinal wall to blood.
7. Contractile protein, myosin and actin regulate muscle contraction.
8. It helps in the regulation of water balance in our body. Low blood protein level causes oedema.
9. Proteins act as buffers in blood and maintain the optimum pH.
10. Antibodies are made up of proteins. It helps to fight against the infection.
11. Protein also a good source of energy. It provides 4 Kcal per gram of protein.

DIGESTION OF PROTEIN

Stomach

Pepsin is present in gastric juice and hydrolyses the peptide bonds in protein molecules. It hydrolyses the dietary protein into polypeptides.

Pepsin

Dietary protein ⟶ large Polypeptides

Small intestine

The polypeptides formed in stomach are digested in small intestine by trypsin, chymotrypsin secreted in the pancreatic juice. Polypeptides are peptides, dipeptides or tripeptides.

Polypeptide **Trypsin & Chymotrypsin** ⟶ Peptides + Amino acids

The carboxypeptidase hydrolyses the peptides in to amino acid.

Peptides **Carboxypeptidase** ⟶ Amino acids

The intestinal mucosa also contains dipeptidase and tripeptidase which hydrolyses dipeptides and tripeptides, respectively into amino acids.

Dipeptides **Dipeptidase** ⟶ Amino acids

Tripeptides **Tripeptidase** ⟶ Amino acids

Absorption

The end product of digestion of protein is amino acids. The amino acids are absorbed in small intestine and transported to the liver via hepatic portal vein.

DEFICIENCY DISEASES

Protein Energy Malnutrition (PEM) / Protein Calorie Malnutrition (PCM)

Protein Energy Malnutrition (PEM) is a form of malnutrition resulting from inadequate protein and calorie intake. It is the primary nutritional problem in India. Due to the “food gap” between the intake and requirement. It causes childhood morbidity and mortality.

- Kwashiorkor (deficiency in protein intake)
- Marasmus (deficiency in protein and calorie intake)
- Marasmic Kwashiorkor (both Kwashiorkor and Marasmus symptoms)

Kwashiorkor

The word Kwashiorkor was first introduced by Dr. Cicely Williams. The term 'Kwashiorkor' means 'the disease which the child gets when the next baby is born'.

Signs and symptoms of Kwashiorkor

1. Growth failure and wasting of muscles.
2. Behavioural changes like irritability and apathy are common.
3. Oedema which is the accumulation of fluid in the tissues making them soft and spongy.
4. Scaly pigmentation of skin (crazy paveoment dermatosis)
5. Changes in hair (dry and sparse and red colour).
6. Loss of appetite, vomiting, diarrhoea
7. Enlargement of the liver.
8. Anaemia
9. Moon face (full well rounded face)

Marasmus

Nutritional Marasmus is due to the consumption of diet deficient in both protein and energy.

Signs and symptoms

1. Growth retardation
2. Severe muscle wasting
3. Loss of subcutaneous fat
4. Child appear skin and bones
5. Irritability and apathy
6. Frequent diarrhoea
7. Dehydration
8. Oedema and fatty liver are absent

Marasmic Kwashiorkor

The child shows a mixture of the features of Marasmus and Kwashiorkor. The features of kwashiorkor are severe oedema of feet and legs and also hands, lower arms, abdomen and face. Pale skin and hair, and the child are unhappy. Signs of marasmus include wasting of the muscles of the upper arms, shoulders and chest so that you can see the ribs.

Treatment

- **Resolving life threatening conditions (Hospital Treatment)**

  Hypothermia, hypoglycemia, infection, dehydration, electrolyte imbalance, anaemia and other vitamin and mineral deficiencies should be corrected

- **Restoring nutritional status (Dietary Management)**

  Inexpensive, easily digestible, evenly distributed food throughout the day and increased number of feedings to increase the quantity of food. Daily calorie intake should be 140-150 Kcal/kg and protein intake 3-5 g/kg body weight.

- **Ensuring nutritional rehabilitation (Rehabilitation)**

  Practical nutritional training for mothers

Prevention

1. Promotion of breast feeding
2. Development of low cost weaning
3. Nutrition education and promotion of correct feeding practices
4. Family planning and spacing of births
5. Immunization
6. Food fortification
7. Early diagnosis and treatment

FOOD SOURCE OF PROTEIN

- Pulses, oil seeds, nuts, soybeans, cereals and millets, bread, dry beans, peas, peanut butter when combined with small amounts of egg, cheese, meat, fish and poultry give just as good an assortment of amino acids as a large an amount of animal foods.
- Soya bean is the richest source with 40% protein followed by watermelon seeds (34%), wheat germ (29%) and ground nuts (26%).
- Although cereals have low protein content, as they are being consumed in bulk, good amount of protein is consumed. The quality of protein in rice is better than that of wheat.
- In vegetarian diets, a combination of cereal grains and legumes provide satisfactory biological value.

Recommended dietary allowances (RDA) of protein

The Indian Council of Medical Research recommends 1.0 g/kg/day as the safe level of intake in terms of dietary protein for Indians.

Group	Particulars	Protein (g)
Men	Sedentary work	60
	Moderate work	
	Heavy work	
Women	Sedentary work	55
	Moderate work	
	Heavy work	
	Pregnant women	+23
	Lactation 0-6 months	 +19
	6-12 months	+13
Infants	0-6 months	1.16 g/kg/d
	6-12 months	1.16 g/kg/d
Children	1-3 years (Preschool children)	16.7
	4-6 years	20.1
	7-9 years	29.5
Boys	10-12 years	39.9
Girls	10-12 years	40.4
Boys	13-15 years	54.3
Girls	13-15 years	51.9
Adolescents	16-17 (Boys)	61.5
	16-17 (Girls)	55.5

Protein quality

The quality of a protein is determined by the kind and proportion of amino acid it contains. Proteins that contain all essential amino acids in proportions capable of promoting growth are described as complete protein, good quality protein, or proteins of high biological value.

A good quality protein is digested and utilized well. Egg protein is a complete protein and is considered as a reference protein with the highest biological value. The quality of other proteins is determined based on their comparison with egg protein.

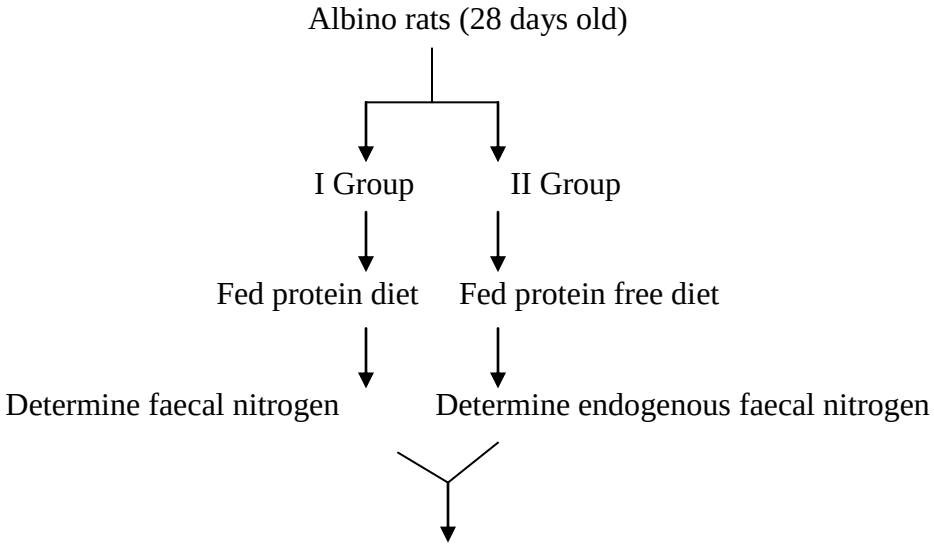
The protein of animal foods like milk, meat, and fish generally compare well with egg in the essential amino acid composition and are categorized as good quality proteins.

Plant proteins are of poor quality, since the essential amino acid composition is not well balanced.

PER, NPR, NPU and amino acid score are most suitable methods of evaluation of quality of protein.

Digestibility coefficient

It refers to the percentage of the ingested protein absorbed into the blood stream after the digestion is completed.

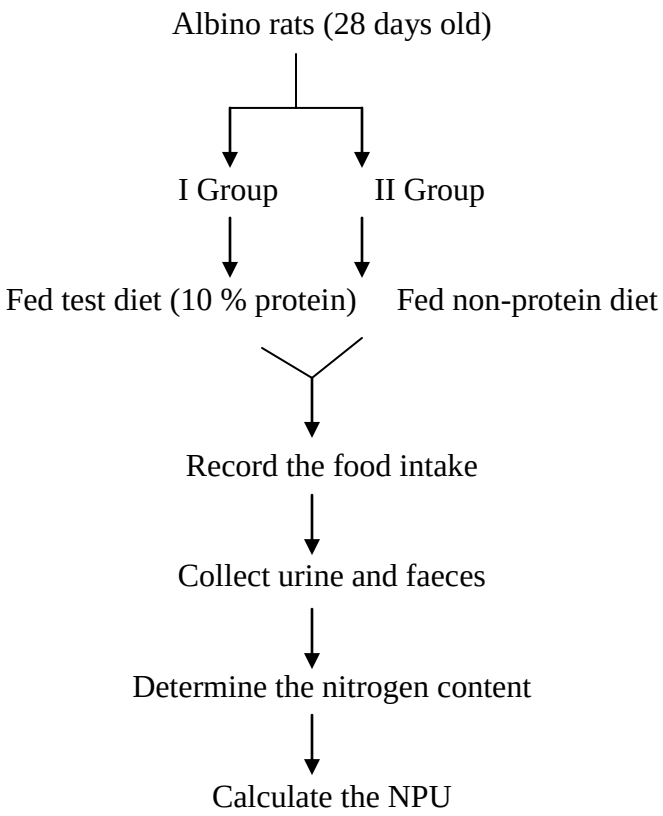


Calculate the Digestibility coefficient

$$\textbf{Digestibility coefficient} = \frac{\textbf{Nitrogen intake – Nitrogen loss in digestion X 100}}{\textbf{Nitrogen intake}}$$

Biological value

Biological value of protein is the percentage of nitrogen that is absorbed and available for use by the body for growth and maintenance. It measures the amount of dietary protein utilised by body for maintenance and growth. The method is briefly as follows:



$$\text{Biological value} = \frac{\text{Nitrogen digested} - \text{Nitrogen lost in metabolism}}{\text{Nitrogen digested}} \times 100$$

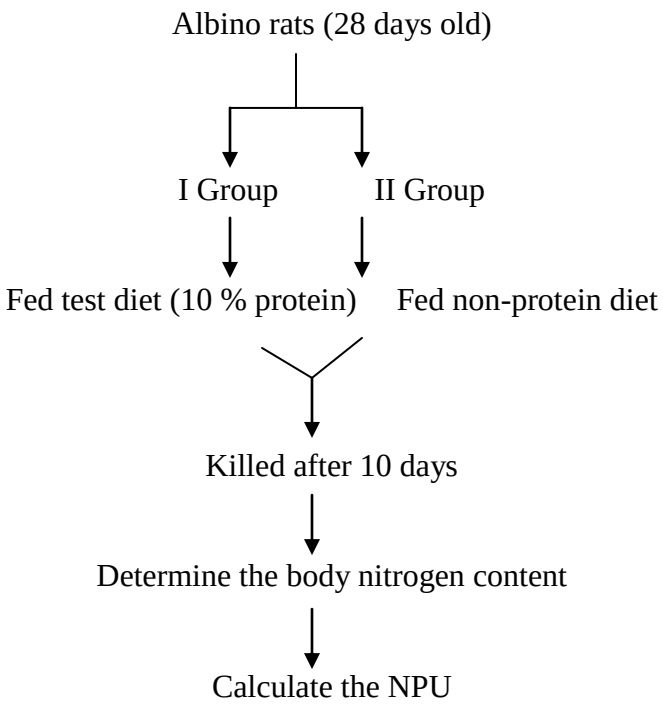
Protein efficiency ratio (PER)

Protein efficiency ratio is defined as the weight gain per gram of protein intake. This method was developed by Osborne, Mendel and Ferry.

$$\text{PER} = \frac{\text{Gain in weight (g)}}{\text{Protein intake (g)}}$$

Net protein utilization (NPU)

Miller and bender (1955) developed a direct method of estimating 'Net Protein Utilisation'. The method is briefly as follows:



$$\textbf{Net protein Utilization} = \frac{\textbf{Nitrogen retained}}{\textbf{Nitrogen intake}} \textbf{ X100}$$

Net protein Ratio (NPR)

This method of assay was introduced by Bender and Doell in 1957. This method is the modification of PER method. An allowance is made for the protein requirements and also for maintenance.

$$NPR = \frac{\text{Gain in weight (g) of the test group + loss in weight (g) of non-protein group}}{\text{Protein intake (g) of test group}}$$

Amino acid score / chemical score

The amino acid score is the ratio between the content of the most limiting amino acid in the test protein to the content of the same amino acid in egg protein (reference protein).

$$\text{Amino acid score} = \frac{\text{Limiting amino acid content in the test protein}}{\text{The same amino acid in egg protein}} \text{ X } 100$$

Nutritional value of food protein

Protein	PER	BV	NPU	Chemical Score	Limiting Amino Acid
Egg	4.5	96	90	100	Nil
Milk	3.0	84	75	65	Sulphur amino acids
Fish	3.0	85	70	60	Tryptophan
Meat	2.7	75	76	70	Sulphur amino acids
Rice	2.2	68	60	60	Lysine, Threonine
Wheat	1.5	58	47	42	Lysine, Threonine
Bengal gram	1.7	58	47	45	Sulphur amino acids

LECTURE 6

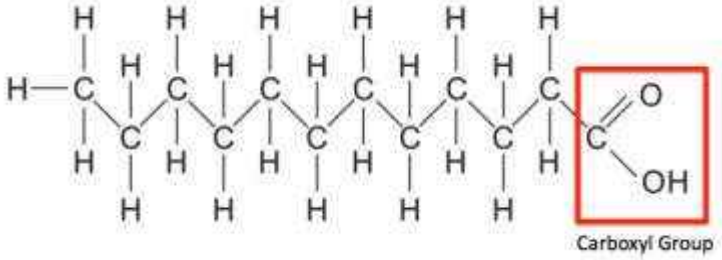
LIPIDS/FATS

Introduction

The word lipid is derived from the Greek word '*lipos*' meaning fat. Fat is one of the three main macronutrients, along with carbohydrate and protein. Fats, also known as triglycerides, are esters of three fatty acid chains and the alcohol glycerol. The terms "lipid", "oil" and "fat" are often confused. "Lipid" is the general term, though a lipid is not necessarily a triglyceride. "Oil" normally refers to a lipid with short or unsaturated fatty acid chains that is liquid at room temperature, while "fat" refer to lipids that are solids at room temperature however, "fat" may be used in food science as a synonym for lipid. Fats, like other lipids, are generally hydrophobic, and are soluble in organic solvents and insoluble in water.

Fatty acids

Fatty acids are the main building blocks of fat. They joined together in groups of three, forming a molecule called a triglyceride. A fatty acid is nothing more than a long C-H chain with a carboxyl group (COOH) on the end. The COOH gives it an acid property.



CLASSIFICATION OF FATTY ACIDS

I. Based on the length of free fatty acid chains

Fatty acid chains differ by length, often categorized as short to very long.

1. **Short-chain fatty acids (SCFA)** are fatty acids with aliphatic tails of five or fewer carbons (e.g. formic acid, Propionic acid, Isobutyric acid, Butyric acid, Isovaleric acid & valeric acid).
2. **Medium-chain fatty acids (MCFA)** are fatty acids with aliphatic tails of 6 to 12 carbons, which can form medium-chain triglycerides (e.g. Caproic acid (C6), caprylic acid(C8), capric acid (C10) and lauric acid (C12))

3. **Long-chain fatty acids (LCFA)** are fatty acids with aliphatic tails of 13 to 21 carbons. (e.g. α Linolenic acid (ALA), Linoleic acid (LA), Eicosapentaenoic acid (EPA), Docosahexaenoic acid (DHA)
4. **Very long chain fatty acids (VLCFA)** are fatty acids with aliphatic tails of 22 or more carbons

II. Based on degree of unsaturation

1. **Saturated Fatty Acids**

Saturated fatty acids have no double bonds. Saturated fatty acids have only single bonds; each carbon atom within the chain has 2 hydrogen atoms. Eg. Butyric acid, Lauric acid, Myristic acid, Palmitic acid, Stearic acid

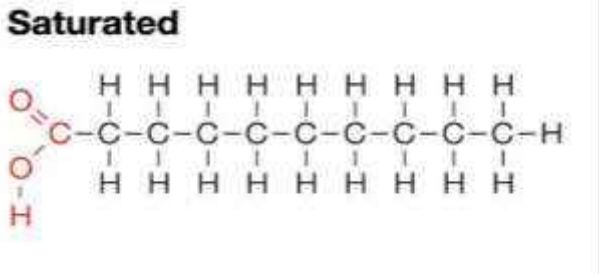
2. **Unsaturated Fatty Acids**

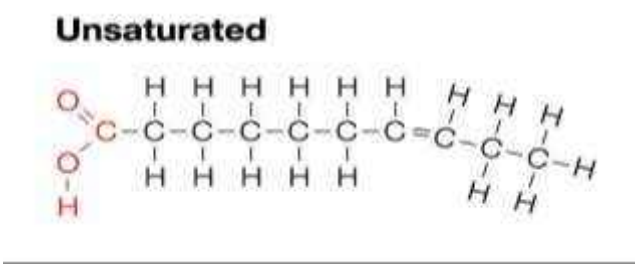
Unsaturated fatty acids have one or more double bonds between carbon atoms. Eg. Oleic acid, Palmitic acid, Erucic acid, Linoleic acid, Linolenic acid and Arachidonic acid.

- **Mono Unsaturated Fatty Acid (MUFA) – one C=C double bond**

  Eg. Oleic acid, Palmitic acid, Erucic acid
- **Poly Unsaturated Fatty Acid (PUFA) – more than one C=C double bond**

  Eg. Linoleic acid, Linolenic acid and Arachidonic acid

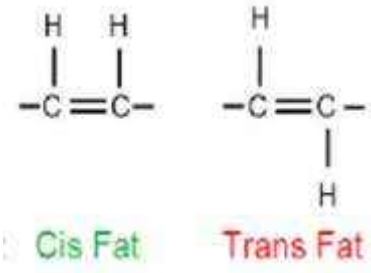




Cis and Trans fatty acids

The two carbon atoms in the chain that are bound next to either side of the double bond can occur in a Cis or Trans configuration.

- **Trans fatty acids -** two hydrogen atoms near the double bonds are on the **opposite sides of the molecule -** makes them straight and more rigid and solid. Trans fats are not found in nature and are the result of human processing (e.g., hydrogenation).



- **Cis fatty acids -** they are on the **same side**, which makes them more flexible and liquid.

III. Based on location of double bonds

1. **Omega−3 fatty acids – (ω−3 fatty acids or n−3 fatty acids) -** PUFAS with a double bond (C=C) at the third carbon atom from the end of the carbon chain. Eg. α-linolenic acid (ALA) (found in plant oils), Eicosapentaenoic acid (EPA) (found in marine oils) and Docosahexaenoic acid (DHA) (found in marine oils)
2. **Omega-6 fatty acids - (ω-6 fatty acids or n-6 fatty acids) -** Omega-6 fatty acids are a family of polyunsaturated fatty acids that have in common a final carbon-carbon double bond in the 6 position. Eg. Linoleic acid and Arachidonic acid.

IV. Based on synthesis in body

- **Essential fatty acid**

Essential fatty acids (EFA) are those which cannot be synthesized by the body and need to be supplied through diet. Linolenic acid, linoleic acid and arachidonic acid are essential fatty acids.

- **Non -essential fatty acids**

Non-essential fatty acids are those which can be synthesized by the body and which need not be supplied through the diet. Palmitic acid, oleic acid and butyric acid are examples of non – essential fatty acids.

CLASSIFICATION OF FATS

I. Based on their chemical composition

1. **Simple lipids or Homolipids** - Esters of fatty acid with various alcohols.
    - **Fats and oils -** Esters of fatty acids with glycerol. e.g., vegetable oil, ghee, butter
    - **Waxes -** Esters of fatty acids and alcohols. In human body it is present in the form of esters of cholesterol e.g., plant wax, insect wax (bees wax)
2. **Compound lipids or Heterolipids -** The compound lipids contain, in addition to fatty acids and glycerol, some other organic compounds.
    - **Phospholipids -** These contain phosphoric acid and nitrogenous base in addition to fatty acids and glycerol. e.g., lecithin and cephalin
    - **Glycolipids -** These lipids contain carbohydrates in combination with fatty acids and glycerol. e.g., cerebrosides
    - **Lipoprotein -** It contains protein subunits along with lipids.

3. **Derived lipids -** Substances derived from simple and compound lipids by hydrolysis are called as derived lipids. Fatty acids, alcohols and sterols are example for derived lipids.

II. Based on degree of unsaturation

1. **Saturated fats -** A saturated fat is a fat in which the fatty acids all have single bonds.

   Examples

   - Butyric acid with 4 carbon atoms (contained in butter)
   - Lauric acid with 12 carbon atoms (contained in coconut oil, palm kernel oil, and breast milk)
   - Myristic acid with 14 carbon atoms (contained in cow's milk and dairy products)
   - Palmitic acid with 16 carbon atoms (contained in palm oil and meat)
   - Stearic acid with 18 carbon atoms (also contained in meat and cocoa bean)

2. **Unsaturated fats -** An unsaturated fat is a fat or fatty acid in which there is at least one double bond within the fatty acid chain.
   a. **Mono Unsaturated fat** – one C=C double bond. Eg. Oleic acid, Palmitic acid, Erucic acid (Olive oil, peanut oil, canola oil, avocados, nuts)
   b. **Poly Unsaturated fat**– more than one C=C double bond. Eg. Linoleic acid, Linolenic acid and Arachidonic acid (corn, safflower, soybean, sunflower oils and fish oils).

FUNCTIONS OF FAT

- It is a concentrated source of energy. One gram of fat gives 9 Kcal.
- Fats are essential for the absorption of fat soluble vitamins like A, D, E, K.
- Fats contain essential fatty acids like linoleic, linolenic and arachidonic acids which are essential for maintaining tissues in normal health.
- Vegetable oils are good source of vitamin E.
- Butter and fish liver oil are good source of vitamin A.
- Fats improve the palatability of the diet
- It gives satiety value, i.e., a feeling of fullness in the stomach
- Phospholipids and other compound lipids are essential constituents of nervous tissue.
- Fats are deposited in the adipose tissue and this deposit serves as the source of energy during starvation.

- Fat deposited in adipose tissue functions like an insulating material against cold and physical injury.
- Cholesterol is essential for maintaining the cell membrane in good condition.
- Cholesterol is act as a precursor for the formation of some steroid hormones

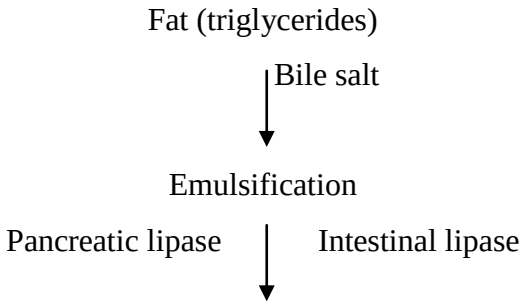
DIGESTION AND ABSORPTION OF FATS

Stomach

Lipase present in stomach is unable to digest fats due to high acidity of gastric juice. So fat is not digested in stomach.

Small intestine

Bile salt is essential for the digestion and absorption of fat. It helps to emulsify fats before digestion. Pancreatic lipase and intestinal lipase hydrolyze the fat into diglycerides, monoglycerides and fatty acids. Phospholipase and cholesterol esterase hydrolyze phospholipids and cholesterol



Absorption of fats

The products of digestion pass into the cells of the intestinal wall, where synthesis of new glycerides takes place. The lipids pass through the lacteals of the small intestine to thoracic duct and then to the blood stream in the form of fine particles known as chylomicrons. A greater part of the cholesterol present in the diet is absorbed while phytosterols present in vegetable fats and oils are not absorbed.

The oxidation takes place through the tricarboxylic acid cycle. Fat is also synthesized in the body from carbohydrates by a complex mechanism.

LIPIDS IN BLOOD

LIPID PROFILE

	DESIRABLE	BORDERLINE	HIGH RISK
Cholesterol	<200 mg/dl	200-239 mg/dl	240 mg/dl
Triglycerides	<150 mg/dl	150-199 mg/dl	200-499 mg/dl
HDL cholesterol	60 mg/dl	35-45 mg/dl	<35 mg/dl
LDL cholesterol	60-130 mg/dl	130-159 mg/dl	160-189 mg/dl
Cholesterol/ HDL ratio	4.0	5.0	6.0

DEFICIENCY OF FATS

Adult

'Phrynoderma' or **'Toad skin'** is the essential fatty acid deficiency symptoms. It is characterised by the presence of horny popular eruption on limbs, backs and buttocks.

Infant

Perianal irritation and changes in skin (dryness, thickening, desquamation).

Symptoms of dietary fat deficiency

- **Cancer** is a condition where the cells of the body start to grow in a rapid, uncontrollable way. Monounsaturated fats and omega 3 essential fatty acids (EFAs) have been linked with the prevention of breast cancer, colon cancer and prostate cancer.
- **Diabetes** is a condition where the body struggles to control blood glucose levels. This causes damage to the blood vessels and a number of vital organs. Diet rich in omega 3 EFAs reduced risk of diabetes.
- **Fat Soluble Vitamin Deficiency -** Polyunsaturated fats support the absorption of fat soluble vitamins. Reducing dietary fat intake significantly can cause to become deficient in these essential vitamins which can lead to a number of unpleasant symptoms including anemia (a low red blood cell count), vision problems, weak bones and skin problems
- **Fatty Liver -** Polyunsaturated fats reduce the deposit of fatty droplets on the liver.
- **Growth Problems** - Both saturated fats and omega 6 EFAs support the development of strong healthy bones.

- **Hair, Nail and Skin Problems** - Polyunsaturated fats promote the formation of healthy hair, nail and skin cells. Not getting enough can have an adverse effect on skin and lead to problems such as acne, dermatitis (inflammation of the skin), psoriasis (a skin disease which causes dry, red, scaly patches to develop) and wrinkles. It can also affect hair and nails leading to dry hair, hair loss and impaired nail growth.
- **Heart Disease** – Reducing fats from diet leads to atherosclerosis (hardening of the arteries), increases triglyceride levels and also reduces blood levels of high density lipoprotein (HDL) cholesterol (a type of cholesterol which removes plaques from the artery walls). Both atherosclerosis and low HDL cholesterol restrict the flow of blood to heart whilst high triglyceride levels have also been linked with an increased heart disease risk.
- **High Blood Pressure** - High blood pressure is a condition which can damage blood vessels and vital organs. It also increases risk of a heart attack or a stroke. Regular consumption of omega 3 EFAs can lower blood pressure whilst not getting enough can have the opposite effect and lead to high blood pressure.
- **Inflammation** - Both monounsaturated fats and polyunsaturated fats reduce inflammation in the body. Failing to eat enough of these fats can lead to a number of inflammatory disorders including arthritis.
- **Macular Degeneration** - Omega 3 EFAs are used by body to create retinal tissue and can combat age related macular degeneration.
- **Mental Problems** - Omega 3 EFAs support the production of signal pathways in brain and also act in a protective capacity.
- **Stroke** - High blood pressure is one of the major contributors to a stroke and as discussed above omega 3 EFA deficiencies cause an increase in blood pressure

FOOD SOURCES

Foods in general contain two types of fat namely “visible fats” and “invisible” or “hidden” fats.

Visible fats - Visible fats are fats extracted from the following sources.

- **Oil seeds:** coconut, corn, cornseed, groundnut, mustard, palm, rice bran, safflower, seasame, soyabean, sunflower and hydrogenated vegetable oils (vanaspathi).
- **Animal fats:** Butter and Ghee.

- **Fish oils:** Shark and cod liver oils.

Invisible or hidden fats:

Invisible or hidden fats are those which form an integral part of foods and are therefore not visible. It includes the fats present in the cells and cell walls and cell membranes of both plant and animal tissues. Almost everything we eat as listed below carries some invisible fats.

- Plant food – Cereals, millets, vegetables, spices, nuts and oil seeds, coconut, avacado.
- Animal food – Milk and milk products (curd, cream, cheese), flesh foods, (mutton, beef, pork, chicken) organ meats (brain, liver, kidney), fish, shrimp, prawn.

REQUIREMENTS

The ICMR recommended allowances for fat for Indians is given in table

S. No	Group	Requirement (g/day)	Total calories from fat (%)
1	Infants (0-1 year)	19	25-30
2	children	25-30	15-20
3	**Adolescents (13-17 year)** Boys	45-50	
	Girls	35-40	
4	Man	25-40	10-20
5	Woman	20-30	

RANCIDITY

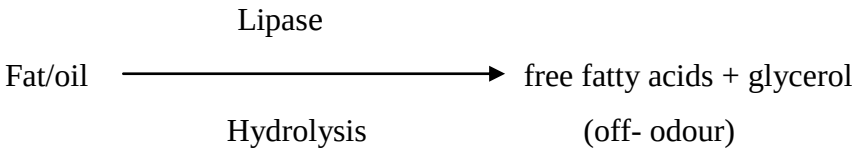
The **development of off-flavours or unpleasant odour in fats** is known as rancidity. There are three main types of rancidity. Rancidity is a term generally used to denote "a condition of unpleasant odours and flavours in foods resulting from deterioration in the fat or oil portion of a food"

- Hydrolytic rancidity
- Oxidative rancidity and
- Ketonic rancidity

Hydrolytic/ lipolytic rancidity

The fats and oils are hydrolysed by the enzyme lipase and yield free fatty acids and glycerol. It results in off-odour development. This is known as hydrolytic rancidity. In case of

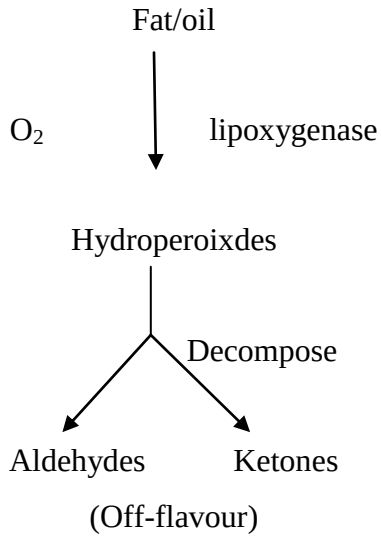
butter and coconut oil, butyric acid and other low molecular weight fatty acids are set free by hydrolysis by lipase. The odours of these acids contribute to the smell of rancid butter.



Oxidative rancidity

The fat and oil are contact with the atmospheric oxygen in the presence of lipoxygenase enzyme. It results in the formation of hydroperoxides. Further, it breaks into aldehydes and ketone which is responsible for off-flavour. This is called as oxidative rancidity.

This is the common type of rancidity observed in all fats and oils. The oxidation takes place at the unsaturated linkage. Certain metals, e.g. copper, hasten the onset of oxidative rancidity.



Ketonic rancidity

This type is most frequently encountered as a result of action of fungi such as *Aspergillus niger, pencillium glaucum* on coconut or other oil seeds. They tallowy odour developed may be due to aldehydes and ketones formed by the action of the enzymes present in the fungi on oils.

Prevention of rancidity

Fats can be protected against the rapid development of rancidity by controlling the conditions of storage.

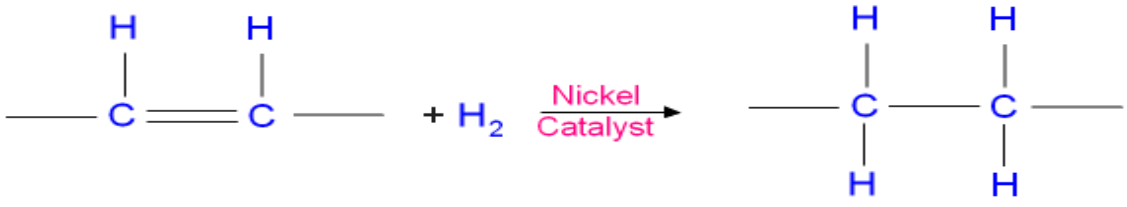
- Storage at refrigerator temperature prevents rancidity.

- Rays of light catalyse the oxidation of fats by the use of coloured glass containers that absorb the active rays, fats can be protected against spoilage.
- Certain shades of green bottles and wrappers and yellow transparent cellophane wrappers are effective in preventing rancidity.
- Vacuum packaging also helps to retard the development of rancidity by excluding oxygen.
- Antioxidants naturally present in the food such as vitamin C, beta carotene and vitamin E protect against rancidity.
- Antioxidants like Butylated Hydroxy Anisole (BHA), Butylated Hydroxy Toluene (BHT), Tertiary Butyl Hydroquinone (TBHQ) and Propyl Gallate (PG) can also be added to prevent rancidity.
- Substances like citric acid may be used along with antioxidants in foods as synergists. A synergist increases the effectiveness of an antioxidant but is not as effective an agent when used alone.
- Some Chelating agents are bind or chelate the metals and prevent the oxidation process. Chelating agents are sometimes called sequestering agents.

Hydrogenation

Hydrogenation means adding hydrogen to a substance. Liquid vegetable oils that are unsaturated will react with hydrogen at about 60 °C in the presence of a nickel catalyst. This is an example of an addition reaction where hydrogen adds across the double bond leaving only single bonds. Hydrogenation raises the melting point above room temperature and makes the liquid oil become solid in a process called hardening. The solid product is used as a margarine or vanaspathi. Unsaturated glycerides present in the plant oil can be converted to more saturated glycerides.

The picture below shows hydrogenation of a double bond.



Liquid fat (more unsaturated fatty acids) —addition of H_2, Nickel→ Semisolid/solid fat (more saturated fatty acids)

Determination of Energy Value of Foods

The energy or calorific value of food depends on the quantity of carbohydrates, fats and proteins present in them. Energy value of food can be determined by oxidizing a known weight of food in an instrument called bomb calorimeter and measuring the heat produced in energy unit. The energy value of food can be expressed in terms of kilocalories. One kilocalorie is the quantity of heat required to raise the temperature of 1 kg of water through 1°C.

Bomb calorimeter – Principle

The energy value of food is usually determined using the instrument bomb caloriemeter. It consists of a heavy steel bomb, with a platinum or gold plated copper lining and a cover held tightly in place by means of a sample.

A bomb calorimeter is a type of constant – volume calorimeter used in measuring the heat of combustion of a particular reaction. Bomb calorimeters have to withstand the large pressure within the calorimeter as the reaction is being measured. Electrical energy is used to ignite the fuel, as the fuel is burning, it will heat up the surrounding air, which expands and escapes through a tube that leads the air out of the calorimeter. When the air escaping through the copper tube it will also heat up the water outside the tube. The temperature of the water allows for calculating calorie content of the fuel.

Procedure

A weighed amount of a sample is usually pressed into pellet forms and placed in a capsule with in the bomb crucible which is then, closed except for the oxygen value charged with oxygen to a pressure of about 25 pounds and immersed in a weighed amount of water. The water is constantly stirred and its temperature taken at intervals of 1 minute by means of a differential thermometer capable of being read to thousand with of a degree. After the temperature of water has been determined, the sample is ignited by means of oxygen present. It undergoes rapid and complete combustion. The heat liberated is absorbed by the water in which the burnt is immersed and the result raise in temperature readings and also continued through an after period in older that the radiation correction may be calculated and the absorbed raise multiplied by the total heat capacity of the apparatus and the water which it is immersed gives the total heat liberated in the bomb. Then this heat arising from accessory combustion must be detected to obtain the number of calories arising from the combustion of the sample.

Calculation

Weight of sample taken = 2g

Weight of water in the outside vessel = 3000 g

Water Equivalent of the calorimeter = 500g

Initial temperature of water = 24° C

Final temperature of water = 26° C

Rise in temperature = **(Final temperature of water – Initial temperature of water)**

= 26 -24 = 2° C

Heat gained by water and calorimeter = (**Weight of water + Equivalent weight of water x Rise in temperature)**

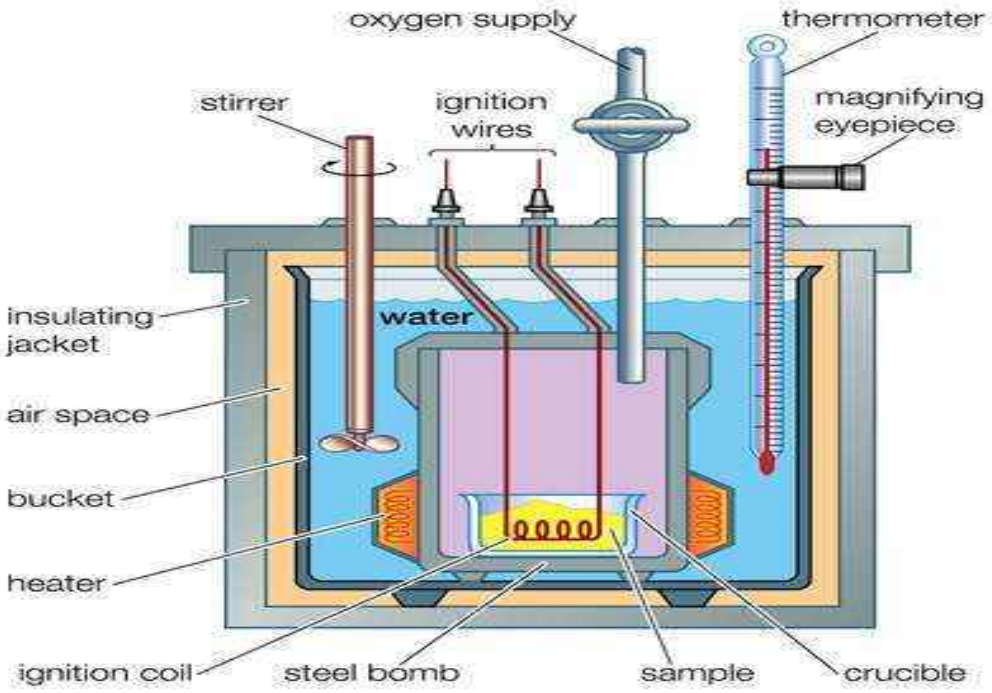
= 3000 + 500 X 2 = 7000

Energy value of given sample = 7000 Cal or 7 KCal

Energy value of given sample per g = **Heat gained by water and calorimeter / weight of the sample**

= 3500 Cal or 3.5 KCal

Bomb Calorimeter



LECTURE 7

VITAMINS

Vitamins may be defined as organic compounds occurring in small quantities in the different natural foods and necessary for the growth and maintenance of good health in human beings and certain experimental animals

Classification of vitamins

Vitamins are classified as follows:

Fat soluble vitamins

1. Vitamin A
2. Vitamin D
3. Vitamin E and
4. Vitamins K

Water soluble vitamins

- Vitamin B complex
    1. Thiamine (Vitamin B_1)
    2. Riboflavin (Vitamin B_2)
    3. Niacin (Vitamin $B_{3)}$
    4. Panthothenic acid (Vitamin B_5)
    5. Pyridoxine (vitamin B_6)
    6. Biotin (Vitamin B_7)
    7. Folic acid (Vitamin B_9)
    8. Vitamin B_{12}
- Vitamin C (Ascorbic acid)

FAT SOLUBLE VITAMINS

VITAMIN A

Vitamin A only present in foods of animal origin. Vitamin A activity is also possessed by carotenoids present in plants. So carotenoids (**β- carotene**) are called as **provitamin A or precursor of vitamin A**. Vitamin A – **Antixerophthalmic vitamin**

FUNCTIONS

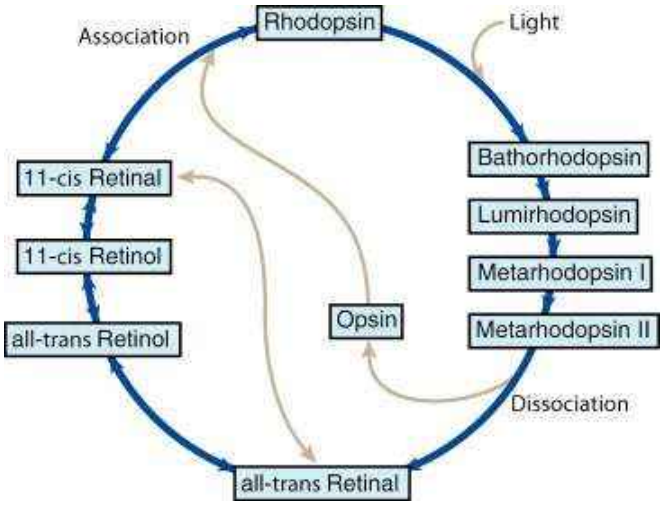
1. Vitamin A is very important for good vision.
2. Maintaining the health of the epithelial tissues of eyes and skin.
3. Supports reproduction and growth.
4. It helps to strengthen immune system.
5. Beta carotene acts as antioxidant and protecting the body against cancer.
6. It is essential for normal bone formation.

Role of vitamin A in Visual cycle

In the retina, retinaldehyde functions as the prosthetic group of the light – sensitive opsin proteins, forming rhodopsin (in rods) and iodopsin (in cones). Any one cone cell contains only one type of opsin and is sensitive to only one colour. In the pigment epithelium of the retina, all – trans – retinol is isomerized to 11-cis-retinol and oxidized to 11 – cis-retinaldehyde. This reacts with a lysine residue in opsin forming the holoprotein rhodopsin.

As shown in the figure the absorption of light by rhodopsin causes isomerization of the retinaldehyde from 11 cis to all trans and a conformational change in opsin. This results in the release of retinaldehyde from the protein and the initiation of a nerve impulse. The formation of the initial excited from of rhodopsin, bathorhodopsin occurs within picoseconds of illumination. There is then a series of conformational changes leading to the formation of metarhodopsin II, which initiates a guanine nucleotide amplification cascade and then a nerve impulse. The nerve impulse generated by the optic nerve is conveyed to the brain where it can be interpreted as vision. The final step is hydrolysis to release all – trans retinaldehyde and opsin. The key to initiation of the visual cycle is the availability of 11 – cis – retinaldehyde, and hence Vitamin A. by a series of reactions all trans retinol is converted to 11 cis – retinal which reassociates with opsin to from Rhodopsin. In deficiency, both the time taken to adapt to darkness and the ability to see in poo light are impaired.

Visual Cycle



DEFICIENCY:

The important deficiency status due to lack of vitamin A in the diet are

Night blindness

In early stages of vitamin A deficiency, the individual cannot see well in dim light. Difficulty in reading or driving the car in dim light is experienced. In advanced deficiency, the subject cannot see objects in dim light. Night blindness is fairly common in regions where the vitamin A intake is inadequate.

Xerosis conjunctivae

The conjunctiva is dry, thickened, wrinkled and pigmented. This is due to the keratinisation of the epithelial cells. The pigmentation gives the conjunctiva a smoky appearance. This condition is extremely common among all age groups in India and other developing countries where the vitamin A intake is inadequate.

Xerosis cornea

When dryness spreads to cornea, it takes on a dull, hazy, lusterless appearance. This is due to the keratinisation of the epithelial tissue covering the cornea.

Bitots spots

Grayish glistening, white plaques formed of desquamated thickened conjunctiva epithelium, usually triangular in shape and firmly adhering to the conjunctiva are frequently found in children having other signs of vitamin A deficiency.

Keratomalacia

When xerosis of the conjunctiva and cornea is not treated, it may develop into the condition known as 'keratomalacia'. The corneal epithelium becomes opaque and ulceration and bacterial invasion of the cornea bring about its destruction resulting in blindness.

DIETARY SOURCES

Sheep liver, butter, ghee, egg, milk, curds, liver oils of shark, green leafy vegetables, carrots, mango and papaya are the rich source of vitamin A.

REQUIREMENTS

Requirements/RDA of Vitamin A

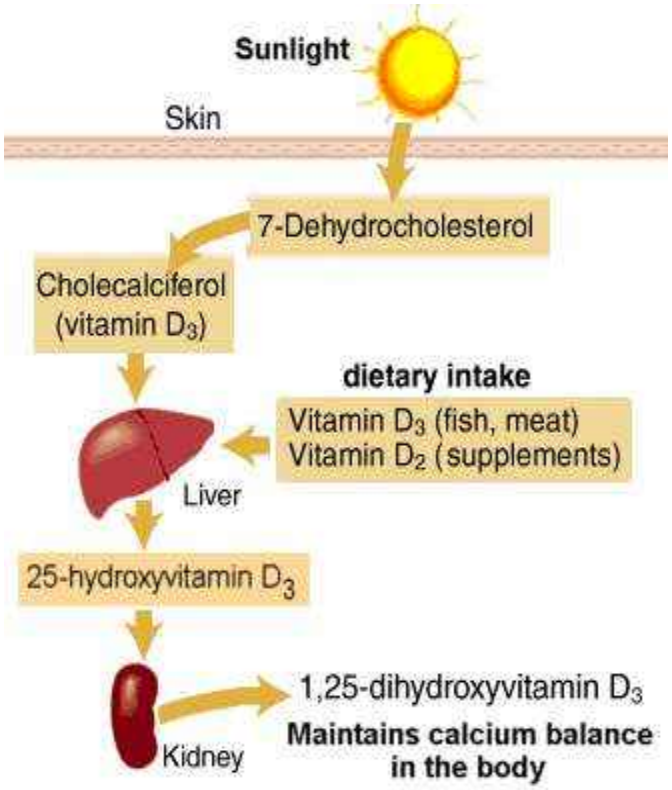
Age Group	Requirements (µg)
Children	400
Adolescents & adult	600
Pregnancy	800
Lactation	950

VITAMIN D

Vitamin D can be synthesized in the body in adequate amounts by simple exposure to sunlight, even for 5 minutes per day is sufficient. The 7-dehydrocholesterol is present in skin and act as precursor of vitamin D. It is essential for bone growth and calcium metabolism. It acts as a hormone in the body by facilitating calcium absorption and deposition in the bone.

FORMS

- Vitamin D2 – Erogcalciferol
- Vitamin D3 - Cholecalciferol



Vitamin D in the body

Vitamin D is a fat soluble vitamin that exists in various forms. The animal form is vitamin D3 (cholecalciferol) and the plant form is vitamin D2 (ergocalciferol). Vitamin D2 and D3 are not biologically active; they must be modified in the body to have any effect. The active form of vitamin D is indeed a hormone and is known as 1,25-dihydroxyvitamin D3 [1,25(OH)2D3] or calcitriol. Both vitamin D2 and D3 have been commercially synthesized and both forms seem to be effective at maintaining blood levels of vitamin D in the body.

Step 1: cholesterol to 7-dehydrocholesterol, which is a precursor of vitamin D3.

Step 2: When we are exposed to UVB radiation, 7-dehydrocholesterol in the skin is converted to vitamin D3.

Step 3: Vitamin D3 must then be hydroxylated in the liver and the kidneys to become active. At this point, it can exert its endocrine effect.

FUNCTIONS

1. Vitamin D helps in the absorption of calcium and phosphorous.
2. It helps in the formation of bone and teeth.
3. Vitamin D helps to maintain the calcium and phosphorous levels in the body by stimulating absorption in the gastro intestinal tract, retention by the kidney.

DEFICIENCY

Vitamin D deficiency causes the disease rickets in children and osteomalacia in adults.

- **Rickets (Children)**

It is a disease in which there is weakness and abnormalities in bone formation. Rickets primarily affects children. Box like head, bowed knee, swelled wrist, pigeon breast are the symptoms of rickets.

- **Osteomalacia (Adult rickets)**

The unmineralised condition of bone (soft bone) is called Osteomalacia. It occurs in women in *Purdah*, not exposed to sunshine. It causes pain in pelvis, lower back, legs and frequent bone fractures.

- **Osteoporosis (old age)**

Osteoporosis is a medical condition in which the bones become brittle and fragile from loss of tissue, typically as a result of hormonal changes, or deficiency of calcium or vitamin D.

SOURCES

Sunlight, Cod liver oil, shrimp, liver, butter, yolk, cheese, milk, spinach and cabbage are the rich source of vitamin D.

REQUIREMENTS

Requirements/RDA of Vitamin D

Age Group	Requirements (mg)
Infant & preschool children	10
Older children & Adult	5
Expectant & Nursing mother	10

VITAMIN E

Vitamin E is essential for normal reproduction in several species of animals and also in human beings. It is otherwise known as Tocopherol/Antisterility vitamin.

FUNCTIONS

The important functions of vitamin E are as follows :

i. Prevents peroxidation of polyunsaturated fatty acids in tissues and cell membranes. In vitamin E deficiency PUFA undergo peroxidation and yellow and brown pigments are formed.
ii. It protects red blood cells from haemolysis by oxidising agents.
iii. In vitamine E deficiency, there is degeneration of cellular and sub-cellular membranes which are known to be rich in PUFA.
iv. Vitamin E offers definite protection to liver injury due to carbon tetrachloride poisoning.
v. Vitamin E along with an activator present in microsomal supernatant is able to prevent the respiratory decline in isolated mitochondria.

DEFICIENCY

Vitamin E deficiency in animals causes several disorders such as reproductive failure, liver necrosis, muscular dystrophy etc. They are briefly discussed below.

Reproductive failure

In the female rat fed on vitamin E deficient diets, foetal death and resorption take place. The primary metabolic defect which causes death and subsequent resorption of the embryo is now known. There are degenerative changes in the uterus. However, the vascular system of the embryo also undergoes degeneration. In the male rat, irreversible testicular degeneration occurs in vitamin E deficiency.

Liver necrosis

Necrosis of the liver produced in rats fed on yeast is prevented by vitamin E

Erythrocyte haemolysis

In vitamin E deficient animals, the erythrocytes readily undergo haemolysis when treated with hydrogen peroxide. This is prevented by vitamin E.

Anaemia

Monkeys fed on vitamin E deficient diets develop anaemia. This is due to lack of haemotopoiesis in the bone marrow rather than from excessive red cell destruction. Administration of vitamin E cures the anaemia.

Muscular dystrophy

Rabbits, guinea pigs, monkeys and chicks fed on vitamin E deficient diets develop muscular dystrophy which is cured and prevented by vitamin E.

SOURCES

Wheat germ, Vegetable oils, sunflower seeds, almonds, safflower oil, Sesame oil and mustard oil, eggs, butter are the rich source of vitamin E.

REQUIREMENTS

Requirements/RDA of Vitamin E

Age Group	Requirements (mg)
Infant & preschool children	3-4
Adult	10
Pregnancy	12
Lactation	13

VITAMIN K

Dam and Schonhyder (1934) found that chicks fed on purified diets containing all vitamins known at that time developed haemorrhagic condition which was cured by Lucerne (alfalfa) and decayed fish meal. Vitamin K is recognized as the antihaemorrhagic factor due to its vital role in blood clotting mechanism.

Forms

- Vitamin K1 – Phyloquinone
- Vitamin K2 - menaquinone

FUNCTION

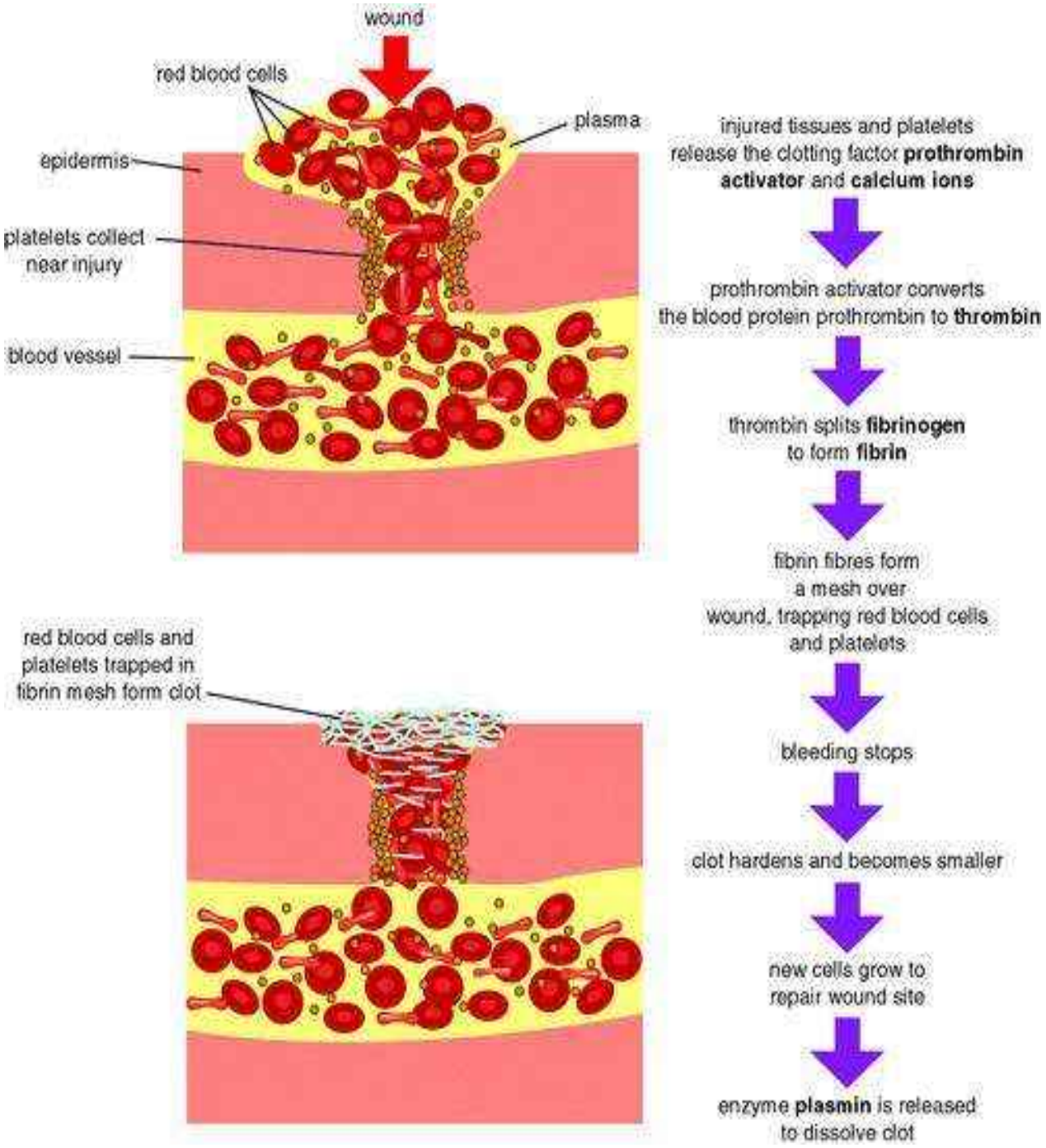
Vitamin K is essential for blood coagulation. Vitamin K administration leads to an increase in the prothrombin level. In vitamin K deficiency, prothrombin level is reduced.

Blood clotting mechanisms:

- When we get cut blood vessels around the wound immediately constrict to reduce blood loss
- The platelets in the blood exposed to air become sticky and clump together to plug the wound
- Thrombokinase & other clotting factors are released by platelets.
- In the presence of of calcium ions thrombokinase converts prothrombin into thrombin. Prothrombin a plasma preotin synthesized in the liver and requires vitamin K.

- Thrombin converts soluble plasma protein fibrinogen into insoluble fibrin fibres which forms ameshworks of threads over the wound
- As the blood flows out, erythrocytes & platelets are trapped in the fibrin fibres and blood clot forms it dries to from scab.

Blood clotting mechanisms



DEFICIENCY

Vitamin K deficiency leads to a lowering of prothrombin level and increased clotting time of blood. This may lead to haemorrhagic conditions. Vitamin K deficiency can occur in the following ways:

i. Inadequate intake of vitamin K

ii. Inadequate intestinal absorption.

Inadequate intake of vitamin K

Inadequate intake of vitamin K by the mother may causes the haemorrhagic disease of the new born. The infants have a low prothrombin level and they recover rapidly when vitamin K is administered by injection.

Inadequate intestinal absorption

In adequate intestinal absorption of vitamin K may result from.

a) lack of bile in the intestine due to defective secretion of bile as in liver disorders intestinal obstruction and

b) Poor absorption due to diarrhea or dysentery.

SOURCES

Vitamin K occurs in plant foods while vitamin K_2 occurs in microganism. The best sources of vitamin K, are the green leafy vegetables eg: alfalfa, spinach, cabbage, etc.

REQUIREMENTS

Requirements/RDA of Vitamin K

Age Group	Requirements (mg)
Infant	12-20
Adult	70-140
Pregnancy	200
Lactation	150
Adolescents	50-100

LECTURE 8

WATER SOLUBLE VITAMINS

The various members of the vitamin B complex are not related either chemically or physiologically, yet they have many features in common:

(a) All of them except lipoic acid are water-soluble.

(b) Most of them, if not all, are components of coenzymes that play vital roles in metabolism

(c) Most of these can be obtained from the same source, i.e., liver and yeast.

(d) Most of them can be synthesized by the intestinal bacteria.

Coenzyme derivatives of water-soluble vitamins

Vitamin	Coenzyme form
Vitamin B_1 (Thiamine)	Thiamine pyrophosphate (TPP)
Vitamin B_2 (Riboflavin)	Flavin mononucleotide (FMN) Flavin adenine dinucleotide (FAD)
Vitamin B_3 (Niacin)	Nicotinamide adenine dinucleotide (NAD) Nicotinamide adenine dinucleotide phosphate (NADP)
Vitamin B_5 (Pantothenic acid)	Coenzyme A (CoA)
Vitamin B_6 (Pyridoxine)	Pyridoxal phosphate (PALP), Pyridoxamine phosphate (PAMP)
Vitamin B_7 (Biotin)	Biocytin
Vitamin B_9 (Folic acid)	Tetrahydrofolic acid (THFA)
Vitamin B_{12} (Cyanocobalamin)	Deoxyadenosyl cobalamin
Vitamin C (Ascorbic acid)	Not known

VITAMIN B_1 (THIAMINE)

Thiamine was the first member of the vitamin B group to be identified and hence given the name vitamin B1. Thiamine was first isolated by Jansen (1934) in Holland from rice polishings. On account of its curing action against beriberi, it is commonly known as **antiberiberi factor.** It is also known as **antineuritic factor** or heat-labile factor. In Europe, it is also designated **aneurin.**

FUNCTIONS:

- ❖ Used in metabolism of carbohydrates for energy, electrolyte balance, muscle and nerve function, and hydrochloric acid production in the stomach.
- ❖ Oxidation of pyruvic acid.

DEFICIENCY

Vitamin B1 deficiency leads to polyneuritis in animals and beriberi in human beings.

Polyneuritis in birds renders them unable to fly, walk or even stand. Rats develop, among other symptoms, a brachycardia (slowing of the heart rate).

The major symptoms may follow

One of the following 3 courses (and accordingly beriberi is of 3 types) :

1. **Wet beriberi**

   Oedema in the legs, enlargement of the heart and palpitation and breathlessness.

2. **Dry beriberi**

   Loss of appetite, tingling and numbness of the legs and wasting of muscle and difficulty in walking.

3. **Infantile beriberi**

   The early symptoms are restlessness, sleeplessness and loss of appetite. Palpitation and breathlessness and loss of appetite develop, as the disease advances, due to the enlargement of the heart. Death may occur suddenly if treatment is delayed.

SOURCES

- **Rich sources:** dried yeast, rice polishing and wheat germ are rich sources.
- **Good sources:** whole cereals, legumes, oilseeds and nuts are good sources.
- **Fair sources:** milled cereals, vegetables, fruits, milk, meat and fish are fair sources.

REQUIREMENTS

Requirements/RDA of Vitamin B_1

Age Group	Requirements (mg)
Infant	0.2-0.3
Preschool children	0.5
Adolescents & Adult	1.0-1.7
Pregnancy	1.6
Lactation	1.7

VITAMIN B_2 (RIBOFLAVIN)

Riboflavin or vitamin B2 was first isolated in 1879 from milk whey which is an essential dietary factor for rats. Since it was first isolated from milk, vitamin B2 is also known as lactoflavin. Originally, it was also known as ovoflavin (from eggs) and hepatoflavin (from liver). Its synthesis was done by Richard Kuhn and Paul Karrer. It is popularly called as the "yellow enzyme" because of its colour.

FUNCTIONS

- Riboflavin plays an important role in many enzyme system involved in the metabolism of carbohydrates, fats and protein.
- Helps in energy production, making niacin, red blood cell formation, and human growth.

DEFICIENCY

- **Angular Stomatitis:** Lesions at the angles of the mouth is called Angular Stomatitis
- **Glossitis:** The inflammation of the tongue with magenta colour is called glossitis
- **Cheilosis**: Dry chapped appearance of the lip with ulcers is called cheilosis.
- **Scrotal lesions:** scrotal dermatitis – adults.
- Common - anemia, mouth sores, sore throat, swelled mucous membranes, and skin disorders.

SOURCES

Liver, dried yeast, egg powder, milk, skim milk powder, dairy, eggs, green leafy vegetables, nuts, meat, legumes, and enriched flour.

REQUIREMENTS

Requirements/RDA of Vitamin B_2

Age Group	Requirements (mg)
Infant	0.3 - 0.4
Preschool children	0.6
Adolescents	1.2
Adult	1.4 - 2.1
Pregnancy	2.1
Lactation	2.0

VITAMIN B_3 (NIACIN)

Vitamin B_3 refers to nicotinic acid and was named as pellagra preventive (PP) factor by an Austrian- American physician of the U. S. Public Health Service, Joseph Goldberger (1920) because of its curing action on pellagra (After Goldberger's death, vitamin B_3 was sometimes called vitamin G in his honour).

The vitamin role on nicotinic acid was first recognized by Conrad Elvehjem and D. Wayne Woolley of Wisconsin University in 1937. As this vitamin has a curing action against black tongue disease in dogs, it is also called as anti black tongue factor. It was first isolated by Funk in 1911.

FUNCTIONS

- Nicotinic acid is essential for the normal functioning of the skin, intestinal tract and nervous system.
- Nicotinamide is a component of two coenzymes (NAD and NADP) which are essential for the metabolism of carbohydrates, fats and proteins.

DEFICIENCY

A deficiency of niacin causes pellagra in man and blacktongue in dogs. Pellagra (of Italian origin, *pellis* = skin; *agra* = rough) is characterized by 3 "Ds", namely **d**ermatitis of the exposed parts, **d**iarrhea and **d**ementia (mental symptoms).

- Glossitis (redness & soreness of tongue) and diarrhoea -loose stools a day with blood and mucous.
- Dermatitis- skin is red and slightly swollen later the skin becomes dry, less red and the surface desquamates.
- Delirium is the commonest severe mental disturbance is acute pellagra. Dementia more frequently seen in the chronic cases.

Niacin equivalent of tryptophan present in dietary proteins

60 mg of tryptophan yield 1mg of niacin in addition to the preformed niacin present in food stuff, the tryptophan present will also provide additional niacin.

SOURCES

Dried yeast, liver, rice polishing Pork, turkey, fish, beef, peanut butter, legumes, enriched and fortified grains.

REQUIREMENTS

Requirements/RDA of Vitamin B_3

Age Group	Requirements (mg)
Infant	650-710 µg /Kg
Preschool children	8
Adolescents	14
Adult	16-21
Pregnancy	16
Lactation	20

VITAMIN B_5 (PANTOTHENIC ACID)

This was first isolated by **Roger J. Williams** in 1938 from yeast and liver concentrates. On account of its wide distribution, he named it as pantothenic acid (*pantosG* = everywhere). The coenzyme form of this vitamin (coenzyme A or CoA-SH) was isolated and its structure determined by **Fritz A. Lipmann.** The chemical synthesis of this coenzyme was, however, described by **Khorana** in 1959. This vitamin is sometimes called as **filtrate factor** or the **yeast factor**.

Functions

- Pantothenic acid in the form of coenzyme A takes part in the metabolisms of carbohydrates and fats.
- •It is essential for the oxidation of pyruvic acid.

DEFICIENCY

In human beings, no definite deficiency syndrome has been ascribed to pantothenic acid, probably because of the ubiquitous nature of this vitamin and because of the fact that a little amount of this vitamin can perhaps be synthesized in the body. Its correlation with achromotrichia (premature greying of the hair) has been described in the case of man, sometimes. But it seems too much to hope that grey hair can be averted by attention to diet.

SOURCES

Although widespread in nature, yeast, liver and eggs are the richest sources of it. The vegetables (potatoes, sweet potatoes, cabbage, cauliflower, broccoli) and fruits (tomatoes, peanuts) and also the skimmed milk, wheat bran, whole milk and canned salmon are some of the less important sources. In most animal tissues and microorganisms, it occurs as its coenzyme.

Requirements

- 5—10 mg per day for children to adolescence.

VITAMIN B_6 (PYRIDOXINE)

The name vitamin B6 was also named as adermin or antidermatitis factor. Vitamin B6 group includes 3 compounds : pyridoxine, pyridoxal and pyridoxamine. Pyridoxine was first isolated, in 1938, from yeast and liver. Later, Snell (1942) discovered the other two compounds. Pyridoxal (PAL) and pyridoxamine (PAM) also occur in nature as their coenzymes, namely, pyridoxal phosphate (PALP) and pyridoxamine phosphate (PAMP), respectively.

FUNCTIONS

Pyridoxine plays major role in protein metabolism.

2. It is essential for the growth of infant.
3. It aids in the formation of synthesis of haemoglobin.
4. Pyridoxine is essential for deamination and decarboxylation.
5. Vitamin B_6 is involved in the conversion of tryptophan into niacin.
6. It aids in the conversion of linoleic acid to arachidonic acid.
7. It aids in the release of glycogen from liver and muscle.

DEFICIENCY

- Vitamin B_6 deficiency or apyridoxosis in rats leads to the development of acrodynia, a disease of dermatitis on ears, mouth and tail and accompanied by edema and scaliness of these structures.
- In human infants, vitamin B_6 deficiency results in convulsions, anaemia, dermatitis and gastrointestinal disorders such as nausea and vomiting. However, this deficiency is rare. Moreover, tryptophan metabolism is also disturbed.
- In adults, the vitamin B_6 deficiency is normally not found because the intestinal bacteria are capable of synthesizing vitamin B_6.
- In B_6-deficient anaemia, the RBCs are microcytic and hyperchromic. There are increased serum iron concentrations, saturation of iron-binding protein, hemosiderin deposits in bone marrow and liver, and failure of iron utilization for hemoglobin synthesis.
- Diseases with malabsorption, such as celiac syndrome, may contribute to vitamin B6 deficiency

SOURCES

- **Rich sources:** Dried yeast, rice polishings, wheat germ and liver are rich sources.
- **Good sources***:* Whole cereals, legumes (pulses), oilseeds and nuts, egg, milk, meat and fish and green leafy vegetables are good sources. Pyridoxine is adequately available in human and cow's milk.

REQUIREMENTS

Requirements/RDA of Vitamin B_6

Age Group	Requirements (mg)
Infant	0.3-0.4
Preschool children	0.6
Adolescents	2.0
Adult	2.0
Pregnancy	2.5
Lactation	2.5

VITAMIN B_7 (BIOTIN)

In 1927, Boas made an important observation that when raw egg white was incorporated as then main source of protein in the diets of rats, they developed deficiency symptoms characterized by dermatitis, loss of hair and muscular in coordination.

In 1935, Fritz Kögl, a Dutch biochemist, isolated in crystalline form from 250 kg of dried egg yolks about 1 mg of a *'bios*' factor (growth promoting factor) necessary for yeast and named it as "biotin". Four years later, Szent-Györgyi *et al* conclusively proved that biotin is synonymous to the "antiegg white injury factor" which is responsible for the cure of egg white injury, induced in rats and other animals by feeding them with raw egg white.

Anti-Egg white factor/ injury

The raw egg white contains a biotin-antagonist protein, *avidin*, which combines with biotin in a firm linkage to form a compound that cannot be absorbed by the intestine and is therefore, excreted. It is also called as coenzyme R because it is a growth factor for the nitrogen-fixing bacterium, Rhizobium.

FUNCTIONS

- ❖ Biotin is essential for normal gestation and lactation in experimental animals.
- ❖ It helps to maintain the skin and the nervous system in sound condition.

- ❖ It is essential for the synthesis of malonyl COA from acetyl COA and oxaloacetic acid from pyruvic acid.

DEFICIENCY

- ❖ In most animals including man, intestinal bacteria synthesize appreciable amounts of biotin. It is because of this reason that biotin-deficiency in human beings, fed on biotinfree diets, cannot be produced.
- ❖ However, biotin-deficiency may be induced by sterilization of intestine and by feeding with raw egg white.
- ❖ Avidin, the egg white protein, inactivates biotin by eliminating it from an otherwise complete diet. Such a deficiency in man leads to dermatitis, loss of hair, decrease in weight and edema.
- ❖ The lesions on skin appear with changes in posture and gait. These disorders may lead to death.
- ❖ Heating egg white destroys the avidin and prevents the so-called egg white injury.

SOURCES

Biotin has a wide range of distribution both in the animal and the vegetable kingdoms. Yeast, liver, kidney, milk and molasses are among the richest sources; peanuts and eggs have lesser amounts. Biotin occurs in nature usually in combined state as biocytin. It is a bound form of biotin, linked as a peptide with the amino acid lysine.

Requirements

The intestinal bacteria synthesize biotin in such appreciable amounts that the amount excreted in urine exceeds the intake. That is why the RDA for this vitamin has not been established. However, about 10 mg per day of biotin is sufficient for an adult.

VITAMIN B_9 (FOLIC ACID)

The potent factor was obtained from spinach leaf and this led to its nomenclature as folic acid, FA (*foliumL* = leaf). The official name of this vitamin is folacin. This is also known as **liver Lactobacillus casei factor** as it was isolated from liver and was shown as necessary for the growth of lactic acid bacteria.

Functions

- ➢ It helps in the formation of haem group of haemoglobin.

- It is essential for the maturation of red blood cells.
- It is act as key compounds for the formation of Purines which are essential constituents of living cells.
- It is essential for the formation Thymine which is essential compound of DNA.
- It is necessary for the conversion of phenylalanine into tyrosine.

DEFICIENCY

Megaloblastic anaemia

- Megaloblastic anaemia in infants and children.
- In man, the folic acid deficiency leads to megaloblastic anemia, glossitis and gastrointestinal disorders. Pregnant women and infants are also particularly vulnerable. Folic acid deficiency is a major feature of tropical sprue, in which there is a general deficiency in absorption of many nutrients from the small intestine.

SOURCES

Important sources are liver, kidney, tuna fish, salmon, yeast, wheat, dates and spinach. Root vegetables, sweet potatoes, rice, corn, tomatoes, bananas, pork and lamb contain little folid acid. With improper cooking, folacin contents are destroyed, like thiamine.

REQUIREMENTS

Requirements/RDA of Folic acid

Age Group	Requirements (mg)
Infant	25
Preschool children	80
Adolescents	200
Adult	200
Pregnancy	500
Lactation	300

VITAMIN B_{12} (CYANOCOBALAMNE)

In 1926, two American physicians, George Minot and George William Murphy discovered that patients suffering from pernicious anemia could be cured by feeding them with about half a pound of liver a day. This **anti-pernicious anemia factor** (APA factor) was, later, isolated in crystalline form in 1948 independently by E. Lester Smith in England and by Edward Rickes and Karl Folkers in the United States. It was then named as **vitamin B12 or**

cyanocobalamin. It is the last B-vitamin to be isolated and is also known as *Factor X* or *L.L.D. factor*. The coenzyme form of this vitamin (deoxyadenosyl cobalamin or cobamide coenzyme) was first isolated by Barker of California. It contains cobalt. It is sensitive to sun light.

FUNCTIONS

- Promotes the maturation of red blood cells.
- It acts on the narrow elements and is involved in the formation of white blood cells and blood platelets.
- It cures the neurological symptoms of pernicious anaemia.
- It acts as a coenzyme in the synthesis of methionine.
- DNA synthesis

DEFICIENCY

Pernicious anaemia:

- Immature red blood cells with normal amounts of haemoglobin, soreness and inflammation of tongue, numbness and tingling in fingers and toes.
- The rare disease juvenile (or congenital) pernicious anemia springs up due to an inability to secrete gastric intrinsic factors.
- It is prominent at 9 months to 10 years of age. As the anemia becomes severe, irritability, anorexia and listlessness occur.
- This typical deficiency disease, adult pernicious anemia (= anemia caused by failure of erythrocyte formation), is characterized by R.B.Cs. becoming abnormally large and fewer in number (1–3 million per cubic millimeter instead of the normal 4–5 million).
- The patient weakens, loses its weight and the nervous system is also gradually affected because there occurs demyelinization of the large nerve fibres of the spinal cord.
- All these changes ultimately lead to death.

SOURCES

- *Vitamin B12 has been found only in animals* ; the chief source is liver, although it is also present in milk, meat, eggs, fish, oysters and clams. Under certain dietary conditions, vitamin B12 may be synthesized by the intestinal microorganisms.
- In general, cyanocobalamin is not present in plant foods except in Spirulina, a blue-green algae.

REQUIREMENTS

Requirements/RDA of Vitamin B_{12}

Age Group	Requirements (mg)
Infant	0.2
Preschool children	0.2-1.0
Adolescents	1.0
Adult	1.0
Pregnancy	1.2
Lactation	1.5

VITAMIN C (ASCORBIC ACID)

The chemical name for Vitamin C is ascorbic acid. It was discovered in 1747 by the British physician Lind and demonstrated that citrus fruit juices prevented and cured scurvy. It is also called **cevitamin.** It has a curing action against scurvy and hence popularly called as **antiscorbutic factor.**

FUNCTIONS

1. Ascorbic Acid is essential for formation of cement substances and collagen which is found in blood vessels teeth and bones.
2. It helps in the biosynthesis of non-essential amino acids.
3. It is essential for the conversion of phenlalanine to tyrosine and for the oxidation of tyrosine.
4. It is required for normal wound healing.
5. Vitamin C is an excellent anti-oxidant and prevents oxidation of cells.
6. Vitamin C is required for carnitine synthesis which helps in the transport of fatty acids in the cell.
7. Vitamin C is essential for the synthesis of norepinephrine a neurotransmitter.
8. It activates hormones.
9. Drug detoxifying metabolic systems in the body requires Vitamin C for its optimal activity
10. Ascorbic acid reduces the ferric iron to ferrous iron and thus helps in the absorption of iron.

DEFICIENCY

Infantile scurvy: The early symptoms of the disease are : (i) loss of appetite and (ii) listlessness. The infant cries when its legs and arms are moved. Swelling is observed at the ends of long bones. Haemorrhages occur under the skin. Gums are swollen. Convulsions may occur resulting in death of the infant.

Scurvy in adults: The disease is characterized by general weakness, spongy bleeding gums, loose teeth with desorbed dentine, swollen tender joints and haemorrhages in various tissues and under the skin.

SOURCES

Amla, drumstick leaves, guava, cashew fruit, agathi, cabbage, bitter gourd, oranges, and tomatoes are the rich source of vitamin C.

REQUIREMENTS

Requirements/RDA of Vitamin C

Age Group	Requirements (mg)
Infant	25
Preschool children	40
Adolescents & Adult	40
Pregnancy	60
Lactation	80

RECOMMENDED DIETARY ALLOWANCES FOR INDIANS

Group	Particulars	Body wt.	Net energy	Protein	Visible Fat	Calcium	Iron	Vit. A µg/d		Thiamin	Riboflavin	Nicotinic acid	Pyridoxin	Ascorbic acid	Free folic acid	Vit. B_{12}
								Retinol	β-carotene							
		kg	Kcal/d	g/d	g/day	mg/d	mg/d			mg/d	mg/d	mg/d	mg/d	mg/d	µg/d	µg/d
Man	Sedentary work		2425							1.2	1.4	16				
	Moderate work	60	2875	60	20	400	28	600	2400	1.4	1.6	18	2.0	40	100	1
	Heavy work		3800							1.6	1.9	21				
Woman	Sedentary work		1875							0.9	1.1	12				
	Moderate work	50	2225	50	20	400	30	600	2400	1.1	1.3	14	2.0	40	100	1
	Heavy work		2925							1.2	1.5	16				
	Pregnant woman	50	+300	+15	30	1000	38	600	2400	+0.2	+0.2	+2	2.5	40	400	1
	Lactation															
	0-6 months	50	+550	+25	45	1000	30	950	3800	+0.3	+0.3	+4	2.5	80	150	1.5
	6-12 months		+400	+18						+0.2	+0.2	+3				
Infants	0-6 months	5.4	108/kg	2.05/kg		500		350	1200	55µg/kg	65µg/kg	710µg/kg	0.1	25	25	0.2
	6-12 months	8.6	98/kg	1.65/kg						50µg/kg	60µg/kg	650µg/kg	0.4			
Children	1-3 years	12.2	1240	22			12	400	1600	0.6	0.7	8	0.9		30	
	4-6 years	19.0	1690	30	25	400	18	400		0.9	1.0	11		40	40	0.2-1.0
	7-9 years	26.9	1950	41			26	600	2400	1.0	1.2	13	1.6		60	
Boys	10-12 years	35.4	2190	54	22	600	34	600	2400	1.1	1.3	15	1.6	40	70	0.2-1.0
Girls	10-12 years	31.5	1970	47			19			1.0	1.2	13				
Boys	13-15 years	47.8	2450	70	22	500	41	600	2400	1.2	1.5	16	2.0	40	100	0.2-1.0
Girls	13-15 years	46.7	2060	65			28			1.0	1.2	14				
Boys	16-18 years	57.1	2640	78	22	500	50	600	2400	1.3	1.6	17	2.0	40	100	0.2-1.0
Girls	16-18 years	49.9	2060	63			30			1.0	1.2	14				

LECTURE 10
MINERALS

Minerals are inorganic substances and the body contains about 24 minerals, all of which must be provided by the diet. These include calcium, phosphorus, potassium, sodium, chlorine, magnesium, iron, manganese, copper, iodine, cobalt, zinc, aluminium, arsenic, bromine, fluorine, nickel, chromium, cadmium, selenium, silicon, vanadium, and molybdenum. These minerals are necessary for the following different functions:

1. As constituents of bones and teeth, e.g., calcium, phosphorus and magnesium,
2. As constituents of body cells of soft tissues such as muscles, liver, etc., e.g., phosphorus.
3. As soluble salts which give to the body fluids and cell contents, their composition and stability which are both essential for life, *e.g.*,
    a) Iron and copper – formation of haemoglobin,
    b) Iodine – formation of thyroxine,
    c) Zinc – constitution of an enzyme, e.g., carbonic anhydrase and a hormone e.g., insulin,
    d) Cobalt – constituents of a vitamin, e.g., vitamin B_{12}

CALCIUM

Calcium is an essential element required for several life processes. The function and requirement of Calcium and Phosphorous are closely linked. Over 99% of the Calcium and Phosphorous is present in the bones and the remaining 1% in the body fluids. The Calcium and Phosphorous are present in the ratio of **2:1 in our body**. Adult body contains about 1000 to 1200 g of calcium. The **diet** should have calcium and phosphorous in ratio of **1:2.** Vitamin D is essential for the absorption calcium.

Functions

1. It is essential for the formation of bone.
2. It is essential for the formation of teeth.
3. Calcium is essential for the clotting of blood.
4. It is essential for the contraction of the heart and skeletal muscle.
5. It regulates the excitability of the nerve fibres.

Food Sources

Milk is the best natural source and skim milk powder is very rich source (1.37 per cent) of calcium. Ragi is the cheapest natural source of calcium, containing about 0.3 to 0.36 per cent. Though sesame seeds are rich in calcium (1.45 per cent) the calcium is present as calcium oxalate mostly in the skin. Green leafy vegetables are one of the cheapest natural sources of calcium containing about 0.44 to 1.13 per cent. Small fish eaten along with bones is an excellent source of calcium.

Deficiency Diseases

Adults: Osteoporosis

Osteoporosis in adults is a condition in which decalcification of the bone occurs due to calcium deficiency in the diet. Fractures of the brittle bones occur even after minor accidents. Pain due to fractures of vertebrae may radiate round the trunk, to the buttocks or down the legs. Healing of fractures is not impaired in osteoporosis and pain subsides as soon as the fractures are healed.

Children:

1) decreased rate of growth; (2) negative calcium balance; (3) loss of calcium from bone leading to the development of osteoporosis; (4) hyperplasia (a diffuse overgrowth) of parathyroid glands; and (5) hypeirritability and tetancy leading to death

Requirements/RDA of Calcium

Age Group	Requirements (mg)
Infant	500
Preschool children	600
Adolescents	800
Adult	600
Pregnancy	1200
Lactation	1200

PHOSPHORUS

Adult body contains about 400-700 g (approximately 1% of the body weight) of phosphorus. The Calcium and Phosphorous are present in the ratio of 2:1 in our body.

Functions

1. Phosphorus is essential for the formation of bones and teeth.
2. It is essential for the formation of phospholipids. Phoshpolipids are major components of cell membrane.
3. It is essential for carbohydrate metabolism as phosphorylation of glycogen.
4. Phosphates play an important role as buffers to prevent changes in acidity of the body fluids.
5. In the DNA and RNA, phosphate is an essential part of the nucleic acids.

Food Sources

Phosphorus is widely distributed in both plants and animal foods. Diets rich in protein and calcium provide sufficient phosphorus. Eggs, milk, meat, fish and flour are excellent sources of phosphorus. Whole grain cereals are good source but fruits and vegetables as whole are low in phosphorus content.

Deficiency

A deficiency of this element is very rare in human beings because diets having sufficient cereals are seldom inadequate in this nutrient. The deficiency is common in animals and causes stiff joints and bones become fragile and break easily.

Requirements/RDA of Phosphorus

Phosphorus requirements depend on the availability of calcium present in the diets. It depends upon the calcium: Phosphorus ratio. The optimal calcium phosphorus ratio for infants and children is 1:1 and or adults 2:1.

IRON

Those elements which are required in traces are known as trace element. Iron is a trace element, as its presence in small amount is very significant. Most of the iron in the body is present as hemoglobin. Most of the body's iron is found in complex forms bound to proteins either as porphyrin or heme compounds or as ferritin. Free inorganic iron occurs in small amount. About 70% of iron is in circulating hemoglobin, 4% into myoglobin. 25% is stored in liver, spleen, kidney and 1% in plasma and various oxidative enzymes.

Functions

1. Iron essential for the formation of haemoglobin.

2. It forms a part of the myoglobin in muscles which makes oxygen available for muscle contraction.

3. Iron present in cytochrome oxidase enzyme. It is very essential of the respiratory chain.

As part of enzymes iron catalyzes many important reactions in the body. Examples are

- Conversion of beta carotene to active form of Vitamin A.
- Synthesis of collagen and neurotransmitters.
- Detoxification of drugs in the liver.
- It is essential for the transport and storage of oxygen.

Food Sources

Vegetable sources are cereals, beans, bajra, letils, vegetable leaves, spinach, dried peach and apricot, pomegranate, jaggary and mollasses. Non-vegetable sources are liver, kidney, brain, egg yolk, meat and fish etc.

Deficiency diseases

Nutritional Anaemia

In healthy individual blood contains approximately 14gm of haemoglobin per 100 ml. Sometimes the level goes down and a low level of haemoglobin in blood is called anaemia. This can be due to deficiency of one or more essential nutrients (iron, calcium, copper, vitamin C, folic acid vitamin B_{12} protein etc). Iron deficiency anaemia follows a specific sequence. The iron reserves getting depleted and transferring (transport form of iron) level of blood goes on increasing and plasma iron is reduced. The blood cells are pale, less in number and small in size. Deficiency of iron may be caused in a number of ways. The diet may not provide enough or it may be in unabsorbable form. This is usually by blood loss but occasionally iron may be excreted in urine and cause haemosidernuria.

Signs and Symptoms of Nutritional Anaemia

1. Reduced Haemoglobin level (less than 12 g /dl).
2. Paleness of skin.
3. Inside of the lower eyelid is pale pink.
4. Finger nails becoming thin and flat (spoon shaped nails).

5. Fatigue, breathlessness on exertion.

6. Sleeplessness, dimness of vision.

Requirements/RDA of Iron

Age group	Requirements (mg)
Adult Men	17
Adult Women	21
Pregnant women	35
Lactating women	21
Adolescents: Boys	28
Girls	26

IODINE

Iodine is an essential constituent of the thyroid hormone produced by the thyroid glands. It occurs as free iodide ions or as protein bound iodine in our body. About 15 – 23 mg of iodine is present in the adult human body.

Functions

1) Iodine plays an important role as it is incorporated into thyroxine hormone. It is responsible for the rate of oxidation with in the cells and in doing so determines the rate of metabolism.
2) Thyroxine is related with the tissues of nerves and muscles.
3) It helps in increasing heart rate.
4) Iodine helps in converting glycogen to glucose. In this way it increase blood sugar level. Liver is affected because there is a possibility of emptying of glucose stores.

Sources

Sea foods like fish, shellfish and fish liver oils are the richest sources. The iodine content of vegetables, fruits and cereals depends upon the iodine of the soil. Iodized salt supplies sufficient iodine intake as it is used by all.

DEFICIENCY DISEASES

Goitre

Deficiency of iodine results in a disease known as goiter. Goiter is an enlargement of the thyroid gland. This is when due to the deficiency of iodine; the secretion of thyroxine hormone is

also reduced, so the gland cells try to produce iodine, in large quantity which leads to multiplication of cells which results in the enlargement of the gland that forms goitre.

Cretinism

The retardation in physical and mental growth of children due to severe iodine deficiency is known as cretinism. The symptoms of cretinism are:

1) Low basal metabolic rate.
2) Muscular flabbiness and weakness
3) Dry skin
4) Rough hair
5) Enlarge tongue
6) Thick lips retardation of skeletal development
7) Mental retardation

Myxedema

Iodine deficiency in adults results in Myxedema. Face becomes thick and looks puffy. Facial expression is poor and person becomes inactive.

Hypothyroidism

This condition may be due to a tumor of thyroid gland which produces and excess of thyroid hormone or it may be due to Grave's disease (Toxic Goiter) an increased production of the hormone due to over active thyroid gland. Hyperthyroidics shows hyper activity, tremors, a rapid heart rate, high blood pressure, bulging eyes, accelerated growth with loss of weight and emotional instability. If the symptoms do not subside with medicines surgery is required.

Requirements/RDA of Iodine

Age group	Requirements (µg)
Adult	100-200
Pregnant women	+25
Lactating women	+50
Infant	40-50

Goitrogens

Goitrogens are substances present in foods which cause goitre. These substances react with iodine present in the food and making it unavailable for absorption. Foods like cabbage, cauliflower, raddish contain goitrogens.

MAGNESIUM

The human body contains minute quantities of magnesium in the bones along with calcium and phosphorus and the rest in soft tissue. An adult human body has about 25g, of magnesium. Normal human serum contains about 2 to 3 mg/100ml.

Functions

1. It acts as activator of several enzymes, e.g., alkaline phosphatase.
2. It is required as a co-factor for oxidative phosphorilation.
3. Magnesium is present in certain enzyme, e.g., co-carboxylase

Effect of deficiency

- Vomiting and intestinal fistula.
- Diabetic acidosis is associated with raised serum level of magnesium.
- Neuromuscular irritability, tremors, cramps, twitching and convulsions
- Confusion, disorientation and visual hallucinations

Source

Cereals, pulses, almond, cashew, peanut, mik, pork etc.

Requirements/RDA of Magnesium

Age Group	Requirements (mg)
Infant	30-45
Preschool children	50
Adolescents	235
Adult	310-340
Pregnancy	310
Lactation	310

POTASSIUM

Approximately 97% of the potassium in the body is available with in the cells and remaining is distributed in the extracelluar fluid. Plasma contains very small amount of potassium but large amount of it is present in red blood cells.

Functions

1. It plays important role in the regulation of the acid base balance in the cells.
2. It is essential for the muscular contraction.

3. It is essential for the growth and builds up of tissues as it is required for the synthesis of protein.

Deficiency

Less intake of food during starvation and surgery is responsible for the deficiency of potassium. Excessive loss might occur during vomiting and diarrhea for long duration. In severe burns and prolonged fever potassium deficiency occurs. Patient might complains of nausea, vomiting, muscle weakness, low blood pressure and tachycardia.

Source

Meat, fish and poultry foodstuffs are good sources of potassium. Fruits, vegetables and whole grain cereals are especially high in potassium. Banana, potatoes, tomato, carrots, grape fruit juice contains potassium.

SODIUM

Sodium chloride or common salt is the taste-giver in our diet. Extracellular fluid contains sodium for the maintenance of normal osmotic pressure and water balance. Approximately 50% of the body's sodium is present in the extracellular fluid, 40% in bones and 10% or less in intracellular fluid.

Functions

1. It regulates acid-base balance of the body.
2. It regulates the osmotic pressure of plasma.
3. Na ions play an important part in the absorption of monosaccharide and amino acids from small intestine.
4. Na ions play an important part in maintaining the heart beat.

Effect of Deficiency

Sodium deficiency occurs when the intake is poor or excessive amount is lost. Sodium loss through excessive sweating causes sodium deficiency unless salt intake is not increased. Vomiting, severe diarrhea ,.addison's disease are also responsible for sodium deficiency there is also giddiness, cramps in muscles, collapsed, veins, low blood pressure, dryness of mouth, inelastic skin and oliguria.

Sources

Fresh meat, fish, poultry, egg and milk are the sources of sodium. Foods from plant origin are poor in sodium. Salt used during cooking is the commonest source of sodium.

Requirements/RDA of Sodium

Age group	Tropical (g)	Temperate (g)
Adult : light work	10	5
Moderate work	15-20	8
Heavy work	25-30	15
Children	5	3-5
Adolescents	10-25	6-8

ZINC

In an adult 2 to 3 gms of zinc is present. It is present in certain parts of the eye, certain glands and in their secretion. Liver, muscles, bone and hair contain some of the remaining zinc. Blood, especially the RBC, has some zinc. In small amounts it is present in tissues and bone cells.

Functions

1. Zinc is essential for the brain development of children
2. It is essential for the spermatogenesis and development of primary and secondary sex organs
3. It is essential for quick wound healing.
4. It is essential for normal taste sensation.
5. Zinc is a constituent of enzymes such as carbonic anhydrase, alkaline phosphatase, and lactic dehydrogenase.
6. It is a constituent of the hormone insulin.
7. It plays a major role in the synthesis of DNA and proteins.

Sources

Sea foods, meat, poultry and eggs are good sources of zinc. Cereals, legumes and nuts contain considerable amounts. Fruits and vegetables are poor sources. On milling eighty per cent of zinc is lost.

Deficiency Diseases

Acrodermatitis enteropathica (severe dermatitis in infants), emotional disorder, impaired taste (hypogeusia) and smell, delayed genital development, immunological defects, impaired glucose tolerance.

Requirements/RDA of Zinc

Infants require 3 to 5 mg. for children 10 to 15 mg and during pregnancy and lactation 20 to 25 mg of zinc are required.

FLUORINE

Fluorine occurs in the form of fluoride in nature. It is present in soils, water, plants and animals.

Functions

1. Fluorine in the teeth helps to protect them from decay.
2. Fluoride gives stability to bone and enamel tissues.

Toxic effects of excess of fluorine

1. Skeletal Fluorosis

Intake of fluorine higher than the 10 mg/litre of drinking water leads to skeletal fluorosis (genuvalgum- knock knees). In skeletal fluorosis there is hypercalcification of the bone of the spine, pelvis, and limbs.

2. Dental Fluorosis (Mottled Enamel)

Intake of fluorine higher than the 3-5 mg/litre of drinking water leads to dental fluorosis. Chalky white patches, brown staining are the symptoms of dental fluorosis.

Source

Soil and water are the major source of fluorine.

Requirements/RDA of Fluorine

Age group	Requirements (mg)
Adult	1.5 – 4.0
Adolescents	1.5-2.5
Infant	0.1-1.0

CHLORINE

Chlorine in the body exists in the form of chloride, an essential electrolyte mineral. It is best known as the other half of sodium-chloride or table salt. It makes up about 0.15% of body weight (around 115 g for an average adult) and is found mainly in the fluid surrounding cells, alongside sodium. A small percentage of about 15% of chloride in the body is located inside cells, with the highest amounts in red blood cells. Chloride is also present in very small amounts

in bones. Most of our chloride intake is from table salt. It is easily absorbed by the intestines. It is eliminated through the kidneys in a finely tuned mechanism that regulates acid-base balance by either removing or retaining chloride.

Functions

This electrolyte mineral works with the other electrolytes potassium and sodium to maintain the proper balance of body fluids, as well as their pH balance. Chloride is also an essential component of digestive juices, as it is needed with hydrogen to form stomach hydrochloric acid.

- keeps the amount of fluid within and around cells in balance
- helps regulate the pH (acid-alkali / acid-base) balance of body fluids
- maintains proper blood volume and pressure
- critical constituent of hydrochloric acid, a key component of gastric juice secreted by the stomach that is vital for maintaining the normal acidic environment needed by pepsin, and aids digestion and absorption of many nutrients including iron and vitamin B12
- may help conserve potassium

Deficiency

Deficiency of chloride, or when blood levels of it drop too low, is known as hypochloremia. It is rare, as chloride is part of table salt which is present in most foods. In fact, people are prone to consuming more chloride than is really needed, due to salt-laden diets. Hypochloremia can occur however, for a variety of reasons that include:

- heavy sweating, as large amounts of sodium and chloride can be lost in perspiration
- excessive fluid loss due to prolonged diarrhea or vomiting, or overuse of coffee or laxatives or diuretics
- over-hydration
- burns
- congestive heart failure
- certain kidney disorders
- Addison's disease
- most often seen in infants on chloride-deficient formulae

Chloride deficiency symptoms

- loss of appetite
- muscle weakness
- lethargy
- dehydration
- deficiency leads to alkalosis, a condition in which body fluids have excess base (alkali), that can result in dangerously high blood pH and excessive loss of potassium in urine (which in turn causes hypokalemic metabolic alkalosis with symptoms that include loss of control of muscle function which might lead to breathing and swallowing difficulties).

Sources

Chloride is found in almost all foods in the salt that is added, often in large amounts, during processing or cooking. Foods high in chloride are table salt or sea salt (sodium chloride), salt substitutes such as potassium chloride, seaweed (such as dulse and kelp), olives, rye, vegetables like celery, lettuce, tomatoes, preserved meats such as bacon, ham, sausages, processed or canned or fast foods that are high in salt. Other chloride food sources include cheese, vegetables, yeast extract as potassium chloride found in most foods.

Requirements/RDA of chlorine

Age group	Requirements (g)
Adult	2.3
Adolescents	2.3
Infant	0.2-10.6
Children	1.5-2.0

LECTURE 11
WATER

Water is vital for human existence. We can live without food for extended periods of time, but without water will result in death. Water is colourless, calorie less compound of hydrogen and oxygen that virtually every cell in the body needs to survive. Water is closer being a universal solvent than any other compound.

Water is the largest constituent of the body, about 60-70 per cent of the total body weight consisting of water. The water content of soft tissues ranges from 70-80 per cent while that of bone about 20 per cent. Body water is distributed as follows:

- Inside the cells of tissue – intracellular water (50 per cent)
- Outside the tissue cells – extra cellular water (20 per cent).

The extracellular water is further subdivided into

- Water in blood plasma (about 4 per cent)
- Interstitial water – water in tissue space (9 per cent)
- Lymph in the lymphatic vessels (7 per cent).

Additional minor divisions of extracellular water are cerebrospinal fluid and aqueous humour in the anterior chamber of the eye).

Functions of water

1. It is an essential constituent of all the cells of the body.
2. It helps in transporting nutrients into the cells and removes waste products.
3. Water is a medium for most biochemical reactions.
4. It is a valuable solvent for electrolytes, non-electrolytes, hormones, enzymes and vitamins.
5. Plays a vital role in the maintenance of body temperature.
6. It is constituents of several enzymes and hormones. Saliva is about 99.5 percent water.
7. Water helps in maintaining the form and texture of the tissues.
8. Water is essential for the maintenance of acid base and electrolyte balance.
9. Water forms good source of minerals like Calcium, Magnesium, Fluoride, Iron and Iodine.

Water intake

Water intake varies widely depending on the climate; As drinking water, about 1500-5000 ml and in food 1000-2000 ml. Water is taken in food and also as drinking water. In addition, water is formed in the tissues by the oxidation of hydrogen present in fats, carbohydrates and proteins.

Water loss

Water is lost continuously from the body in the following ways:

(i) via kidney as urine (1000-1500 ml)

(ii) via the skin in the form of insensible perspiration and as sweet (800-5200 ml)

(iii) via the lungs in the expired air (400 ml)

(iv) via the large intestines in the faeces (100-200 ml)

(v) Lactating women in the milk.

In a normal individual, the water intake is approximately equal to water lost from the body and the water content of the body is maintained fairly constant.

Requirements

Requirements of water vary with climate, activities and surface area of the body. Normal intake of water ranges between 8-10 glasses (2500-3000 ml) per day.

Dehydration

Extra cellular and intra cellular fluid losses through vomiting, diarrhoea, excessive sweating is called dehydration.

Oral Rehydration Therapy

It is the administration of fluid orally to prevent or correct dehydration.

Oral Rehydration Salt (WHO and UNICEF formula)

To be dissolved in one litre of potable drinking water.

Sodium chloride (NaCl) -3.5 g,

Sodium bicarbonate ($NaHCO_3$) - 2.5 g,

Potassium chloride (KCl) -1.5 g and

Glucose - 20 g

DIETARY FIBRE

Dietary fibre is defined as that portion of plant material ingested in the diet that is resistant to digestion by gastro intestinal secretions. It consists of hemicellulose, cellulose, lignins, oligosaccharides, pectins, gums and waxes.

TYPES OF FIBRE

Soluble Fibre

Soluble fibres are indigestible but soluble in water, e.g., Gums, pectins, mucilages.

Insoluble Fibre

Insoluble fibres are indigestible and insoluble in water e.g., Cellulose, hemicelluloses, lignin.

Functions of Fibre

1. It control the blood glucose level by slows down the absorption of glucose from food.
2. Soluble dietary fibre reduces the serum cholesterol level.
3. Fibre intake reduces the blood pressure and reduces the risk of heart diseases.
4. Insoluble fibre helps to prevent constipation.
5. Insoluble fibre helps to prevent colon cancer and haemorrhoids.
6. It controls the appetite by giving feeling of fullness to stomach.
7. It helps in elimination of wastes.
8. Dietary fibre intake reduces the risk of atherosclerosis.

Sources

Types and Source of Fibre

Types of Fibre	Major Food Sources
Soluble fibres	
Gums, pectins, mucilages	Citrus fruits, apple, oats, barley, legumes
Insoluble fibres	Whole wheat products, wheat bran, whole grain breads, cereals and vegetables like green peas, beans, cabbage.
Cellulose, hemicelluloses,	
Lignin	Skin of vegetable and fruits

Fibre Content of Foods

Cereals

- Whole cereal flours contain about 8 to 15% of total dietary fibre.

Pulses

- Whole pulses (with cuticle) contain about 32 to 40% of dietary fibre.
- Dhals (decuticled split pulses) contain from 8 to 10%.

Nuts

- Groundnut (without skin) contains about 9.3% of dietary fibre.

Vegetables

- The dietary fibre content of vegetables on fresh basis may vary from 1.3 to 7-8% and on dry basis from 9.6 to 38.4%.

Fruits

- The dietary fibre content of fruits range from 1.4 to 3.6% on the fresh basis and 9.6 to 14.4% on dry basis.

RDA for Dietary Fibre

Recommended dietary allowances for fibre is 30 g/day or 12g/1000 Kcal for normal healthy adult. American diabetic association recommends 25-38 g of dietary fibre per day per persons suffering from diabetes.

LECTURE 12
PRESERVATION OF FOOD BY TEMPERATURE

Preservation

Preservation are the measures employed to prevent (or) minimize all such undesirable changes takes place due to spoilage which caused by both internal and external organisms.

Need for preservation of foods

Freshly harvested or prepared products are highly attractive in appearance and possess good taste and aroma, but deteriorate rapidly if kept for sometimes.

This is due to

- Fermentation caused by moulds, yeasts and bacteria
- Enzymes present in the products may affect the colour and flavour adversely
- Chemicals present in the processed food react with one another and spoil its taste and aroma
- Air coming in contact with the product, may react with the glucosidal materials present in it and render the product bitter
- Traces of metal from the equipment may get into the product and spoil its taste and aroma.

Importance / advantages of food preservation

- Prevention of wastage of fresh produce.
- To extent the shelf life of the produce.
- To develop value added products.
- Self employment opportunities.
- Provides convenient and preferable forms to the consumer (jam, jellies, RTS, squash,etc.).
- Foreign exchange earnings from the export of processed foods and it will improve our national economy.
- Available in the form of ready to serve and hence (fuel consumption) energy is saved.

I. Preservation of low temperature

Microbial growth and enzyme reactions are retarded in foods stored at low temperature. The lower the temperature, the greater the retardation. Low temperature can be produced by

a. Cellar storage (about 15°C)

The temperature in cellar (underground rooms) where surplus food is stored in many villages is usually not much below that of the outside air and is seldom lower than 15°C. It is not enough to prevent the action of many spoilage organisms or of plant enzymes. Root crops, potatoes, cabbage, apples, onions and similar foods can be stored for limited periods during the winter months.

b. Refrigerated (or) chilling (0 to 5°C)

Chilling temperature are obtained and maintained by means of ice or mechanical refrigeration. It may be used as the main preservative method for foods or for temporary preservation until some other preservative process is applied. Most perishable foods, including eggs, dairy products, meats, sea foods, vegetables and fruits, may be held in chilling storage for a limited time with little change from their original condition. Enzymatic and microbial changes in the foods are not prevented but are slowed considerably.

c. Freezing (18 to 40°C)

At temperature below the freezing point of water, growth of microorganisms and enzyme activity are reduced to minimum. Most perishable foods can be preserved for several months. Fruits, vegetables, juices and fleshy foods (meat poultry fish and sea foods) can be preserved in this method.

II. Preservation by high temperature

Application of heat to the foods leads to the destruction of microorganisms. The specific treatment varies with:

a. The organisms that has to be killed.
b. The nature of the food to be preserved and
c. Other means of preservation that may be used in addition to high temperature.

High temperatures used for preservation are usually:

(1) Pasteurization temperature – below 100°C

(2) Heating at about 100°C and

(3) Sterilization temperature above 100°C.

a. Pasteurization–below 100°C

Pasteurization is a heat treatment that kills part but not all the microorganisms present and the temperature applied is below 100°C. The heating may be by means of steam, hot H_2O, dry heat or electric currents and the products are cooled promptly after the heat treatments.

Methods of pasteurization

- HTST method - High temperature and short time (above 70°C)
- LTH method - Low temperature and higher time (or) Holding method (60-70°C)

b. Heating at about 100°C

A temperature of approximately 100°C is obtained by boiling a liquid food, by immersion of the container of food in boiling water or by exposure to flowing steam. Acid foods, e.g., sauerkraut may be preheated to a temperature somewhat below 100°C, packaged hot and not further heat processed. Blanching fresh vegetables before freezing or drying involves heating briefly at about 100°C.

c. Sterilization-above 100°C

By this method all microorganisms are completely destroyed due to high temperature. The time and temperature, necessary for sterilization vary with the type of food. Temperatures above 100°C can only be obtained by using steam pressure sterilizers such as pressure cookers and autoclaves.

Vegetables like green peas, okra, beans, etc. being non acidic and containing more starch than sugar, require higher temperature to kill the spore forming organisms. Continuous heating for 30-90 min. at 116°C is essential for their sterilization. Before using, empty cans and bottles should also be sterilized for about 30 min. by placing them in boiling water.

Difference between pasteurization and sterilization

Pasteurization	**Sterilization**
1. Partial destruction of microorganism 2. Temperature below 100°C 3. Normally used for fruits	1. Complete destruction of microorganism 2. Temperature 100°C and above 3. Normally used for vegetables

Food Irradiation

Food irradiation (the application of ionizing radiation to food) is a technology that improves the safety and extends the shelf life of foods by reducing or eliminating microorganisms and insects. Irradiation is sometimes referred to as "**cold pasteurization**" since the result achieved is similar to heat-based pasteurization but without the heat.

Purpose of Irradiation

Prevention of Foodborne Illness – irradiation can be used to effectively eliminate organisms that cause foodborne illness, such as *Salmonella* and *Escherichia coli (E. coli).*

Preservation – irradiation can be used to destroy or inactivate organisms that cause spoilage and decomposition and extend the shelf life of foods.

Control of Insects – irradiation can be used to destroy insects in or on tropical fruits

Delay of Sprouting and Ripening – irradiation can be used to inhibit sprouting (e.g., potatoes) and delay ripening of fruit to increase longevity.

Sterilization – irradiation can be used to sterilize foods, which can then be stored for years without refrigeration.

Sources of Irradiation

There are three sources of radiation approved for use on foods.

Gamma rays are emitted from radioactive forms of the element cobalt ($Cobalt^{60}$) or of the element Cesium ($Cesium^{137}$).

X-rays are produced by reflecting a high-energy stream of electrons off a target substance (usually one of the heavy metals) into food.

Electron beam (or e-beam) is similar to X-rays and is a stream of high-energy electrons propelled from an electron accelerator into food.

Doses

- Radiation dose of upto 1 Mrad is not hazardous.
- Low doses (<1 kGy): control of insects and parasites, delay of ripening
- Medium doses (1 - 7 kGy): reduction of foodborne pathogens, extension of shelf life
- High dose (> 25 kGy): sterilization

Mode of action

Energy waves passing through the food break molecular bonds in the DNA of bacteria, other pathogens, and insects. These organisms die or, unable to reproduce, their numbers are held

down. Food is left virtually unchanged, but the number of harmful bacteria, parasites and fungi is reduced and may be eliminated.

Labeling of Irradiated Foods

Any irradiated food, or food containing an irradiated ingredient must carry the word “Irradiated” in a prominent position either as part of the main label or next to the ingredient that has been irradiated. It may also (optional) show the international icon for irradiated food called the “Radura” symbol along with the statement “Treated with radiation” or “Treated by irradiation” on the food label.



Foods suitable for Irradiation

The main types of food that have been classified as safe to irradiate are meat, seafood, fruit, vegetables, herbs and spices.

Effects of irradiation on the nutritional content and safety of food

The Food and Drug Administration has evaluated the safety of this technology and has found irradiation to be safe under a variety of conditions and has approved its use for many foods. Irradiation does not significantly reduce nutritional quality or significantly change food taste, texture or appearance. Irradiated foods do not become radioactive. Irradiation can produce changes in food, similar to changes caused by cooking, but in smaller amounts.

LECTURE 13
PROCESSING OF PUFFED, FLAKED AND EXTRUDED PRODUCTS

The snack food is one of the most important areas of the food industry. Designing snack foods today can be a complex process to meet changing consumers taste and expectations and elusive search for something unique that also appeals to a wide variety of people. As a simplest, inexpensive and quickest traditional method of dry heat application for preparation of weaning food formulations and ready-to-eat snacks products, popping and puffing have been practiced since hundreds of years.

Popping

Popping is a process in which kernels are heated until internal moisture expands and pops out through the outer shell of the kernel. Popping is defined as direct heating of the moistened grain (18 to 20 %) at a temperature of 190° C to 250°C in a closed container until moisture inside the grain vaporizes and expands the kernel causing it to "pop" or explode. During the popping of popcorn the pericarp acts as a pressure vessel and popping occurs at about 177°C, which is equivalent to a pressure of 135psi inside the kernel. During popping the material practically gets sterilized and most of the seed microflora is destroyed. Popping also improves the digestibility of starch as it involves gelatinization of starch and degradation of dietary fibres.

A variety of popped cereals like popcorn, popped rice, popped sorghum, popped ragi and popped pulses were marketed and consumed as snacks by all segments of population. While popcorn is popular among urban elite, other popped cereals are more popular among middle and lower income segments of the population in regions where the specific cereal is predominantly grown. Popcorn is an important snack food made from corn and is growing popular day by day. The microwave popcorns are becoming popular with the increasing availability of microwave ovens at home scale. The popcorn coated with different ingredients such as butter, hydrogenated oil, sugar syrup, salt, savors, etc., are sold in market in small packs.

Flow chart for preparation of popping of millets

Millets (Samai/Varagu)

↓

Soaking in water (1:1.5, 1-2hrs) (Moisture 19%)

↓

Draining

↓

Tempering (2-4hrs)

↓

Popping in roasted fine sand (BS 60 mesh sieve) (270°C)

↓

Popping

Puffing

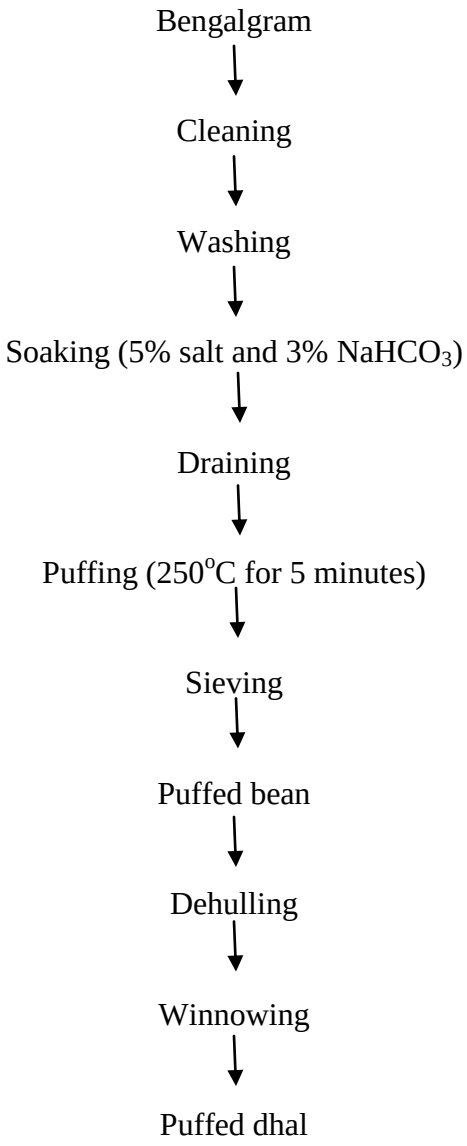
Puffing is a popular Ready - to - eat snack product is obtained by puffing cereals and pulses. Puffing is a process where, sudden release of water vapour and expansion of pre-gelatinized kernel takes place. Superheated vapour is produced inside the grains by instantaneous heating, which cooks the grain and expands the endosperm while escaping with great force through the micropores of the grain structure. Most of the water in the kernel is superheated at the moment of popping and provides driving force for expanding the kernel once pericarp ruptures. It is defined as heating of the dry grain at a temperature of 250°c in hot sand. Puffed rice and puffed bengal gram are more common in India.

Puffed Rice

Puffed rice is very popular in many countries as a cereal breakfast component or as a light food. It is a whole grain puffed product from parboiled milled rice. It is prepared from hydrothermally treated or pregelatinized milled rice by heating in high temperature air, oil and sand or by gun puffing method. Puffed rice is ready for consumption and easily digestible. It is commonly used in snacks, cereals drinks, Ready- to- Eat (RTE) breakfast cereals and infant foods. During puffing, rice kernels increase their volume several times and a fully heat-treated crisp, porous, ready-to-eat product is created. Regardless of the puffing process two important parameters should be taken into account: the selection of an appropriate sort of rice, and the use

of a proper hydro-heat treatment of raw rice. The optimum moisture content for puffing of rice for puffing expansion in most of the studies found to be 13-14%. Salt solution is invariably added to milled rice before it is heat expanded in the industry.

Flow chart for preparation of puffed pulses (Bengal gram)



Flaking

A popular form of ready-to-eat breakfast cereal is the crisp flake. Before flaking, grains and pulses are softened by being partially cooked in steam. They are then pressed or rolled into flakes which are subsequently dried. Such partially-cooked cereals may be used as quick-cooking or ready-to-eat foods. Flaked grains are eaten crisp and need to be packaged in moisture-

proof containers. The moisture content must be below 7 per cent; otherwise the product becomes soggy and tough.

Flow chart for preparation of flaking of millets

Dehulling

↓

Soaking in water (3- 4 hrs)

↓

Steaming/ pressure cooking

(to effect complete gelatinization of starch)

↓

Drying (upto18% moisture)

↓

Flaking (using machine)

(to requisite thickness between heavy duty roller)

↓

Drying at 70° C for 1 hr

↓

Packing

Extrusion

Extrusion is a process of partial gelatinization of the flour and extruding in small orifices under high pressure and temperature. Extrusion is a process that combines several unit operations, including mixing, kneading, shearing, heating, cooling, shaping and forming. It involves compressing and working raw materials e.g. flours, starches, proteins, salt, sugar and other minor ingredients, to form a semi-solid mass under a variety of controlled conditions and, then forcing into to pass through a restricted opening such as a shaped hole or slot at a predetermined rate. Heat is applied directly by steam injection or indirectly through a heated barrel or the conversion of mechanical energy. The final process temperature in the cooking extruder can be as high as 200°C but the residence time is relatively short i.e. 10-60 seconds. Thus extrusion cooking is also called a high temperature short time (HTST) process. This

process of HTST, extrusion bring gelatinization of starch, denaturation of proteins, modification of lipids and inactivation of enzymes, microbes and many antinutritional factors.

Flow chart for the preparation of noodles

Maida: Millet flour blend

↓

100:0 90:10 80:20 70:30

↓

Mixing with salt

↓

Steaming for 10 minutes

↓

Cooling

↓

Kneading for 10 min (300 ml / 1 kg flour) water

↓

Extruding

↓

Sun drying (one day) / cabinet drying at 70 ^{o}C for 4 hrs

↓

Packing

↓

Storing

Examples of extruded foods

Types of product	Examples
Cereal-based products	Expanded snack foods RTE and puffed breakfast cereals Soup and beverage bases, instant drinks Weaning foods Pre-gelatinised and modified starches, dextrins Crisp bread and croutons Pasta products Pre-cooked composite flours
Sugar-based products	Chewing gum Liquorice Toffee, caramel, peanut brittle Fruit gums
Protein-based products	Texturised vegetable protein (TVP) Semi-moist and expanded pet foods and animal feeds and protein supplements Sausage products, frankfurters, hot dogs Surimi Caseinates Processed cheese

LECTURE 14
PRESERVATION OF FOOD BY SUGAR AND SALT

I. Preservation by using sugar

Sugar absorbs most of the available water with the result that there is very little water for the growth of microorganisms hence their multiplication is inhibited, and even those already present die out gradually. Dry sugar does not ferment. Thus sugar acts as a preservative by osmosis and not as a true poison for microorganisms. Fruit syrup, jam, jelly marmalade, preserve, candy, crystallized fruit and glazed fruit are preserved by sugar.

Jam

Jam is a product made by boiling fruit pulp with sufficient quantity of sugar to a reasonable thick consistency, firm enough to hold the fruit tissues in position. Apple, sapota, papaya, plums, mango, grapes, jack, pine apple, banana, guava and pears are used for the preparation of jam. It can be prepared from one kind of fruit or from two or more kind of fruits. In its preparation about 45% of fruit pulp should be used for every 55% of sugar. The FPO or FSSAI specification of jam is 68.5% TSS, 45% of fruit pulp and 0.5 – 0.6% acid (citric acid) per 100g of prepared product.

Preparation of jam

- Select matured and ripened fruits.
- To remove the dust and dirt.
- Remove the skins from the flush of the fruits.
- Cut the peeled fruits into small portions.
- Make it into pulp either by use of hand pulper (or) by mixie.
- Take the pulp and sugar (required amount) in a vessel and heat it over fire by stirring continuously, till the final TSS reaches 68.5^{o}bx.
- Remove from the fire and allow cooling to room temperature.
- Pack it in a sterilized bottle and cover it with airtight cover.
- Label it with correct information.

Sheet or flake test

The end point of the jam is judged by sheet (or) flake test. A small portion of jam is taken out during boiling in a spoon (or) wooden ladle and cooled slightly. It is then allowed to drop. If

the product falls off in the form of a sheet (or) flakes instead of flowing in a continuous stream (or) syrup, it means that the end point has been reached and the product is ready. Otherwise, boiling is continued till the sheet test is positive.

Problems in jam production

Crystallization

If the percentage of invert sugar is less than 30 % cane sugar may crystallize out on storage and if it is more than 50% the jam will become a honey like mass duo to formation of small crystals of glucose. Corn syrup or glucose may be added along with cane sugar to avoid crystallization.

Sticky or gummy jam

Because of higher percentage of TSS, Jams tend to become gummy or sticky. This problem can be solved by addition of pectin or citric acid or both.

Premature setting

This is due to low TSS and high pectin content in the jam and can be prevented by adding more sugar.

Surface graining and shrinkage

This is caused by evaporation of moisture during storage of jam. Storing in cool place can reduce it.

Microbial spoilage

Moulds may spoil the jam during storage. It is also advisable to add 40 ppm of SO_2 in the form of KMS or 200 ppm of benzoic acid.

Jelly

Jelly is prepared by boiling the fruit with or without water, straining, mixing the strained and clear juice extract with sugar and boiling the mixture to a stage at which it will set to a clear gel. A perfect jelly should be transparent, well set, but not too stiff and should have the original flavour of the fruit. It should be of attractive colour and should keep its shape when removed from the mould. When cut, it should retain its shape and show a clean-cut surface. It should be tender enough to quiver, but not flow.

The FPO or FSSAI specification for jelly is 65% TSS, 45% of fruit extract and 0.5 – 0.75% acid (citric acid) per 100g of prepared product. Guava, sour apple, plum, karonda, wood apple, papaya and jack fruit are rich in pectin and generally used for the preparation of jelly.

Preparation of jelly

- Select sound fully matured ripe fruit (not over ripened or under ripened).
- Wash thoroughly with water to remove any adhering dirt.
- Peel the skin and make it into pulp and add water at the rate of 1½ times and boil it for 30 min. with citric acid.
- Filter the boiled pulp through the muslin cloth.
- Check the pectin strength of the extracted pectin by using alcohol test method.
- Add required sugar to the extract and cook by stirring continuously till it reaches the TSS of 65° bx.
- Remove from the fire and cool, it to room temperature.
- Pack it in a sterilized bottle and cover it airtight.
- Label with correct information.

Alcohol test

One teaspoonful of strained extract is taken in a beaker and 3 teaspoonful of methylated spirit are poured gently down the side of the beaker, which is rotated for mixing and allowed to stand for a few minutes. Observations are

a. If extract is rich in pectin, a single transparent lump or clot will form. An equal amount of sugar is to be added to the extract for preparation of jelly.
b. If extract contains moderate amount of pectin, the clot will be less firm and fragmented. Three-fourth, amount of sugar is to be added.
c. If extract is poor in pectin, numerous small granular clots will be seen. Half the amount of sugar is added.

Add required sugar to the extract and cook by stirring continuously till it reaches the TSS of 65° bx. Remove from the fire and cool, it to room temperature. Pack it in a sterilized bottle and cover it airtight. Label with suitable information.

Problems in Jelly Making

Failure to set

This may be due to addition of too much sugar, lack of acid or pectin, cooking below or beyond end point, prolonged cooking.

Cloudy or foggy jellies

This is due to the following reasons use of non-clarified juice or extract, use of immature fruits, over cooking, over cooling, non-removal of scum.

Formation of crystals

It is due to excess of sugar

Syneresis or weeping of jelly

The phenomenon of spontaneous exudation of fluid from a gel is called syneresis or weeping. This may due to excess of fluid, too low concentration of sugar, insufficient pectin, premature gelation and fermentation.

Squash

Squash is a type of fruit beverage .The FPO or FSSAI specification for squash is fruit juice 25 per cent, TSS 45° brix, acidity 1.0 per cent, preservative 350 ppm of SO_2 (or) 700 ppm of potassium metabisulphite or 600 ppm of sodium benzoate. The dilution is 1:4. Lime, mango, orange and pine apple are used for making squash commercially using KMS as preservative or fruits viz. jamun, passion fruit, raspberry, strawberry, grape fruit with sodium benzoate as preservative.

Preparation of squash

- Select ripe and firm fruits
- Washing and Peeling of outer skin
- Cut into small pieces
- Extraction of juice by crushing
- Strain the juice and Juice measuring
- Preparation of sugar syrup (Sugar + water + acid heating to dissolve)
- Strain and Cool the syrup
- Mix with juice
- Addition of preservative
- Bottling and Capping

Marmalade

Marmalade is a fruit jelly in which slices of the fruit or its peel are suspended. The term is generally used for products made from citrus fruits like oranges and lemons in which shredded

peel is used as the suspended material. The FPO or FSSAI specification for marmalade is TSS – 65%, and fruit juice – 45 % of the prepared product. Citrus marmalades are classified into

- Jelly marmalade – it is prepared from the clarified pectin extract.
- Jam marmalade - The pectin extract is not clarified and whole pulp is used.

Preparation of marmalade

- Select ripe fruits
- Washing and peeling outer portion
- Cutting the peel in to fine shreds and boiled for 10 -15 mins
- Cutting the thick slices of peeled fruit or crushing into pulp
- Boiling (2-3 times its weight of water for 40 – 60 min)
- Straining extract
- Test for pectin content (alcohol test)
- Addition of sugar
- Cooking to 103 -1 05 ^{o} C
- Addition of prepared shreds
- Boiling till jellying point (test for end point - sheet / drop /temperature test)
- Cooling and add flavour
- Fill in sterilized bottles, seal and store at ambient temperature**.**

II. Preservation by using salt

Sodium chloride is an indispensible component of food. At lower concentrations it contributes significantly to the flavour. At higher concentrations it exhibits an important bacteriostatic action. Salt at a concentration of 15-25%, is sufficient to preserve most products. It inhibits enzymatic browning and discolouration and also acts as an antioxidant, salt in the form of brine is used for canning and pickling of vegetables and curing of meat. Salt has been reported to have the following effects;

- It causes high osmotic pressure and hence plasmolysis of cells
- It dehydrates foods by drawing out and tying up moisture as it dehydrates microbial cells.
- It ionizes to yield the chloride ion, which is harmful to organisms
- It reduces the solubility of O_2 in the moisture
- It sensitizes the cell against CO_2

- It interferes with the action of proteolytic enzymes.

Pickling process

Pickling is the result of fermentation by lactic acid forming bacteria which are generally present in large numbers on the surface of fresh vegetables and fruits. These bacteria can grow in acid medium and in the presence of 8-10 % salt solution whereas the growth of a majority of undesirable organisms is inhibited. There are two methods for pickling

Dry salting method

Alternate layers of vegetables and salt 20-30 g of dry salt /Kg vegetables are kept in a vessel which is covered with a cloth and a wooden board and allowed to stand for about 2 hrs. During this period due to osmosis sufficient juice comes out from the vegetables to form brine. As soon as the brine is formed the fermentation process starts and CO_2 begins to evolve. The salt content is now increased gradually so that by the time pickle is ready, salt concentration reaches 15% when fermentation is over gas formation ceases.

Fermentation in brine

Steeping of the vegetable in a salt solution of predetermined concentration for a certain length of time is called brining. This type of treatment is adopted in the case of cucumbers and similar vegetables which do not contain sufficient juice to from brine with dry salt. Brine can be prepared by dissolving in common salt in water and filtering it through the cloth to remove insoluble impurities. The remaining process is similar to that of dry salting method.

III. Preservation by using chemicals

Preservative

A preservative is defined as any substance which is capable of inhibiting, retarding or arresting the growth of microorganisms. Microbial spoilage of food products is also controlled by using chemical preservatives. The inhibitory action of preservatives is due to their interfering with the mechanism of cell division, permeability of cell membrane and activity of enzymes. Two important chemical preservatives are permitted to beverages according to the FPO (1955).

- Sulphur dioxide
- Benzoic acid

Sulphur dioxide (SO_2)

It is widely used throughout the world in the preservation of juice, pulp, nectar, squash, crush, cordial and other products. It has good preserving action against bacteria and moulds and

inhibits enzymes, etc. in addition it acts as an antioxidant and bleaching agent. Potassium metabisulphite ($(K_2S_2O_5)$ is commonly used as a stable source of SO_2.

Benzoic acid

It is only partially soluble in water hence its salt, sodium benzoate is used. The bacterial action of benzoic acid is increased in the presence of CO_2 and acid. It is more effective against yeasts tan against moulds. It does not stop lactic acid and acetic acid fermentation.

Permissible limits of preservatives in food products (FPO)

Sulphur dioxide	**Quantity (ppm)**
Fruit pulp	2000-3000
Fruit juice concentrate	1500
Dried fruits	2000
squash	350
Jam, marmalade, preserve	40
RTS	70
Pickles	100
Dehydrated vegetables	2000
Benzoic acid	**Quantity (ppm)**
squash	600
Jam, jelly, maramalde	200
Pickles	250
Tomato sauce	750

LECTURE 15
FOOD PACKAGING AND NUTRITION LABELING

Food Packaging:

In today's society, packaging is pervasive and essential. It surrounds, enhances and protects the goods we buy, from processing and manufacturing, through handling and storage, to the final consumer. Without packaging, materials handling would be a messy, inefficient and costly exercise and modem consumer marketing would be virtually impossible. The packaging sector represents about 2% of Gross National Product (GNP) in developed countries and about half of all packaging is used to package food.

Definition of packaging:

Packaging has been defined as a socio scientific discipline which operates in society to ensure delivery of goods to the ultimate consumer of those goods in the best condition intended for their use.

The Packaging Institute International (PII) defines packaging as the enclosure of products, items or packages in a wrapped pouch, bag, box, cup, tray, can, tube, bottle or other container form to perform one or more of the following functions: containment, protection, preservation, communication, utility and performance. If the device or container performs one or more of these functions, it is considered a package.

Other definitions of packaging include a co-ordinated system of preparing goods for transport, distribution, storage, retailing and end-use, a means of ensuring safe delivery to the ultimate consumer in sound condition at optimum cost, and a techno-commercial function aimed at optimizing the costs of delivery while maximizing sales.

Package, Packaging, Packing:

It is important to distinguish between the words "package," "packaging" and "packing." The package is the physical entity that contains the product. Packaging was defined above and in addition, is also a discipline. The verb "packing" can be defined as the enclosing of an individual item (or several items) in a package or container.

Levels of Packaging:

A **primary package** is the one which is in direct contact with the contained product. It provides the initial, and usually the major protective barrier. Example: Metal cans, paperboard

cartons, glass bottles and plastic pouches, aerosal spray can, Beverage can, cushioning envelopes, plastic bottles, skin pack.

A **secondary package** contains a number of primary packages. It is outside the primary packaging perhaps used to group primary packages together. It is the physical distribution carrier and is sometimes designed so that it can be used in retail outlets for the display of primary packages. Ex. Corrugated case, Boxes

A **tertiary package** is made up of a number of secondary packages. It is used for bulk handling. Example being a stretch-wrapped pallet of corrugated cases.

A **quaternary package** is frequently used to facilitate the handling of tertiary packages. This is generally a metal container up to 40 m in length which can be transferred to or from ships, trains, and flatbed trucks by giant cranes. Certain containers are also able to have their temperature, humidity and gas atmosphere controlled. This is necessary in particular situations such as the transportation of frozen foods, chilled meats and fresh fruits and vegetables.

Functions of packaging:

Contain

- portion control (profitability)
- consistency
- company reputation
- consumer expectation
- consumer convenience

Protect

- contamination
- maintain quality
- legislation (Codex, local legislation)
- product consistency
- company reputation

Inform (labelling)

- nature of the contents
- legislation, Codex, and other codes
- nutrition
- instructions for use

- elimination of fraud
- storage requirements

Attract

- advertise that this product is satisfying and fun and healthy

Classification of Packages:

Packages can be classified as 1) traditional or natural and 2) fabricated or modern packaging materials based on the availability of the materials.

Traditional or natural packaging materials are – Bamboo basket, fiber or leaf mats, Leather containers of animal skin, clay containers, gunny bags, cloth bags, Arecanut and teak leaves sheath.

Modern packaging materials can be divided into rigid, semi rigid and flexible materials.

- **Rigid containers** are – metal drums, metal barrels, glass bottles, glass jars, wooden boxes, wooden crates, plastic bottles, plastic drums, plastic crates, paper drums, plywood containers.
- **Semi rigid containers** are – aluminium collapsible tube, plastic collapsible tube, composite container, paper based cartons.
- **Flexible container** is plastic bags.

Types of Packaging Materials:

1. Paper based packaging materials
2. Metal packaging materials
3. Glass packaging materials
4. Plastic packaging materials
5. Edible and bio based packaging materials

Paper as Packaging Material:

Paper derives its name from the reedy plant “papyrus”, which the ancient Egyptians used to produce the world's first writing material by beating and pressing together thin layers of the plant stem. Paper is widely used as a packaging material because of its stiffness and printability.

The main **advantages of paper as packaging material** are – Good stiffness, good absorbent, good creaseability, good printability, low density, not brittle, biodegradable, low cost. The main **disadvantages** are – poor tensile strength, poor wet strength, tear easily, no barrier property without coating.

Types of Paper:

- Kraft paper
- Bleached paper
- Greaseproof Paper
- Glassine paper
- Vegetable parchment
- Waxed paper

Paper is divided into two broad categories:

- Fine papers, generally made of bleached pulp, and typically used for writing paper, bond, ledger, book, and cover papers, and
- Coarse papers, generally made of unbleached kraft softwood pulps and used for packaging.

Paperboard Products:

Paper is generally termed board when its grammage exceeds 224 g m^{-2}. Multiply boards are produced by the consolidation of one or more web plies into a single sheet of paperboard, which is then subsequently used to manufacture rigid boxes, folding cartons, beverage cartons and similar products. One advantage of multi-ply forming is the ability to utilize inexpensive and bulky low-grade waste materials (mostly old newspapers and other postconsumer waste papers) in the inner plies of the board where low fiber strength and the presence of extraneous materials (e.g., inks, coatings, etc.) have little effect on board properties. However, multi-ply boards containing postconsumer waste papers are not used for food contact purposes.

Paperboard Grades:

- Linerboard
- Foodboard
- Folding Boxboard (Cartonboard)
- Chipboard
- Folding Cartons
- Beverage Cartons
- Molded pulp containers

Metal packaging materials

Four metals are commonly used for the packaging of foods: steel, aluminium, tin and chromium. Today, materials like tinplate and aluminum have become universally adopted for the manufacture of containers and closures for foods and beverages, largely due to several important qualities of these metals. These include their mechanical strength and resistance to working, low toxicity, superior barrier properties to gases, moisture and light, ability to withstand wide extremes of temperature and ideal surfaces for decoration and lacquering.

Advantages

Metal cans have a number of advantages over other types of container, including the following:

- They provide total protection of the contents
- They are convenient for ambient storage and presentation
- They are tamperproof.

Disadvantages

- The high cost of metal and the high manufacturing costs make cans expensive.
- They are heavier than other materials, except glass, and therefore have higher transport costs

Protective and Decorative Coatings:

- They protect the metal from the contents
- They avoid contamination of the product by metal ions from the container
- They facilitate manufacture
- They provide a basis for decoration and product dentification
- They form a barrier to external corrosion or abrasion

Protective coatings:

For most containers, the enamel is applied to the metal in the flat before fabrication. Many types of internal enamel are available for food containers including oleoresinous, vinyl, vinyl organosol, acrylic, alkyd, polybutadiene, phenolic and epoxyphenolic

Decorative Coatings:

Although the primary purpose in decorating the external surface of a metal container is to improve its appearance and assist its marketability, it also significantly improves the container's external corrosion resistance.

Aluminum foils and containers:

Aluminum foil is a thin-rolled sheet of alloyed aluminum varying in thickness from about 4 to 150 μm. Foil can be produced by two methods: either by passing heated aluminium sheet ingot between rollers in a mill under pressure and then rerolling on sheet and plate mills until the desired gauge is obtained, or continuously casting and cold rolling. Aluminum foil is essentially impermeable to gases and water vapor when it is thicker than 25.4 μm, but it is permeable at lower thicknesses due to the presence of minute pinholes

Glass packaging materials

Glass has been defined by the American Society for Testing and Materials (ASTM) as "an amorphous, inorganic product of fusion that has been cooled to a rigid condition without crystallizing". The two main types of glass container used in food packaging are bottles (which have narrow necks) and jars (which have wide openings).

Advantages:

Glass containers have the following advantages:

- They are impervious to moisture, gases, odours and micro-organisms
- They are inert and do not react with or migrate into food products
- They are suitable for heat processing when hermetically sealed
- They are re-useable and recyclable
- They are resealable
- They are transparent to display the contents
- They are rigid, to allow stacking without container damage.

Disadvantages:

- Higher weight which incurs higher transport costs than other types of packaging
- Lower resistance than other materials to fractures, scratches and thermal shock
- More variable dimensions than metal or plastic containers
- Potentially serious hazards from glass splinters or fragments in foods.

Plastic packaging material:

Plastics are organic polymers which are long chain molecules obtained by addition or condensation of one or more monomers. Polymerization of single repeating unit gives homo polymers and addition of more than one monomer results in co-polymer.

Plastics have become a major packaging material, along with paper, metal and glass. Plastics are used mainly for consumer packages in the form or wraps, pouches, cartons, bags, tubes, bottles, jars and boxes. In transport, they are used in the form of sacks, stretch films for wrapping tray-loads, containing unit packs or for entire pallet loads. The advent of snack foods, convenience foods and prepared foods has been possible to a great extent due to the availability of plastic packaging materials.

Advantages:

- Barrier to water vapour and gases
- Light weight
- Good strength
- Design flexibility
- Resistance to breakage
- Machinability-high speed filling using form fill and seal techniques
- Glossy and transparent
- Colouring is possible
- High tensile strength
- High tear strength
- High printability
- High level lamination
- Low cost

Disadvantage:

- The disadvantage of plastic is the disposability i.e. it is difficult to get it disintegrated into soil.

Polyethylene:

It is the polymer of ethylene. It accounts for the biggest proportion of the plastics used in packaging. It is good barrier to water vapour but less to oxygen; has high permeability to volatile oils and swells in contact with fats and oil. It gives very good heat seals and easily coated to other materials and serves as a laminated layer. It is used as bags, liners, bottles

1. Low Density Polyethylene (LDPE):
2. Linear Low Density Polyethylene (LLDPE):
3. High Density Polyethylene (HDPE):

Polypropylene (PP):

Polypropylene is a linear polymer containing little or no unsaturation. PP has low water vapour transmission, medium gas permeability, good resistance to greases and chemicals, good abrasion resistance, high temperature stability, good gloss and high clarity

Polystyrene (PS):

It is produced by the polymerization of styrene. It is transparent, but has low barrier property. The material is used for packing vegetables and fresh meat on trays, yoghurt and other milk products in cups, and for the over wrapping of fruits and vegetables.

Polycarbonate:

It is glass-like, heat resistant, and sterilisable upto 130°C and is available in the form of film, beside the rigid containers, but has very few food packaging applications.

Polyvinyl Chloride (PVC):

There are two types – rigid and plasticized. The rigid form has good moisture and gas barrier properties and resistance to fats. Hence, these are used for packing butter or margarine, and for making transparent bottles for mineral water, edible oils, fruit juices, carbonated beverages and beer, as these bottles can withstand pressure. The plasticised form is used for packing meat, fruits and vegetables and for shrink wrapping. It is also used for the shrink wrapping of pallets.

Polyvinylidene Chloride (PVdC):

PVDC has the lowest water vapour, oxygen and CO2 permeability amongst all commercially used plastic films, besides having resistance to fats and chemicals. PVDC is used as a coating material on polyethylene and other plastics, to improve the gas and moisture barrier properties of the native plastics. It is used for packing dense materials like cheese, poultry etc.

Ethylenevinyl Alcohol (EVOH):

EVOH copolymers offer not only excellent processability but also superior barriers to contaminants such as gases, odors, fragrances and solvents. It is widely used polymer in the manufacture of high barrier containers, as in the manufacture of bulk bags used for aseptic packaging, retort pouches and containers.

Polyethylene Terephthlate (PET):

PET is a linear, transparent thermoplastic polymer. It has low permeability to moisture and gases, but has poor sealing property. Hence, it has to be laminated with PE. PET containers

are used widely for packing mineral water, carbonated and non-carbonated beverages, syrups, edible oils and liquors.

Retort Pouch:

The retort pouch is a flexible package, hermetically sealed on three or four sides and made from one or more layers of plastic or foil, each layer having a specific functionality. The advantages include the ease of carrying, reheating and serving, as well as weight and space saving. Finally, disposal of the used pouch is much simpler than for the metal can as it can be easily flattened. Three-layer pouch structure would consist of an outer layer of 12 μm PET for strength and toughness, a middle layer of 7 to 9 μm of aluminium foil as a moisture, light and gas barrier and an inner layer of 70 to 100 μm of CPP for heat sealability, strength and compatibility with all foods. An additional inner layer of 15 to 25 μm of PA is used when a longer shelf life is required.

Edible and bio based packaging materials

Biodegradable polymers are a newly emerging field in food packaging. These are biodegradable, ecofriendly and are produced from natural or renewable resources. Biodegradable polymers (BDPs) or biodegradable plastics refer to polymeric materials that are 'capable of undergoing decomposition into carbon dioxide, methane, water, inorganic compounds, or biomass in which the predominant mechanism is the enzymatic action of microorganisms, that can be measured by standardized tests, in a specified period of time, reflecting available disposal condition'.

Classification of Biodegradable Polymers

Biopolymers can be classified into two main groups, these two groups being

- The agropolymers obtained by biomass fragmentation processes (polysaccharides, proteins…),
- The biopolyesters obtained
    - by synthesis from bio-derived monomers (polylactic acid – PLA) or
    - by extraction from micro-organisms (polyhydroxyalkanoate – PHA) or
    - by synthesis from synthetic monomers (polycaprolactone – PCL, aromatic and aliphatic copolyesters – PBAT, PBSA…)

Advantages of Biodegradable Polymers

The principal advantages of plastics are their cost, functionality, durability and weight. The advantages of biodegradable plastic are numerous.

- Biodegradable plastics take less time to break down
- Biodegradable plastics are renewable
- Biodegradable plastics are good for the environment
- Biodegradable plastics require less energy to produce
- Biodegradable plastics are easier to recycle
- Biodegradable plastics are not toxic
- Biodegradable plastics reduce dependence on foreign oil

Controlled atmosphere Packaging:

Controlled atmosphere packaging (CAP) is the enclosure of food in a gas impermeable package inside which the gaseous environment with respect to CO_2, O_2, N_2, water vapor and trace gases has been changed, and is selectively controlled to increase shelf life. In this technology, the storage system consists of airtight storage chambers, O_2 regulatory unit, CO_2 absorbing unit equipment for monitoring as well as controlling the chambers and composition. Liquid N_2 generator is commonly installed to flush and chambers with liquid N_2 as and when required for maintaining optimum level of O_2. For maintaining optimum level of CO_2 in the chambers, air in chambers is circulated through CO_2 scrubber frequently. CO_2 absorbing materials such as hydrated lime, calcium or potassium hydroxides are generally used in scrubber. Refrigeration unit is employed for maintaining storage temperature. The storage life of various fruits and vegetables can be increased by 2 to 4 times the normal life by employing Controlled Atmosphere storage technology.

Modified Atmosphere Packaging:

Modified Atmosphere Packaging (MAP) can be defined as the enclosure of food in a package in which the atmosphere inside the package is modified or altered to provide an optimum atmosphere for increasing shelf life and maintaining quality of the food. Modification of the atmosphere may be achieved either actively or passively. Active modification involves displacing the air with a controlled, desired mixture of gases, a procedure generally referred to as gas flushing. Passive modification occurs as a consequence of the food's respiration or the metabolism of micro-organisms associated with the food; the package structure normally

incorporates a polymeric film, and so the permeation of gases through the film (which varies depending on the nature of the film and the storage temperature) influences the composition of the atmosphere that develops.

Advantages:

- Shelf life will be increased by 50 to 400%.
- Reduced economic losses due to longer shelf life.
- Provides a high quality product.
- Centralized packaging and portion control.
- Improved presentation – clear view of product and all –around visibility.
- Little or no need for chemical preservatives.
- Sealed packages are barriers against product recontamination and drip from package.
- Odorless and convenient packages.

Disadvantages:

- Added costs for gases, packaging materials and machinery.
- Temperature control necessary.
- Different gas formulations for each product type.
- Special equipment and training required.
- Increased pack volume adversely affects transport costs and retail display space.
- Loss benefits once the pack is opened or leaks.
- CO_2 dissolving into the food could lead to pack collapse and increased drip.

Gases used in MAP:

The three main gases used in MAP are CO_2, O_2, and N_2, either singly or in combination.

Carbon dioxide: Carbon dioxide is the most important gas in the MAP of foods because of its bacteriostatic and fungistatic properties. It inhibits the growth of any spoilage bacteria, the degree of inhibition increasing with increasing concentration.

Oxygen: Oxygen promotes several types of deteriorative reactions in foods including fat oxidation, browning reactions and pigment oxidation. Most of the common spoilage bacteria and fungi require O_2 for growth. For these reasons, O_2 is either excluded or the level set as low as possible. Exceptions occur where O2 is needed for fruit and vegetable respiration or the retention of color in red meat.

Nitrogen: Nitrogen is an inert gas with no odor or taste. It has a lower density than air and a low solubility in water and other food constituents, making it a useful filler gas in MAP to counteract package collapse caused by CO_2 dissolving in the food. Nitrogen indirectly influences the micro-organisms in perishable foods by retarding the growth of aerobic spoilage microbes but it does not prevent the growth of anaerobic bacteria.

Labelling

Labelling" includes any written, printed or graphic matter that is present on the label, accompanies the food, or is displayed near the food, including that for the purpose of promoting its sale or disposal

Nutrition labelling

Nutrition labelling is information found on the labels of prepackaged foods. The legislated information includes:

- The Nutrition Facts table
- The ingredient list
- Some optional nutrition claims

These give information about the nutritional value of a food. The Nutrition Facts table gives information about calories, 13 core nutrients and percentage daily value (% DV) of nutrients. All of the information in the Nutrition Facts table is based on an amount of food. This amount is always found at the top of the nutrition facts table.

Important functions of labelling:

(i) Describe the Product and Specify its Contents:

A label provides complete information regarding the product. It mainly includes ingredients of the product, its usage, and caution in use, cares to be taken while using it, date of manufacturing, batch number, etc.

(ii) Identification of the Product or Brand:

It is easier to identify a particular product among many with the help of labelling.

(iii) Grading of Product:

When a product has different qualities, labelling helps to find out which pack contains what type of quality

(iv) Help in Promotion of Products:

The fourth function of labeling is to promote sales. Sometimes a consumer gets encouraged to buy a product simply due to attractive label. Nowadays labeling is used as an effective sales promoting tool.

(v) Providing information required by Law:

Another important function performed by labeling is to provide statutory warning required by law. To put 'smoking is injurious to health' on the package of cigarette and 'Chewing Tobacco is Injurious to Health' on the package of Pan Masala are the examples of statutory warning? Similarly, in case of hazardous or poisonous products, appropriate statutory warning need to be put on the label.

Mandatory Labelling of Prepackaged Foods

The following information shall appear on the label of prepackaged foods

The name of the food

The name shall indicate the true nature of the food and normally be specific and not generic. A "coined", "fanciful", "brand" name, or "trade mark" may be used

List of ingredients

Except for single ingredient foods, a list of ingredients shall be declared on the label. The list of ingredients shall be headed or preceded by an appropriate title which consists of or includes the term 'ingredient'. All ingredients shall be listed in descending order of ingoing weight (m/m) at the time of the manufacture of the food. The following foods and ingredients are known to cause hypersensitivity and shall always be declared:

- Cereals containing gluten; i.e., wheat, rye, barley, oats;
- Crustacea and products of these;
- Eggs and egg products;
- Fish and fish products;
- Peanuts, soybeans and products of these;
- Milk and milk products (lactose included);
- Tree nuts and nut products; and
- Sulphite in concentrations of 10 mg/kg or more.

Net contents and drained weight

The net contents shall be declared in the metric system ("Système International" units). The net contents shall be declared in the following manner:

(i) for liquid foods, by volume;

(ii) for solid foods, by weight;

(iii) for semi-solid or viscous foods, either by weight or volume.

Name and address

The name and address of the manufacturer, packer, distributor, importer, exporter or vendor of the food shall be declared.

Country of origin

The country of origin of the food shall be declared if its omission would mislead or deceive the consumer.

Lot identification

Each container shall be embossed or otherwise permanently marked in code or in clear to identify the producing factory and the lot.

Date marking and storage instructions

If not otherwise determined in an individual Codex standard, the following date marking shall apply:

(i) The "date of minimum durability" shall be declared.

(ii) This shall consist at least of:

- the day and the month for products with a minimum durability of not more than three months;
- the month and the year for products with a minimum durability of more than three months. If the month is December, it is sufficient to indicate the year.

(iii) The date shall be declared by the words:

- "Best before ..." where the day is indicated;
- "Best before end ..." in other cases.

(iv) The words referred to in paragraph (iii) shall be accompanied by:

- either the date itself; or
- a reference to where the date is given.

(v) The day, month and year shall be declared in uncoded numerical sequence except that the month may be indicated by letters in those countries where such use will not confuse the consumer.

LECTURE 16

COMMON FOOD ADULTERANTS AND THEIR DETECTION

Adulteration

Food adulteration is an act of intentionally debasing the quality of food offered for sale either by the admixture or substitution of inferior substances or by the removal of some valuable ingredient. Food adulteration takes into the account not only the intentional addition or substitution or abstraction of substances which adversely affect nature, substances and quality of foods, but also their incidental contamination during the period of growth, harvesting, storage, processing, transport and distribution.

Adulterant means any material which is or could be employed for making food unsafe or sub-standard or mis-branded or containing extraneous matter.

Adulteration of food stuffs is commonly practiced in India by traders. In order to protect the health of the consumer, the Government of India promulgated the prevention of food adulteration act (P.F.A) in 1954 which prohibits the manufacture, sale and distribution of not only adulterated foods but also foods contaminated with toxicants and misbranded foods.

Types of adulterants

Type	Substances added
Intentional adulterants	Sand, marble chips, stones, mud, other filth, talc, chalk powder, water, mineral oil and harmful colour
Incidental adulterants	Pesticide residues, droppings of rodents, larvae in foods
Metallic adulterants	Arsenic from pesticides, lead from water, effluent from chemical industries, tin from cans

Methods for detection of common adulterants in foods

S. No	Food	Adulterant	Detection
1	Milk	Water	The presence of water can be by putting a drop of milk on a polished slanting surface. The drop of pure milk either stops or flows slowly leaving a white trail behind it ,whereas milk adulterated with water will flow immediately without leaving a mark
		Starch	Add a few drops of iodine solution. Formation of blue colour indicates the presence of starch
2	Fat and oil	Argemone oil	Take small quantity of oil in a test tube. Add equal quantity of concentrated nitric acid to a sample and shake carefully. Red to reddish brown colour in acid layer indicates the presence of argemone oil
3	Ghee or butter	Mashed potato, sweet potato & other starches	Add a few drops of iodine solution, which is brownish in colour turns to blue if the starches are present
		Vanaspathi or margarine	Take about one teaspoon full of melted sample of ghee or butter with equal quantity of concentrated hydrochloric acid in a stoppered test tube and add to it a pinch of sugar shake for one minute and let it for five minutes. Appearance of crimson colour in lower layer shows presence of vanaspathi or margarine
4	Asafoetida	Soap stone or earth matter	Little portion of the sample is shaken with water and allowed to settle. If stone of soap or earth matter present will settle at the bottom
5	Chilly powder	Red brick powder, soap stone	To a little powder of chilli add small amount of conc. HCl and ix to the consistency of paste dip the rear end of the match stick into the paste and hold over the flame brick red flame colour due to the presence of calcium slats in brick powder.

6	Turmeric powder	Metanil yellow	Take a turmeric powder in a test tube. Add a few drops of conc. HCl. Instant appearance of violet colour which disappears on dilution with water. If the colour persists, presence of metanil yellow indicated.
7	Pepper	Dry papaya seeds	Papaya seeds can be separated out from pepper as they are shrunken , oval in shape and greenish brown or brownish black in colour.
8	Powdered spices	White powdered stone, chalk powder, yellow soap	Take one gram of powdered spices in a test tube and add 5 ml of carbon tetra chloride solvent. Shake well and leave for some time. Impurities will settle at the bottom while the spice powder will flat on the surface
9	Wheat flour	Chalk powder	Add few drops of hydrochloric acid, effervescence (give off bubbles) indicate the presence of chalk powder
10	Dal whole and spilt	Khesari dal	Add 50 ml of dilute Hydrochloric acid to dal and keep on simmering water about 15 minutes. The pink colour developed indicates the presence of khesri dal.
11.	Pulses	Lead chromate	Shake 5 gram of pulse with 5 ml of HCl. Pink colour indicates lead chromate.
12	Honey	Molasses (sugar solution)	A cotton wick dipped in pure honey when lighted with match stick burns and shows the purity of honey. If adulterated, the presence of water will not allow the honey to burn. If it does it will produce cracking sound.
13	Sugar/ Jaggery	Chalk powder	Stir a spoonful sample of sugar in a glass of water, the chalk powder settle down.

		Washing soda	Add few drops of hydrochloric acid, effervescence (give off bubbles) indicate the presence of washing soda
14	Tea	Exhausted tea	Take a filter paper and spread a few tea leaves. Sprinkle with water to wet the filter paper, if coal tar is present it would immediately stain the filter paper. Wash the filter paper under tap water and observe the stains against light
15	Salt	Chalk powder	Stir a spoonful of sample of salt in a glass of water. The presence of chalk will make solution white and other insoluble impurities will settle down.

LECTURE 17
FOOD LAWS AND REGULATIONS AND QUALITY CONTROL STANDARDS

Food standards, such as those that specify the maximum allowable level of contaminants in food, or the amount of food additives permitted, are an important instrument for the assurance of food safety. This is particularly the case in international trade, where the food commodity being traded is manufactured in another country, and the history and conditions of manufacture may not be fully known. International trade on food is based on agreements achieved at the World Trade Organization (WTO), including the Agreement on Sanitary and Phytosanitary Measures and the Agreement on Technical Barriers to Trade. Under these agreements, the Codex Alimentarius Commission (CAC) has the role of implementing the Joint FAO/WHO Food Standards Program, establishing international food quality and safety standards. These standards are adopted or used as models by many countries in their domestic food legislation and regulations.

Food standards are legislative instruments, often called –food laws, which are enacted to protect consumers against unsafe products, adulteration, and fraud; to ensure quality compliance; as well as to protect honest food producers, processors, distributors, and traders. They also facilitate the movement of goods within and between countries by providing a common lexicon for food quality and safety. Standards are designed for many purposes. The main concern is quality and safety, but any standard/regulation cannot be set to the zenith level of quality; therefore, above-average quality is generally taken into consideration in enforcement on a wider scale.

FOOD SAFETY REGULATORY BODIES

There are various food safety regulatory bodies and international organizations throughout the world, which are responsible for dealing with aspects of food safety and food trade among different countries, and for maintaining biodiversity. Some important bodies are listed below:

1. Codex Alimentarius Commission (CAC)
2. World Health Organization (WHO)
3. Food and Agricultural Organization (FAO)

4. World Trade Organization (WTO)
5. World Organization for Animal Health (WOAH)
6. European Union (EU) standards
7. US Food and Drug Administration (FDA)
8. International Plant Protection Organization (IPPO)
9. Convention on Biodiversity (CBD)
10. International Commission on Microbiological Specifications for Foods (ICMSF)

AMERICAN FOOD STANDARD BODIES

The FDA is a federal agency of the US Department of Health and Human Services, one of the US federal executive departments. The FDA is responsible for protecting and promoting public health through the regulation and supervision of food safety, tobacco products, dietary supplements, prescription and over-the-counter pharmaceutical drugs (medications), vaccines, biopharmaceuticals, blood transfusions, medical devices, electromagnetic radiation emitting devices, cosmetics, animal foods and feeds, and veterinary products. The FDA Food Safety Modernization Act of 2010 (FSMA) is a recent food law that came into existence in 2011. It aims to ensure the US food supply is safe by shifting the focus of federal regulators from responding to contamination to preventing it. The FSMA has given the FDA new authority to regulate the way foods are grown, harvested, and processed. The law grants the FDA a numberof new powers, including mandatory recall authority, which the agency has sought for many years. The FSMA requires the FDA to undertake more than a dozen rule-makings and issue at least 10 guidance documents, as well as a host of reports, plans, strategies, standards, and notices, among other tasks. The law was prompted after many reported incidents of food-borne illnesses during the 2000s. Tainted food has cost the food industry billions of dollars in recalls, lost sales, and legal expenses. Earlier the US Department of Agriculture (USDA) and the FDA were the main actors in food regulation in the United States. These two federal agencies encompass all phases of the food regulatory system: They evaluate, investigate, regulate, inspect,and sanction. The USDA‘s work focused on the safety of meat, poultry, and some egg products, whereas the FDA regulated all other foodstuffs, such as whole eggs, seafood, fruits, vegetables, grain products, and milk.

EUROPEAN FOOD SAFETY AUTHORITY (EFSA)

The European Food Safety Authority (EFSA) is the keystone of European Union (EU) risk assessment regarding food and feed safety. In close collaboration with national authorities and in open consultation with its stakeholders, EFSA provides independent scientific advice and clear communication on existing and emerging risks. The EFSA is an independent European agency funded by the EU budget, which operates separately from the European Commission, European parliament, and EU member states. EU Food Law is a market leading publication on food policy and legislation. It covers the latest legislative developments in the food industry, including health claims, nutrition claims, food safety, food labeling legislation, genetically modified food, food and drink advertising, traceability, food law enforcements, contaminants, and food hygiene proposals, among others. EU Food Law follows legislative developments from food authorities including the EFSA and the European Commission.

CODEX ALIMENTARIUS COMMISSION (CAC)

The Codex Alimentarius is a collection of internationally recognized standards, codes of practice, guidelines, and other recommendations relating to foods, food production, and food safety.

It was established in early November 1961 by the Food and Agriculture Organization of the United Nations (FAO), was joined by the World Health Organization (WHO) in June 1962, and held its first session in Rome in October 1963. The Commission's main goals are to protect the health of consumers and ensure fair practices in the international food trade. The Codex Alimentarius is recognized by the World Trade Organization as an international reference point for the resolution of disputes concerning food safety and consumer protection.

As of 2012, there were 186 members of the Codex Alimentarius Commission: 186member countries and one member organization, the European Union (EU). There were 215 Codex observers: 49 intergovernmental organizations, 150 non-governmental organizations, and 16 United Nations organizations.

The CAC (codex alimentarius is Latin for ‒food code‖) is an intergovernmental body established in 1963 to set guidelines and standards to ensure ‒fair trade practices‖ and consumer protection in relation to the global food trade. The following are the basic objectives of the CAC:

- To protect the health of consumers and ensure fair practices in the food trade
- To promote the coordination of all food standards work undertaken by international governmental and nongovernmental organizations
- To determine priorities, initiate and guide the preparation of draft standards through and withthe aid of appropriate organizations
- To finalize standards elaborated above and publish them in Codex Alimentarius either as regional or worldwide standards together with international standards already finalized by other bodies, wherever this is practicable
- To amend published standards, as appropriate, in the light of new developments The Codex Alimentarius is a collection of internationally adopted food standards and related texts presented in a uniform manner. These food standards and related texts aim at protecting consumers‘ health and ensuring fair practices in the food trade. The Codex Alimentarius includes standards for all the principal foods, whether processed, semi processed, or raw, for distribution to consumers. Materials for further processing into foods should be included to the extent necessary to achieve the purposes of the Codex Alimentarius as defined. The Codex Alimentarius includes provisions with respect to food hygiene, food additives, residues of pesticides and veterinary drugs, contaminants, labeling and presentation, methods of analysis and sampling, and import and export inspection and certification.

INTERNATIONAL STANDARDIZATION ORGANIZATION (ISO)

The ISO is an international quality management accreditation agency. ISO 22000 is a standard developed by them dealing with food safety management. ISO 22000:2005 specifies requirements for a food safety management system whereby an organization in the food chain needs to demonstrate its ability to control food safety hazards in order to ensure that food is safe at the time of human consumption. It enables any establishment:

1. to plan, implement, operate, maintain, and update a food safety management system aimed at providing products that, according to their intended use, are safe for the consumer;
2. to demonstrate compliance with applicable statutory and regulatory food safety requirements;
3. to evaluate and assess customer requirements and demonstrate conformity with those mutually agreed customer requirements that relate to food safety, in order to enhance

customer satisfaction;

4. to effectively communicate food safety issues to their suppliers, customers, and relevant interested parties in the food chain;

5. to ensure that the organization conforms to its stated food safety policy;

6. to demonstrate such conformity to relevant interested parties; and

7. to seek certification or registration of its food safety management system by an external organization, or make a self-assessment or self-declaration of conformity to ISO 22000:2005.

INDIAN FOOD STANDARD LEGISLATION AND REGULATORY BODIES

In 2006, India passed the Food Safety and Standard Act (FSSA), a modern integrated food law to serve as a single reference point in relation to the regulation of food products, including all types of food commodities. The passing of this Indian act is a significant first step; much more has to happen to eliminate the confusing overlap with old laws and regulations. The Food Safety and Standards Authority of India (FSSAI) still needs to make considerably substantive changes in infrastructure and appropriate stewardship for it to match with international standards of the United States and Europe.

1. By the mid-1990s, the food processing sector laws were framed in a veritable grid of regulation, including a multitude of state laws, as well as the following national laws:

a. The Infant Milk Substitutes Feeding Bottles & Infants Food Act, 1992 (Rules: 1993) (regulation of production, supply)

b. Milk and Milk Products Order, 1992

c. Environmental Protection Act, 1986

d. Pollution Control Act, 1986

e. Meat Food Products Order (MFPO), 1973

f. Insecticide Act, 1968

g. Solvent Extracted Oil Control (SEO) Order, 1967

h. Export (Quality Control and Inspection) Act, 1963

i. Essential Commodities Act, 1955

j. Fruits Product Order (FPO), 1955

k. Prevention of Food Adulteration (PFA) Act, 1954 (Rules: 1955)

l. Vegetable Oil Product Control (VOP) Order, 1947

m. Agriculture Produce Act, 1937

2. Some food laws that can be declared voluntarily by the manufacturers of finished productsinclude the following:

a. AGMARK Standards (AGMARK)

b. Codex Alimentarius Standards

c. Bureau of Indian Standards Act, 1986

d. Consumer Protection Act, 1986.

PREVENTION OF FOOD ADULTERATION ACT, 1954

The Prevention of Food Adulteration (PFA) Act was promulgated by Parliament in 1954 to make provision for the prevention of adulteration of food, along with the Prevention of Food Adulteration Rules (1955), which was incorporated in 1955 as an extension to the PFA Act. Broadly, the PFA Act covers food standards, general procedures for sampling, analysis of food, powers of authorized officers, nature of penalties, and other parameters related to food. It deals with parameters relating to food additives, preservative, and coloring issues; packing and labeling of foods; prohibition and regulations of sales; and so on. Like the FPO, amendments to PFA Rules are incorporated with a recommendation made by the Central Committee of Food Standards, which has been setup by the central government under the Ministry of Health and Family Welfare, comprising members from different regions of the country. The provisions of the PFA Act and Rules are implemented by the state governments and local bodies as provided in the rules. The PFA Act had been in place for over five decades, and there was a need for change for various reasons, including the changing requirements of our food industry.

The PFA Act will be repealed from the date to be notified by the central government as per the Food Safety and Standards Act, 2006. Until that date, when new standards are specified, the requirements and other provisions of the PFA Act (1954) and Rules (1955) shall continue to be in force as a transitory provision for food standards. The PFA Division has contemplated the following draft rules/ regulations for the Food Safety and Standards Act of India (FSSAI), 2006, which are in the process of being further scrutinized in consultation with state governments.

Milk and Milk Product Amendment Regulations, 2009

Consequent to the delicensing of the dairy sector in 1991 under the Industrial Development & Regulation Act, the Department of Animal Husbandry, Dairy and Fisheries

promulgated the Milk and Milk Product Order (MMPO; 1992) under section 3 of the Essential Commodities Act, 1955. The objective of the order is to maintain and increase the supply of liquid milk of a desired quality in the interest of the general public, and also for regulating the production, processing, and distribution of milk and milk products. As per the provisions of this order, any person/dairy plant handling more than 10,000 L of milk per day or 500 MT of milk solids per annum needs to be registered with the registering authority appointed by the central government. There is no restriction on the setting up of new dairy units and expanding milk processing capacity; noting the requirement of registration is for enforcing the prescribed Sanitary and Hygienic Conditions, Quality and Food Safety Measures as specified in Schedule V of the MMPO 1992. To comply with the provisions of Paragraph 5(5)(B) of MMPO 1992, two inspection agencies the National Productivity Council and the Export Inspection Council (EIC) of India were notified for annual inspection of registered dairy units, on a rotating basis. Since their inception the central and state registering authorities have registered 803 dairy units with a combined milk processing capacity of 881.50 lakhs liters/day in the cooperative, private, and government sectors upto 31.03.2008. Further, the central registering authority has granted 12 new registrations with a milk processing capacity of 25.0 lakhs liters/day (9 dairy units for milk processing and the remaining 3 units for marketing/trading), enhanced the milk processing capacity of 14 dairy units, and canceled the registration of 10 dairy units during 2008e2009. MMPO 1992 was renamed MMPR (2009), and now it has been subsumed as milk and milk products regulations under section 99 of the FSSA of 2006.

Fruit Product Order (FPO), 1955

The FPO (1955), promulgated under section 3 of the Essential Commodities Act, 1955, with an objective to manufacture fruit and vegetable products while maintaining sanitary and hygienic conditions on the premises and quality standards laid down in the order. It is mandatory for all manufacturers of fruit and vegetable products, including some nonfruit products such as nonfruit vinegar, syrup, and sweetened aerated water, to obtain a license under this order. This order was implemented earlier by the Ministry of Food Processing Industries (now by the FSSAI) through the Directorate of Fruit and Vegetable Preservation, headquartered in New Delhi. The Directorate has five regional offices with headquarters located in New Delhi, Mumbai, Kolkata, Chennai, and Guwahati, as well as a suboffice at Lucknow in northern region. The field officers of the regional offices undertake periodic

inspections of the manufacturing units to ensure maintenance of hygienic conditions in the factories and draw random samples of products from them, as well as from markets, which are analyzed in the laboratories to test their conformity according to the specifications laid out under the FPO. The Central Fruit Product Advisory Committee, comprising officials of concerned government departments, technical experts, and representatives of the Central Food Technology Research Institute, Bureau of Indian Standards, the fruits and vegetable processing industry, and consumer organizations, recommends amendments to the FPO.

Edible Oils Packaging Order, 1998

To ensure the availability of safe and quality edible oils in packed form at predetermined prices to consumers, the central government promulgated on September 17, 1998, an Edible Oils Packaging (Regulation) Order under the Essential Commodities Act, 1955, to make compulsory the packaging of edible oils sold in retail, unless specifically exempted by the concerned state governments.

Vegetable Oil Products Order, 1998

Two earlier order vegetable oil Products (Control) Order, 1947, and Vegetable Oil Products (Standards of Quality) Order, 1975dhave been replaced by a single order called the –Vegetable Oil Products (Regulation) Order, 1998‖ for proper regulation of the manufacture, distribution, and sale of vegetable oil products. The vegetable oil products industry is regulated by this order through the Directorate of Vanaspati, Vegetable Oils & Fats, Department of Food, Public Distribution, Ministry of Consumer Affairs, and Food & Public Distribution.

Meat Food Products Order (MFPO), 1973

Consumption of meat and meat products, and consumers‘ preference for these products, is gradually increasing. In the meat and meat processing sectors, poultry is the fastest growing animal protein in India. Indian consumers prefer to buy fresh meat from the wet market, rather than processed or frozen meats. A mere 6% of poultry meat produced (about 100,000 MT) issold in processed form. Of this, only about 1% undergoes processing into value-added products (ready-to-eat/ready-to cook products). Processing of large animals is largely for the purpose of exports. Meat and meat products are highly perishable and can transmit diseases from animals to humans. Processing of meat products are licensed under the MFPO 1973 now under Ministry of Food Processing Industries.

Solvent-Extracted Oil, De-Oiled Meal and Edible Flour (Control) Order, 1967

This order is basically a quality control order to ensure that solvent-extracted oils in particular do not reach consumers for consumption before they are refined and conform to the quality standards specified in the order for the intended purpose. Standards for the solvent (hexane) that is to be used for the extraction of oil from oil-bearing materials have also been specified to eliminate possible contamination of oil from the solvent used. This order has variousmajor roles:

1. Governing the manufacture, quality, and movement of solvent extracted oils, de-oiled meal,and edible flour
2. Protecting consumers through quality assurance of solvent extracted oils, de-oiled meal, andedible flour
3. Eliminating the possibility of diversion of the oils for unintended uses
4. Prohibiting purchase, offers to purchase, use, or stock for use, any solvent not conforming to the quality standards for extraction of vegetable oils, specifying particulars to be declared on the label affixed to the container.

Food Safety and Standards Authority of India (FSSAI)

The FSSAI was established under Food Safety and Standards Act, 2006, which consolidates various acts and orders that hitherto handled food-related issues in various ministries and departments. The FSSAI was created to develop science-based standards for articles of food and to regulate their manufacture, storage, distribution, sale, and import to ensure the availability of safe and wholesome food for human consumption. The Ministry of Health & Family Welfare, Government of India, is the administrative ministry for the implementation of the FSSAI. Various central acts including the PFA Act 1954, FPO 1955, MFPO 1973, VegetableOil Products (Control) Order 1947, Edible Oils Packaging (Regulation) Order 1988, Solvent Extracted Oil, De-Oiled Meal and Edible Flour (Control) Order 1967, and the MMPO 1992, have been repealed after commencement of the FSSAI Act, 2006. The FSSAI, headquartered in New Delhi, and the state food safety authorities enforce provisions of this act. Food Safety & Standards Regulations (FSSR), 2011, is divided into six parts:

1. Licensing and Registration of Food Businesses
2. Packaging and Labeling
3. Food Product Standards and Food Additives

4. Prohibition and Restriction on Sales
5. Contaminants, Toxins, and Residues
6. Laboratory and Sampling Analysis

Salient Features of FSSAI Act, 2006

Some of the salient features of the Act are:

- FSSAI as a single reference point for all matters relating to Food Safety and Standards, Regulations and Enforcement.
- Integrated response to strategic issues like Novel foods, Health Foods, Nutraceuticals, GM foods, international trade etc.
- Adequate information dissemination on food to enable consumer to make informed choices.
- Graded penalty depending upon the gravity of offences.
- Adequate representation of government, industry organizations, consumers, farmers, technical experts, retailers etc.
- Enforcement of the legislation by the State Governments/ UTs through the state Commissioner for Food Safety, his officers and Panchayati Raj/Municipal bodies.

Establishment of the Authority

Ministry of Health & Family Welfare, Government of India is the Administrative Ministry for the implementation of FSSAI. The Chairperson and Chief Executive Officer of Food Safety and Standards Authority of India (FSSAI) have already been appointed by Government of India. The Chairperson is in the rank of Secretary to Government of India.

Functions performed by FSSAI

- Framing of Regulations to lay down the Standards and guidelines in relation to articles of food and specifying appropriate system of enforcing various standards thus notified.
- Laying down mechanisms and guidelines for accreditation of certification bodies engaged in certification of food safety management system for food businesses.
- Laying down procedure and guidelines for accreditation of laboratories and notification of the accredited laboratories.
- To provide scientific advice and technical support to Central Government and State Governments in the matters of framing the policy and rules in areas which have a direct or indirect bearing of food safety and nutrition.

- Collect and collate data regarding food consumption, incidence and prevalence of biological risk, contaminants in food, residues of various, contaminants in foods products, identification of emerging risks and introduction of rapid alert system.
- Creating an information network across the country so that the public, consumers, Panchayats etc receive rapid, reliable and objective information about food safety and issues of concern.
- Provide training programmes for persons who are involved or intend to get involved in food businesses.
- Contribute to the development of international technical standards for food, sanitary and phyto-sanitary standards.
- Promote general awareness about food safety and food standards.

Export Inspection Council

The EIC was set up by the Government of India under section 3 of the Export (Quality Control and Inspection) Act, 1963 (the 22nd of 1963), in order to ensure sound development of India‘s export trade through quality control and inspection and for matters connected thereto. Basmati rice, honey, fish and fish products, poultry meat products, egg products, and milk products are major food commodities for which the EIC provides certification; it also acts as an advisory body to the central government, which is empowered under the act to identify commodities to be subjected to quality control and/or inspection before export, to establish standards of quality for such identified commodities, and to specify the type of quality control and/or inspection to be applied to such commodities.

Agricultural and Processed Food Products Export Development Authority (APEDA)

The APEDA was established by the Government of India under the Agricultural and Processed Food Products Export Development Authority Act passed by the parliament in December 1985. The act (the second of 1986) came into effect on February 13, 1986, through a notification issued in the Gazette of India: Extraordinary: Part-II (Sec. 3(ii): 13.2.1986). The APEDA replaced the Processed Food Export Promotion Council and is mandated with the responsibility of promoting exports and developing the following scheduled products: meat and meat products, poultry and poultry products, dairy products, confectionery, biscuits and bakery products, honey, jaggery and sugar products, cocoa and its products, chocolates of all kinds, alcoholic and nonalcoholic beverages, cereal and cereal products, groundnuts, peanuts and

walnuts, pickles, papads and chutneys, guar gum, floriculture and floriculture products, and herbal and medicinal plants. In addition to this, APEDA has been entrusted with the responsibility to monitor the importation of sugar.

HACCP

Hazard analysis and critical control points, or HACCP, is a systematic preventive approach to food safety from biological, chemical, and physical hazards in production processes that can cause the finished product to be unsafe and designs measures to reduce these risks to a safe level.

In the early 1960s, a collaborated effort between the Pillsbury Company, NASA, and the U.S. Army Laboratories began with the objective to provide safe food for space expeditions. People involved in this collaboration included Herbert Hollander, Mary Klicka, and Hamed El-Bisi of the United States Army Laboratories in Natick, Massachusetts, Paul A. Lachance of the Manned Spacecraft Center in Houston, Texas, and Howard E. Baumann representing Pillsbury as its lead scientist.

To ensure that the food sent to space was safe, Lachance imposed strict microbial requirements, including pathogen limits (including *E. coli*, *Salmonella*, and *Clostridium botulinum*). Using the traditional end product testing method, it was soon realized that almost all of the food manufactured was being used for testing and very little was left for actual use. Therefore, a new approach was needed.

NASA's own requirements for critical control points (CCP) in engineering management would be used as a guide for food safety. CCP derived from failure mode and effects analysis (FMEA) from NASA via the munitions industry to test weapon and engineering system reliability. Using that information, NASA and Pillsbury required contractors to identify "critical failure areas" and eliminate them from the system, a first in the food industry then. Baumann, a microbiologist by training, was so pleased with Pillsbury's experience in the space program that he advocated for his company to adopt what would become HACCP at Pillsbury.

Definitions

Control point: Any point in a specific food system where loss of control does not lead to an unacceptable health risk.

Critical control point (CCP): Any point or procedure in a food system where control can be exercised and a hazard can be minimized or prevented.

Critical limit: One or more prescribed tolerances that must be met to ensure that a CCP effectively controls a microbiological health hazard.

CCP decision tree: A sequence of questions to assist in determining whether a controlpoint is a CCP.

Corrective action: Procedures followed when a deviation occurs.

Deviation: Failure to meet a required critical limit for a CCP.

HACCP plan: The written document that delineates the formal procedures to be followed in accordance with these general principles.

Hazard: Any biological, chemical, or physical property that may cause an unacceptable consumer health risk (unacceptable contamination, toxin levels, growth, and/or survival of undesirable organisms).

Monitoring: A planned sequence of observations or measurements of critical limits designed to produce an accurate record and intended to ensure that the critical limit maintains product safety.

Risk category: One of six categories prioritizing risk based on food hazards.

Validation: That element of verification focused on collecting and evaluating scientific and technical information to determine whether the HACCP plan, when properly implemented, will effectively control the hazards.

Verification: Methods, procedures, and tests used to determine whether the HACCPsystem is in compliance with the HACCP plan.

HACCP Principles

1. **Assess the hazards and risks associated with the growing, harvesting, raw materials, ingredients, processing, manufacturing, distribution, marketing, preparation, and consumption of the food in question.**

Plan to determine the food safety hazards and identify the preventive measures the plan can apply to control these hazards. A food safety hazard is any biological, chemical, or physical property that may cause a food to be unsafe for human consumption.

2. **Determine the CCP(s) required to control the identified hazards.**

A critical control point (CCP) is a point, step, or procedure in a food manufacturing process at which control can be applied and, as a result, a food safety hazard can be prevented, eliminated, or reduced to an acceptable level.

3. **Establish the critical limits that must be met at each identified CCP.**

A critical limit is the maximum or minimum value to which a physical, biological, or chemical hazard must be controlled at a critical control point to prevent, eliminate, or reduce that hazard to an acceptable level.

4. Establish procedures to monitor the CCP(s).

Monitoring activities are necessary to ensure that the process is under control at each critical control point. In the United States, the FSIS requires that each monitoring procedure and its frequency be listed in the HACCP plan.

5. Establish corrective actions to be taken when there is a deviation identified by monitoring a given CCP.

These are actions to be taken when monitoring indicates a deviation from an established critical limit. The final rule requires a plant's HACCP plan to identify the corrective actions to be taken if a critical limit is not met. Corrective actions are intended to ensure that no product is injurious to health or otherwise adulterated as a result if the deviation enters commerce.

6. Establish procedures for verification that the HACCP system is working correctly.

Validation ensures that the plants do what they were designed to do; that is, they are successful in ensuring the production of a safe product. Plants will be required to validate their own HACCP plans. FSIS will not approve HACCP plans in advance, but will review them for conformance with the final rule.

Verification ensures the HACCP plan is adequate, that is, working as intended. Verification procedures may include such activities as review of HACCP plans, CCP records, critical limits and microbial sampling and analysis. FSIS is requiring that the HACCP plan include verification tasks to be performed by plant personnel. Verification tasks would also be performed by FSIS inspectors. Both FSIS and industry will undertake microbial testing as one of several verification activities.

Verification also includes 'validation' – the process of finding evidence for the accuracy of the HACCP system (e.g. scientific evidence for critical limitations).

7. Establish effective record-keeping systems that document the HACCP plan.

The HACCP regulation requires that all plants maintain certain documents, including its hazard analysis and written HACCP plan, and records documenting the monitoring of critical control points, critical limits, verification activities, and the handling of processing deviations. Implementation involves monitoring, verifying, and validating of the daily work that is

compliant with regulatory requirements in all stages all the time. The differences among those three types of work are given by Saskatchewan Agriculture and Food.

Applications

- Fish and fishery products
- Fresh-cut produce
- Juice and nectar products
- Food outlets
- Meat and poultry products
- School food and services